SK 360°: A CENTRALIZED DIGITAL PLATFORM FOR YOUTH GOVERNANCE IN LIPA CITY

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CHAPTER 1
INTRODUCTION

In the era of rapid digital transformation in the Philippines, the adoption of modern information and communication technologies (ICT) has become essential in improving transparency, coordination, and efficiency in local governance. National initiatives led by the Department of Information and Communications Technology (DICT), such as the eGovPH App and integrated digital governance frameworks, highlight the growing shift toward centralized and technology-driven public service systems. Within this context, local governance institutions are encouraged to adopt digital platforms that support organized documentation, systematic reporting, and effective communication. Aligned with this direction, digital systems that integrate real-time communication, role-based access control, automated reporting, centralized document management, descriptive analytics, visualization dashboards, and KPI-based monitoring play a crucial role in promoting efficient, transparent, and data-driven governance, particularly in youth-led institutions.

The Sangguniang Kabataan (SK) serves as the primary youth governance body in the Philippines, mandated to lead youth participation, strengthen civic engagement, and promote accountability in local communities as stated in the SK Reform Act of 2015 [1]. Youth governance plays an important role in democratic development because institutions like the SK help build young leaders, promote public participation, and support long-term community involvement. National agencies such as the Department of the Interior and Local Government (DILG) emphasize that effective SK operations rely on organized documentation, coordinated reporting structures, and consistent monitoring of youth programs across barangays [2]. With this, having clear, transparent, and systematic processes becomes essential to good governance.

Within cities and municipalities, the SK Federation acts as the central coordinating body that connects all barangay SK councils. It is responsible for consolidating accomplishment reports, monitoring youth activities, issuing announcements, and guiding SK leaders in the implementation of community-based programs. However, many of these workflows are still done manually, relying on paper-based documentation and communication through informal channels such as group chats [3], [4]. These fragmented practices cause late submissions, inconsistent file formats, data loss, difficulty maintaining

organized and traceable records, and limited transparency in monitoring budget allocations, expenses, and liquidation reports across barangays.

These challenges are more evident in Lipa City, which has seventy-two barangays with different administrative capacities and digital practices. The Federation often experiences delays, incomplete submissions, inconsistent report formats, and difficulty consolidating documents across barangays. Per existing policy, financial validation remains the sole responsibility of the barangay SK Chairman prior to submission; the Federation and the proposed system perform no auditing function. This means the Federation's role is limited to receiving, recording, consolidating, and archiving reports without performing financial verification to follow existing administrative procedures. Leadership transitions every three years also create challenges because newly elected SK officials must gain access to previous reports and records. Without a centralized repository, documents become scattered, lost, or inaccessible, making it harder for new councils to continue existing programs.

Studies on digital governance show that local government units can improve efficiency, coordination, and transparency by adopting ICT tools that support existing workflows [5], [6]. While some LGUs already use digital platforms, most are not designed specifically for SK governance and often lack youth-focused features such as standardized accomplishment report formats, organized leadership profiles, centralized communication, and support for leadership transitions [7], [8]. This reveals a technological gap in digital tools made specifically for youth-governance processes in the Philippines.

To address these challenges, this study proposes SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City. The system integrates essential SK functions, including accomplishment and budget report submission and consolidation, federation-level monitoring, document preservation, communication, scheduling, leadership information, and public information access. The system supports file upload for completed accomplishment reports. Through the web application, budget reports may be submitted through direct data entry, which automatically generates standardized budget reports, or through document upload for externally prepared reports. The mobile application supports budget report submission through document upload only. Accomplishment report submission does not require direct data entry, report-category selection, reporting-period selection, or automatic report generation. Budget reports, on the other hand, are organized

through monthly, quarterly, semi-annual, and annual reporting categories to support monitoring and record management.

The system also includes yearly-based monitoring, budget allocation and liquidation management, leadership profile management, announcements, submission-slot management, meeting scheduling, an integrated event calendar, and centralized record preservation for authorized SK officials. Role-based access and administrator-managed account creation are implemented only for the SK Federation President/Admin, SK Chairman, and SK Secretary to ensure secure and organized system operations. Communication is enhanced through real-time chat and virtual meeting features, while the Point System Ranking encourages participation and accountability based on defined documentation, compliance, late-submission, and official-meeting participation criteria. The Dashboard Module also integrates descriptive analytics and monitoring functions that utilize visualization dashboards and Key Performance Indicators (KPIs) to monitor report submission trends, participation levels, completed and pending reports, yearly progress comparisons, and budget utilization through graphical charts and dashboards.

Public users may access selected information such as announcements, calendar information, leadership information, annual budget information, and the published Local Youth Development Plan (LYDP) through the web application without requiring account registration, while report management, rankings, and administrative functions remain restricted to authorized SK officials. The system also allows public users to submit feedback and comments through the web application, with the option to submit anonymously or non-anonymously. Email OTP verification is required before the feedback or comment is accepted. The mobile application is restricted to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts and does not provide access for public users. The system further supports leadership transitions by preserving historical records and leadership information, promoting organized, transparent, and efficient youth governance aligned with national guidelines and the operational needs of the SK Federation of Lipa City.

Overall, this study contributes to the modernization of youth governance through a centralized digital platform designed specifically for SK workflows, supporting transparency, budget accountability, federation-level coordination, descriptive analytics, and data-driven monitoring across barangays in Lipa City. While other LGU platforms

focus on general administrative needs, this system is designed for youth-governance requirements such as report standardization, inter-barangay coordination, communication, and leadership transition support.

1.1 PROJECT CONTEXT

The SK Federation of Lipa City oversees the activities, reports, and coordination of the seventy-two barangay SK councils, as mandated by the SK Reform Act of 2015 [1]. As the coordinating body, the Federation ensures youth programs are documented, monitored, and aligned with national and local governance standards. Despite this responsibility, the current workflow remains manual and fragmented. Barangay SK councils submit physical documents, use inconsistent digital formats, and rely on group chats for communication, causing delayed submissions, incomplete records, duplicated files, difficulty organizing citywide documentation, and limited transparency in tracking budget allocations, liquidation reports, and financial utilization [3], [4]. Variations in barangays' digital capabilities complicate coordination, leading to inefficiencies in report consolidation and monitoring.

These challenges affect the efficiency and transparency of SK operations. Delays in report submission and inconsistencies in documentation make it difficult for the Federation to generate accurate and timely reports required for evaluation and compliance with government agencies. The absence of a centralized system limits the ability to track participation, monitor performance, and maintain records across barangays. As a result, decision-making processes become slower, less data-driven, while coordination relies on manual follow-ups and informal communication channels, limiting the Federation's ability to monitor report completion, participation levels, budget utilization, and barangay performance.

The proposed system centralizes SK functions, including accomplishment and budget report submission and consolidation, federation-level report monitoring and quality documentation review, budget allocation and liquidation management, announcements, meeting scheduling and participation, leadership profile management, Point System Ranking, descriptive analytics, and real-time communication through messaging and virtual meetings to improve coordination, transparency, and monitoring across barangays.

Accomplishment reports are submitted through file upload of completed reports only and do not require direct data entry, report-category selection, reporting-period selection, or automatic report generation. Through the web application, budget reports may be submitted through direct data entry or document upload, while the mobile application supports budget report submission through document upload only. Budget reports are organized according to monthly, quarterly, semi-annual, and annual reporting categories for monitoring and record management. Selected submitted accomplishment reports and organized budget reports may be consolidated through the web application and compiled into consolidated PDF files for federation-level monitoring and documentation.

The Point System Ranking records positive or negative points according to defined documentation, compliance, late-submission, and official-meeting participation criteria among barangay SK councils. Role-based access is implemented for the SK Federation President/Admin, SK Chairman, and SK Secretary. The SK Federation President/Admin creates and manages SK Chairman accounts, while SK Chairmen manage their barangay leadership records, including the SK Secretary, Treasurer, and Council Members. Adding an SK Secretary creates or links the Secretary system account, while Treasurer and Council Member records do not receive system login access. The SK Federation President/Admin manages and schedules official meetings, while authorized SK Chairmen and SK Secretaries may access and participate in permitted meetings, including virtual meetings. The platform also includes an event calendar accessible to authenticated SK accounts according to their permissions.

Public users may access announcements, calendar information, leadership information, annual budget information, and the published Local Youth Development Plan (LYDP) through the web application without requiring account registration. The system provides a web-based feedback and comment function that allows public users to submit feedback either anonymously or non-anonymously, with email OTP verification required before the submission is accepted. Public users do not have access to the Point System Ranking. The mobile application is restricted to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts and does not provide access for public users. The system follows administrative procedures, wherein the financial validation of budget reports is handled by the barangay SK Chairman prior to submission. The platform does not perform budget verification. Its Document Management and

Archiving function instead preserves and provides authorized access to records produced or stored through connected system modules, including Accomplishment Reports, Budget Documents, and Event Records. It does not operate as a separate generic document-upload repository.

The system is supported by descriptive analytics and monitoring functions integrated into the Dashboard Module that utilize visualization dashboards and Key Performance Indicators (KPIs) to present insights such as report submission trends, participation levels, completed and pending reports, yearly performance comparisons, and budget utilization through charts and graphical dashboards based on submitted financial data. These analytics provide the SK Federation with a clearer understanding of performance and engagement across barangays, enabling more informed and data-driven decision-making.

Furthermore, the system preserves historical records and previous-year reports to support leadership transitions, enabling newly elected SK officials to access past submissions, documentation, and yearly performance records for comparison and continuity purposes. The system also supports the transition to a new SK administration by allowing the current President to continue or a succeeding President to be assigned, while outgoing official accounts are deactivated and eligible officials may be reappointed for the succeeding term. This addresses common concerns raised by the Commission on Audit (COA) and the Department of the Interior and Local Government (DILG) regarding record loss and weak documentation during leadership turnover. By maintaining a centralized and accessible repository of records, the system strengthens institutional continuity and reduces the risk of data loss.

This project supports Sustainable Development Goal (SDG) 16, which promotes transparent and accountable institutions, and SDG 17, which emphasizes strengthened coordination and partnerships [9], [10]. By offering a digital platform designed specifically for SK operations, the study addresses a significant gap in existing youth governance systems. Unlike general LGU platforms, the proposed system focuses on SK-specific reporting and coordination needs, making it valuable for improving documentation, communication, and continuity within the SK Federation of Lipa City. Overall, the system contributes to more efficient, transparent, accountable, and data-driven youth governance

practices by supporting centralized coordination, budget transparency, descriptive analytics, and federation-level monitoring within the SK Federation of Lipa City.

1.2 PURPOSE AND DESCRIPTION

The purpose of this study is to develop SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, a web- and mobile-based system designed to support the operational, administrative, and coordination needs of the SK Federation of Lipa City and its seventy-two barangay SK councils. The proposed system aims to centralize accomplishment and budget report submission and consolidation, document management and archival preservation, federation-level report monitoring and quality documentation review, announcements, submission-slot management, scheduling, communication, meeting management, participation and attendance, leadership information, Point System Ranking, dashboard-based descriptive analytics and monitoring, and selected public information access.

Accomplishment reports are submitted through file upload of completed report documents only. Through the web application, budget reports may be submitted through direct data entry or document upload, while the mobile application supports budget report submission through document upload only. Budget reports are organized according to monthly, quarterly, semi-annual, and annual reporting categories for monitoring and record management. The system provides authorized SK officials with role-based access to its functions. Public users may access designated announcements, calendar information, leadership information, annual budget information, and the published Local Youth Development Plan (LYDP) through the web application without requiring account registration. Public feedback and comments require successful email OTP verification before submission is accepted. Public users do not have access to the Point System Ranking. The mobile application is restricted to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts.

The study contributes to a youth-governance-focused digital platform that integrates reporting, documentation, communication, monitoring, and public information functions within a single system. Its contribution lies in adapting digital governance concepts to the specific operational requirements of the SK Federation, particularly the

management and consolidation of accomplishment and budget reports, centralized record preservation, federation-level monitoring of barangay submissions and quality documentation, leadership transition support, Point System Ranking, and descriptive analytics for monitoring submission trends, participation, performance, and budget utilization.

The study may benefit the SK Federation President/Admin by providing centralized monitoring, report consolidation, submission-slot and user management, report review, meeting management, and analytics; SK Chairmen and SK Secretaries by providing organized tools for report submission, communication, scheduling, meeting participation, and access to permitted records; and barangay SK councils by providing a structured platform for coordinating and maintaining their records. Public users and community stakeholders may benefit from improved access to selected announcements, calendar information, leadership information, annual budget information, and the published LYDP through the web application. The project may also serve as a reference for future researchers and developers interested in applying digital technologies to youth governance and other local government coordination processes.

1.3 OBJECTIVES OF THE STUDY

General Objective

The general objective of this study is to design and develop SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City that improves report management, budget transparency, communication, coordination, public transparency, and data-driven monitoring for better decision-making. Furthermore, the study aims to evaluate the system based on the ISO/IEC 25010 software quality attributes, including functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Specific Objectives

Specifically, this study aims to:

1. To develop a centralized digital platform for the submission, management, consolidation, federation-level monitoring, and archival preservation of accomplishment and budget-related records of SK councils in Lipa City.

    1.1. To support completed accomplishment report file uploads and the organization of budget reports according to monthly, quarterly, semi-annual, and annual reporting categories to improve monitoring and record organization.

    1.2. To implement yearly-based monitoring for budget-related reports and records for comparison, monitoring, and evaluation purposes.

2. To develop a budget management and liquidation module that records budget allocations, expenses, and financial utilization to promote accountability, transparency, and better governance among SK councils.

    2.1 To allow SK officials to manage budget reports through direct data entry or document upload on the web application and document upload on the mobile application.

3. To establish centralized announcements, scheduling, communication, meeting participation, Point System Ranking, public information and feedback, and account and leadership transition management.

    3.1. To provide public users with access to selected public information, including the published LYDP, and allow OTP-verified feedback.

4. To implement a Point System Ranking based on documentation, compliance, late submissions, and official meeting participation.

5. To integrate descriptive analytics and monitoring functions that include visualization dashboards and Key Performance Indicators (KPIs) for monitoring submission performance, participation trends, completed and pending reports, budget utilization, and yearly progress.

6. To evaluate the developed system based on the ISO/IEC 25010 software quality attributes in terms of functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

1.4 SCOPE AND DELIMITATION

Scope of the Project

This study focuses on the design and development of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, which supports the actual operational processes of the Sangguniang Kabataan. The system covers the digital submission of barangay SK accomplishment reports, wherein authorized SK officials upload completed accomplishment report files for centralized storage, review, and consolidation. Accomplishment report submission does not require direct data entry, report-category selection, reporting-period selection, or automatic report generation. Through the web application, budget reports may be submitted through direct data entry or document upload, while the mobile application supports budget report submission through document upload only. Budget reports are organized according to monthly, quarterly, semi-annual, and annual reporting categories for proper recording, monitoring, liquidation tracking, and centralized archiving. The web-based Report Consolidation Module allows the SK Federation President/Admin to consolidate selected accomplishment reports and monthly, quarterly, semi-annual, and annual budget reports into PDF files for federation-level monitoring and documentation, including available quality documentation for barangay submissions. Through the mobile application, authorized SK officials can access Mobile Consolidation to monitor barangay submission statuses using total, submitted, and pending counts, apply year and reporting-period filters, search barangay records, and check available quality documentation. The mobile function supports monitoring and viewing only and does not generate consolidated PDF files. The platform also supports announcements, meeting management and participation, event coordination through an integrated calendar, real-time chat and virtual meetings, and participation monitoring through a Point System Ranking. Meeting attendance may include face-to-face and valid virtual participation, with finalized attendance reflected in the Point System Ranking. It also includes leadership profile management, reappointments and term transition support, allowing the succeeding administration to retain or assign the SK Federation President while preserving previous leadership records and deactivating outgoing official accounts. Role-based access and administrator-managed account creation apply only to the SK Federation President/Admin, SK Chairman, and SK Secretary. The Module Management function allows the SK Federation President/Admin to manage submission slots and their Upcoming, Open, Past Deadline, and Closed states, with late submissions accepted while

a slot remains Past Deadline until manually closed. The same module includes Public Portal Documents, where the SK Federation President/Admin can publish or replace the Local Youth Development Plan (LYDP). Public users may access selected announcements, calendar information, leadership information, annual budget information, and the published LYDP through the web application without registration. Public users do not have access to the Point System Ranking. Public feedback and comments may be submitted anonymously or non-anonymously through the web application after successful email OTP verification. The Document Management and Archiving function preserves and provides authorized access to records generated or submitted through connected system modules, including Accomplishment Reports, Budget Documents, and Event Records. Financial validation of budget reports remains the responsibility of the barangay SK Chairman prior to submission; the system does not perform financial verification. The mobile application is restricted to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts. The system is developed exclusively for the SK Federation of Lipa City within the academic research scope.

The platform emphasizes federation-level consolidation, descriptive analytics, and KPI-based monitoring to support efficient oversight and data-driven decision-making across all barangays in Lipa City. By incorporating a Point System Ranking, descriptive analytics dashboards, and mobile accessibility for authorized SK officials, it aims to enhance participation, monitoring, accountability, and engagement in SK activities. Overall, SK 360° supports transparency, coordination, budget accountability, institutional memory, and data-driven monitoring in local youth governance. It promotes a more organized and systematic approach to managing SK operations by integrating multiple functions into a single platform and encourages accountability and compliance with reporting and documentation requirements among Barangay SK councils.

Delimitation of the Project

This study is limited to a web and mobile application designed exclusively for the SK Federation of Lipa City. The system includes accomplishment report submission, budget reporting with monthly, quarterly, semi-annual, and annual navigation, budget allocation and liquidation management, announcements, meeting schedules, leadership

profiles, Point System Ranking, dashboard-based descriptive analytics and monitoring, an integrated event calendar, and real-time communication. It excludes automated budget auditing, predictive analytics, artificial intelligence-based analysis, comprehensive project evaluation tools, and integration with external LGU and COA databases. Accomplishment reporting is limited to uploading completed report files and excludes direct data entry, report-category selection, reporting-period selection, and automatic report generation. For budget reporting, direct data entry is available only on the web application, while mobile supports document upload only. Financial verification remains the responsibility of the barangay SK Chairman before submission. Mobile Consolidation is limited to monitoring and viewing and does not generate consolidated PDF files. Document Management and Archiving preserves and retrieves records from connected modules and is not a separate generic document-upload repository.

The study also excludes expansion to other municipalities or LGU departments, inter-city or nationwide SK coordination, online payment gateways, SMS broadcasting, advanced analytics beyond descriptive insights, offline operation, automated election modules, and extensive customization. A past start year is not accepted when creating a succeeding SK term. The leadership transition feature does not conduct or determine elections; it only manages system accounts and official records after leadership changes are determined. The development is limited to authorized SK officials in Lipa City. Public users may view selected announcements, calendar information, leadership information, annual budget information, and the published LYDP, and submit OTP-verified feedback through the web application only. Public users cannot access the Point System Ranking and do not need an account. The mobile application is restricted to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts. The system supports coordination, documentation, consolidation, monitoring, and record-keeping but does not replace mandatory hard-copy submissions required by the DILG or COA. These functions support organized financial documentation, role-based access, preservation of historical records, continuity across administrations, and monitoring of authorized SK activities within the defined current academic and operational system scope.

These delimitations keep the project focused and aligned with academic constraints while supporting centralized reporting and consolidation, budget transparency, descriptive analytics, and federation-level monitoring within the SK Federation of Lipa City.

CHAPTER 2
REVIEW OF RELATED LITERATURE, STUDIES AND SYSTEMS

2.1 RELATED LITERATURES/STUDIES

This section presents a thematic discussion of existing foreign and local literatures and studies related to youth governance, Sangguniang Kabataan (SK) operations, the use of digital systems in local governance, budget transparency, descriptive analytics, and the application of engagement technologies such as point-based ranking and real-time communication tools. The review highlights recurring issues, theoretical perspectives, and technological approaches that are relevant to the proposed project.

Youth Governance and the Sangguniang Kabataan

Republic Act No. 10742, also known as the Sangguniang Kabataan Reform Act of 2015 [1], was enacted to strengthen youth participation, transparency, and accountability in local governance. While the law granted fiscal autonomy and mandated leadership training for SK officials, several authors emphasize that operational challenges remain, particularly in documentation, coordination, and reporting processes. Studies indicate that the effectiveness of youth councils is strongly influenced by administrative capacity, leadership continuity, and access to organized digital systems that support governance functions beyond traditional manual practices. These findings suggest that legal reforms alone are insufficient without parallel technological support mechanisms that improve documentation, reporting, budget monitoring, public transparency, and data-driven decision-making.

Challenges in SK Operations and Documentations

Local studies reveal that many SK councils still rely on manual and semi-digital methods such as paper-based records, spreadsheet-based files, printed budget documents, liquidation records, and communication through social media group chats. Rosales [3] found that inconsistent documentation practices and delayed submission of accomplishment reports negatively affect program planning, monitoring, and evaluation. Similarly, Padillo and Vallespin [4] reported that fragmented communication and the absence of standardized reporting tools contribute to low perceptions of SK effectiveness and reduced youth participation. These findings suggest that procedural inefficiencies

weaken the intended outcomes of the SK Reform Act, particularly in report monitoring, financial documentation, transparency, and accountability.

Further, Flores et al. [11] emphasized that leadership transitions every three years often result in the loss of institutional memory due to poor record-keeping and the absence of centralized repositories. Damit [8] also highlighted that despite expanded mandates, SK councils remain constrained by limited organizational capacity and weak monitoring mechanisms. More recent assessments, such as multi-sectoral evaluations in Taguig City [14] and qualitative studies on advocacy organization in Pagadian City (Lastimado et al. [13]), continue to highlight persistent concerns over accountability, logistical barriers, resource constraints, and coordination issues in SK operations, reinforcing the need for structured digital support in documentation, consolidation, and inter-barangay collaboration. These studies collectively underscore the need for digital support systems that can institutionalize processes, preserve historical records, and ensure continuity across leadership terms.

Digital Governance and ICT in Local Government

Foreign and local literature on e-governance supports the adoption of information and communication technologies to improve transparency, efficiency, coordination, budget monitoring, and data-driven decision-making in local governments. Siar [6] observed that early LGU websites in the Philippines provided limited interactivity and failed to support operational workflows, focusing mainly on information dissemination. More recent frameworks from the United Nations Development Programme [5] emphasize that integrated digital platforms can enhance coordination, crisis response, and sustainable development when aligned with existing governance structures. Systems like Kabataan-Konek [15], a barangay-level web-based youth information management tool with data analytics, demonstrate progress in youth profiling and basic monitoring but remain limited to single-barangay scopes without federation-level integration or advanced engagement features. This limitation highlights the gap between isolated digital initiatives and scalable, city-level youth governance platforms that support federation-level consolidation, descriptive analytics, KPI monitoring, budget transparency, and public access.

Post-2020 initiatives underscore the potential of digital platforms for youth empowerment in the Philippines. For example, the Skill Our Future platform, launched by

UNDP in partnership with Microsoft and others [18], offers free online courses on digital literacy and employability skills to bridge the digital divide among underserved youth. While not governance-focused, it complements SK tools by building digital readiness for civic participation, aligning with the need for accessible platforms in youth-led institutions like the proposed SK 360°

Point System Ranking in Digital Governance and Youth Engagement

Recent foreign and local studies highlight the potential of performance-based monitoring mechanisms to improve motivation, participation, accountability, and task compliance in digital platforms. Point System Ranking refers to the use of quantifiable points, rankings, leaderboards, and performance indicators to monitor participation, timely submission, report completeness, and compliance in governance-related systems. These mechanisms are supported by gamification and gameful engagement studies, which suggest that the use of points and rankings can encourage healthy competition, accountability, and sustained engagement, particularly in youth-oriented digital platforms [12]. In governance-related applications, point-based monitoring mechanisms may encourage users to complete required tasks and participate actively without relying on monetary incentives. Systematic reviews of gameful civic engagement have also reported positive outcomes in motivation, civic learning, and enjoyment in e-participation platforms. In the context of SK operations, the Point System Ranking of SK 360° is designed to encourage barangay SK councils to submit reports on time, complete required documentation, participate in programs, and maintain consistent engagement through measurable performance points. This approach supports the need for motivational mechanisms that can promote sustained youth participation and involvement in governance activities [7].

Real-Time Communication and Video Conferencing in Governance Systems

The use of real-time communication tools such as instant messaging and video conferencing has become an essential component of modern digital governance systems. Studies indicate that these technologies improve coordination, reduce delays in decision-making, and minimize the need for frequent physical meetings. Video calls enable remote collaboration, especially for organizations managing multiple locations or jurisdictions. Integrated tools like chat and video conferencing are increasingly recommended for

distributed governance bodies to overcome logistical barriers, as seen in post-pandemic adaptations by Philippine local government units (LGUs), where platforms facilitated continued public service delivery, virtual consultations, and crisis communication amid movement restrictions. These technologies are particularly relevant for rural and geographically dispersed barangays, where transportation costs and connectivity challenges may hinder frequent in-person coordination. For youth governance bodies like the SK Federation, real-time chat and virtual meeting features support efficient coordination among barangays, especially in clarifying report submissions, budget-related concerns, event schedules, and federation-level announcements, facilitate virtual meetings, and ensure continuous communication despite time and logistical constraints.

Youth-focused studies further reinforce this need. The YouthLed Philippines and Social Weather Stations [7] national survey reported low trust and engagement among Filipino youth and recommended the use of digital platforms to increase accessibility, participation, and transparency in youth institutions. Collectively, these literatures and studies establish that manual SK workflows, weak digital infrastructure, and lack of engagement mechanisms hinder governance performance and youth involvement.

Data Privacy Considerations in Youth Governance Platforms

A critical consideration in developing digital platforms for youth governance is compliance with data privacy regulations, particularly in handling personal information of public users, SK officials, and system administrators, as in the Philippines, Republic Act No. 10173 (Data Privacy Act of 2012) [16] mandates safeguards for personal data processed by government entities, including LGUs, requiring transparency, legitimate purpose, proportionality, and rights such as access, correction, and erasure for data subjects, while imposing obligations on personal information controllers (e.g., government agencies or platform operators) to implement security measures, notify breaches, and ensure lawful processing, with penalties for unauthorized access or disclosure. For SK-focused platforms like the proposed SK 360° in Lipa City, implementing encryption, role-based access, OTP verification for public posting, and consent mechanisms is essential to protect sensitive data (e.g., leadership profiles, reports, and chat logs) while aligning with national standards and promoting secure, trustworthy public participation and governance engagement, which is particularly vital in a federated system involving

multiple barangays, where real-time features like chat and virtual meetings could involve sharing personal details.

In summary, both foreign and local sources consistently point to documentation inefficiencies, coordination gaps, leadership transition issues, and limited engagement in SK operations, while also emphasizing the potential of ICT solutions, Point System Ranking, descriptive analytics, and real-time communication tools to address these challenges. These insights directly inform the design, features, and long-term vision of the proposed system, particularly its centralized reporting, budget transparency, Point System Ranking, descriptive analytics, KPI monitoring, real-time communication, and web-based public viewing features, ensuring that it is responsive, transparent, inclusive, and adaptable to evolving governance and monitoring needs. The system also supports public participation through web-based feedback and commenting features, where public users may submit anonymous or non-anonymous feedback with email OTP verification required before the submission is accepted. The proposed SK 360° platform addresses these gaps by integrating accomplishment report submission through document upload, automated budget reporting, budget allocation and liquidation management, federation-level consolidation, Point System Ranking, descriptive analytics, KPI-based monitoring, real-time communication, web-based public viewing access, and historical preservation with leadership transition support into a centralized platform designed with data privacy and security considerations tailored to the SK Federation of Lipa City. In terms of report submission, accomplishment reports are submitted through document upload only. For budget reports, the web application supports either direct data entry or document upload, while the mobile application supports document upload only. The centralized structure also helps reduce scattered records and improves access to important information among authorized SK officials. Its web-based public features further support transparency by allowing community members to view selected information without requiring account registration. At the same time, the mobile application remains limited to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts to maintain proper role-based access. By preserving historical reports and leadership records, the system also supports continuity during transitions between SK administrations. This holistic approach overcomes fragmentation in the literature, positioning SK 360° as a scalable model to enhance youth participation, accountability, and sustainable continuity.

2.2 RELATED SYSTEMS

Table 1 Gap Analysis Table of Related Systems and Applications

Feature(s)	SKIES	NYC-SK	Kabataan - Konek	E-Task	SK 360°
Country	**Philippines**	**Philippines**	**Philippines**	**Philippines**	**Philippines**
Application Type	**Web**	**Web**	**Web**	**Web**	**Mobile and Web**
Accomplishment Report Submission	Partial	No	No	Partial	Yes
Budget Document Uploads and Automation	No	No	No	Yes	Yes
Role-Based Access Control	Yes	Limited	Limited	Yes	Yes
Announcements & Calendar	Limited	Yes	No	Yes	Yes
Real-Time Chat / Virtual Meetings	No	No	No	No	Yes
Point System Ranking	No	No	No	No	Yes
Descriptive Analytics and KPI Monitoring	Yes	Limited	Yes	Limited	Yes
Leadership History & Records	Partial	Partial	No	No	Yes
Federation-Level Consolidation	Partial	No	No	No	Yes

Table 1 presents the gap analysis of related systems and applications.

Several existing systems and applications have been developed to support youth governance, Sangguniang Kabataan (SK) operations, and local youth information management. These platforms vary in scope, functionality, and level of integration, with some focusing on specific areas such as youth profiling, performance monitoring, document management, public information, or transparency. Digital governance studies also indicate that information and communication technologies can support improved coordination, transparency, and efficiency in local governance [5], [6].

The SK Information and Evaluation System (SKIES) is a technology-enabled platform designed for SK councils, Local Youth Development Officers, and local government units. It provides functions for managing youth profiles, monitoring youth concerns, supporting Comprehensive Barangay Youth Development Plan preparation, organizing documents, and managing user roles and permissions [19]. Although SKIES provides youth data management, document organization, and governance-related monitoring functions, it does not provide the same combination of accomplishment report submission through document upload, budget and liquidation management, Point System Ranking, real-time communication, and federation-level report consolidation proposed in SK 360°.

NYC-SK, through the Sangguniang Kabataan Web Portal, provides SK, Local Youth Development Officer, and Local Youth Development Council members with access to information, templates, forms, report submission, learning materials, announcements, and other resources for SK governance [20]. While the platform supports information dissemination, report submission, training, and communication between the National Youth Commission and SK stakeholders, its reviewed functions do not provide the same integrated combination of budget and liquidation management, Point System Ranking, descriptive analytics, KPI-based monitoring, real-time chat and video communication, and federation-level consolidation proposed in SK 360°.

Kabataan-Konek is a barangay web-based youth information management system with data analytics [15]. It provides functions related to youth information management and data analysis. However, its barangay-level orientation differs from the federation-level scope of SK 360°. The proposed system is intended to support coordination, reporting, monitoring, and consolidation across the seventy-two barangay SK councils of Lipa City.

The e-TASK (Enhancing Transparency and Accountability of Sangguniang Kabataan) system is designed to promote transparency and accountability within SK operations [21]. While it addresses transparency-related requirements, its reviewed system information does not indicate the same integrated combination of real-time communication, Point System Ranking, descriptive analytics, KPI-based monitoring, organized monthly, quarterly, semi-annual, and annual budget reporting, and leadership-history preservation proposed in SK 360°. Therefore, SK 360° extends beyond transparency-oriented functions by integrating operational, reporting, monitoring, communication, and continuity-related capabilities within one platform.

The proposed SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City is designed to address the limitations identified across the reviewed systems by integrating major SK governance functions into a centralized web and mobile platform. It provides accomplishment report submission through document upload only, budget reporting through direct data entry or document upload on the web application, and budget document upload on the mobile application. Budget reports are organized through categorized monthly, quarterly, semi-annual, and annual navigation. The system also includes budget allocation and liquidation management, document management and

archiving, real-time chat and virtual meetings, Point System Ranking, descriptive analytics, KPI-based monitoring, visualization dashboards, leadership history preservation and term-transition support, yearly monitoring, and federation-level data consolidation. The platform implements administrator-managed, role-based access control for authorized SK officials. Public users may access selected announcements, calendar information, leadership information, annual budget information, and the published LYDP through the web application without requiring account registration. Public users do not have access to the Point System Ranking. The system's handling of personal information is designed with consideration of the Data Privacy Act of 2012 (RA 10173) [16].

The comparison shows that while existing systems address specific aspects of SK operations, the reviewed platforms do not provide the same combination of reporting automation, budget transparency, liquidation tracking, federation-level consolidation, Point System Ranking, descriptive analytics, KPI-based monitoring, real-time communication, and leadership continuity and term-transition support offered by SK 360°. Existing systems provide valuable solutions for specific functions such as youth information management, evaluation, document organization, report submission, information dissemination, and transparency, but these functions remain distributed across different platforms. This fragmentation creates a need for an integrated system capable of supporting the broader operational requirements of the SK Federation.

Internationally, initiatives such as the DEEP Linking Youth project demonstrate how digital platforms can support youth participation and engagement through technology-enabled approaches [17]. Although its scope and implementation context differ from the Philippine SK setting, it demonstrates the potential of digital tools to strengthen youth participation and engagement. This supports the relevance of developing SK 360° as a centralized platform that combines operational management, monitoring, communication, transparency, and youth engagement within the specific context of the SK Federation of Lipa City.

2.3 SYNTHESIS

The reviewed studies, systems, and related literature demonstrate that digital technologies can support youth governance through improved information management,

documentation, monitoring, communication, transparency, and participation. Existing platforms such as SKIES, NYC-SK, Kabataan-Konek, and e-TASK address different aspects of youth governance. SKIES provides youth data management, issue monitoring, planning, document organization, and role management, while the NYC-SK Web Portal provides information, forms, report submission, learning resources, and communication from the national level to barangays [19], [20]. Kabataan-Konek focuses on youth information management and data analytics [15], while e-TASK promotes transparency and accountability in Sangguniang Kabataan operations [21]. These systems demonstrate the value of applying digital platforms to youth governance but also show that existing solutions have different scopes and areas of specialization.

The reviewed literature also indicates that SK operations continue to encounter challenges involving reporting, documentation, financial processes, participation, and administrative capacity. Previous research has identified difficulties related to compliance, reporting requirements, financial processes, and the capacity of SK officials to perform their responsibilities effectively [11], [13], [14]. These findings reinforce the need for digital tools that do not merely provide information but also support the actual operational processes of SK officials. Digital governance literature also supports the use of ICT to strengthen coordination, transparency, and efficiency in local governance [5], [6].

The identified gap is not the absence of digital platforms for youth governance, but the lack of an integrated platform combining major SK Federation operational requirements. Reviewed systems provide useful functions individually but do not offer the same combination of accomplishment report submission through document upload, budget and liquidation documentation, centralized document archiving, federation-level report consolidation, real-time communication, Point System Ranking, descriptive analytics, KPI-based monitoring, and leadership-history preservation proposed in SK 360°. Thus, the project combines and extends existing functions while addressing the specific operational requirements of the SK Federation of Lipa City.

The proposed SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City is therefore justified by the need for a system specifically designed around the operational structure of the Lipa City SK Federation and its seventy-two barangay SK councils. Unlike platforms focused mainly on youth information, evaluation, training, transparency, or individual governance functions, SK 360° provides a unified environment for

accomplishment report submission through document upload only, budget and financial documentation, document management, communication, scheduling, monitoring, analytics, performance ranking, and federation-level consolidation. For budget reporting, the web application supports direct data entry or document upload, while the mobile application supports document upload only. Authorized SK officials can access essential functions through web and mobile devices, while selected public information, including the published LYDP, is available through the web application only. Public users do not have access to the Point System Ranking. The mobile application is restricted to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts.

The project also supports institutional continuity by preserving historical reports, documents, leadership information, and yearly records across leadership transitions. Because SK leadership and responsibilities change over time, organized records help maintain continuity in programs and administrative processes. This emphasis on continuity, centralized records, and federation-level monitoring distinguishes SK 360° from platforms focused mainly on individual or barangay-level information management.

The proposed system builds upon rather than replaces useful functions demonstrated by previous systems. SKIES shows the value of centralized youth information, document repositories, planning, and role management [19]; NYC-SK demonstrates centralized information, report submission, training, and communication [20]; Kabataan-Konek applies data analytics to youth information management [15]; and e-TASK demonstrates digital tools for SK transparency and accountability [21]. SK 360° combines these concepts in a unified operational context and adds functions based on Lipa City SK Federation requirements.

The project is justified by a specific integration and contextualization gap rather than by claiming that existing youth governance systems are inadequate. A centralized platform is needed to support the Lipa City SK Federation and effectively connect its barangay SK councils through standardized reporting, centralized documentation, communication, monitoring, analytics, and federation-level consolidation. SK 360° addresses this gap through an integrated web and mobile platform designed around these requirements, making it distinct in scope, functionality, and intended organizational context from the reviewed systems.

CHAPTER 3
TECHNICAL BACKGROUND

This chapter presents the detailed technologies to be used in the development of the proposed system. Also, a thorough discussion on the technical description of the proposed system and its components are described.

3.1 DETAILS OF THE TECHNOLOGY TO BE USED

This section discusses the technologies and resources essential to the development of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City. It covers the software, hardware, network, and peopleware components that support the system's functionality, security, scalability, centralized monitoring, descriptive analytics, and real-time coordination features. The selected technologies are compatible with web and mobile platforms and are designed to support SK operations such as accomplishment and budget report submission, budget liquidation tracking, document archiving, communication, scheduling, Point System Ranking, and descriptive analytics. Development tools, frameworks, and database systems were utilized to ensure secure data handling, efficient system performance, KPI-based monitoring, visualization dashboards, and real-time communication. The involvement of system analysts, developers, and designers ensures that the system is properly designed, developed, and maintained to meet the technical and operational needs of youth governance.

In selecting the appropriate technologies, careful consideration was given to their reliability, scalability, and compatibility with the system's requirements. The chosen tools and platforms are capable of supporting multiple users across different barangays, ensuring consistent system performance and accessibility. These technologies also enable seamless integration of core features such as real-time communication, centralized data management, budget transparency monitoring, Point System Ranking, descriptive analytics dashboards, and KPI-based monitoring, which are essential in addressing the operational challenges identified in the study.

Furthermore, the selected technologies are designed to support flexibility and future system enhancements without requiring major modifications to the existing structure. Security measures are also considered through the use of reliable frameworks and controlled access mechanisms to protect sensitive information. Overall, these

technological components contribute to the development of a robust, scalable, and data-driven system suitable for the dynamic operational and monitoring needs of youth governance in Lipa City.

Software/Hardware Requirements

Table 2 Software Requirements for Systems Development

Software Classification	Software Name
Integrated Development Environment (IDE)	Visual Studio Code – used as the primary code editor for developing the web and mobile application due to its flexibility, extension support, and compatibility with multiple programming languages.
Programming Languages and Frameworks	PHP 8.2 (server-side scripting), JavaScript ES6 (client-side scripting), HTML5 / CSS3 (web structure and styling), Tailwind CSS 3.x (responsive and utility-first interface styling), Laravel 12.56.0 (web application framework), Flutter 3.24 (mobile application framework)
Database	MySQL 8.0 – used to store user accounts, barangay profiles, accomplishment reports, budget allocation and liquidation records, announcements, chat records, Point System Ranking data, KPI records, and descriptive analytics data.
Third-Party Libraries	Firebase (2025) (for real-time chat and messaging), Agora API (2025) (for virtual meetings), Bootstrap 5.3 (for responsive web interface design), Tailwind CSS 3.x (for responsive and utility-based web interface design), Chart.js 4.x (for descriptive analytics, KPI monitoring, and graphical dashboard visualization), PHPMailer / Mailgun API (2025) (for email OTP verification and email-based alerts).
Design & Prototyping	Figma – used for designing the user interface and user experience of the web and mobile application, including wireframes, prototypes, and layout planning.
Others	XAMPP 8.2.4 (local development server for backend and database testing), Hostinger (2025) (for live deployment and web hosting), GitHub (2025) (for version control and collaborative code management)

Table 2 lists the software requirements for developing mobile and web applications.

Visual Studio Code (Version 1.94, 2025)

Visual Studio Code is an Integrated Development Environment (IDE) utilized for writing, editing, and managing source code for both web and mobile applications. It was selected for the development of SK 360° due to its compatibility with the programming languages used in the system, including PHP, JavaScript, HTML, CSS, and Dart for Flutter. Its built-in debugging tools, extensive extension support, and integrated version control capabilities enable the development team to efficiently develop, test, and maintain the system. These features contribute to improved development productivity, code consistency, streamlined workflow, and efficient management of integrated web and mobile system components throughout the project lifecycle.

PHP (Version 8.2)

PHP functions as the main server-side scripting language of the system. It handles backend operations such as user authentication, role-based access control, accomplishment and budget report processing, budget liquidation management, document archiving, Point System Ranking computation, and database transactions. PHP was used in SK 360° because it provides reliable server-side processing and secure data handling, which are essential for managing official SK records and ensuring proper system functionality.

JavaScript (Version ES6, 2025)

JavaScript ES6 is used to implement dynamic and interactive features within the web application. It enhances user experience by enabling real-time updates, form validation, asynchronous data handling, and interactive dashboards. JavaScript helps SK 360° deliver smoother user interactions, dashboard visualization, dynamic report filtering, and faster system responses, particularly during report submission, data viewing, and dashboard navigation.

HTML5 / CSS3 (Version 2025)

HTML5 and CSS3 are used to structure and design the web interface of the system. HTML5 defines the layout and content of the application pages, while CSS3 provides styling, responsiveness, and visual consistency. These technologies were used

in SK 360° to ensure the system is accessible, visually organized, and compatible across various devices and web browsers.

Tailwind CSS (Version 3.x, 2025)

Tailwind CSS is used as the utility-first CSS framework for styling the web interface of SK 360°. It allows the development team to build responsive, consistent, and modern user interfaces directly within the application components. Tailwind CSS was used in SK 360° to support faster interface development, flexible customization, and a clean visual design that improves usability across different screen sizes.

Laravel (Version 12.56.0, 2026)

Laravel serves as the primary web application framework for backend development. It simplifies routing, authentication, database management, and API integration through its Model-View-Controller (MVC) architecture. Laravel was used in SK 360° to provide a structured, secure, and maintainable backend system suitable for handling governance-related processes, budget transparency management, descriptive analytics, and sensitive information.

Flutter (Version 3.24, 2025)

Flutter is a cross-platform development framework utilized for building the mobile version of the SK 360° system. For this project, the mobile application is developed and deployed specifically for Android devices. Flutter was selected to extend system accessibility by enabling SK officials to submit reports, view announcements, access calendars, receive notifications, and use other authorized mobile features through Android devices. This enhances system flexibility and allows authorized SK officials to perform essential tasks such as report submission, communication, monitoring, and viewing relevant system information conveniently through the mobile application.

MySQL (Version 8.0, 2025)

MySQL is the relational database management system used to store and manage system data, including user records, barangay information, accomplishment reports, budget allocation and liquidation records, chat logs, Point System Ranking data, KPI records, and descriptive analytics data. MySQL was used in SK 360° to ensure

centralized, secure, and efficient data storage that supports synchronized access across both web and mobile platforms.

Firebase

Firebase is used to support the real-time chat functionality of SK 360°. It enables authorized SK officials to send and receive messages through the system while providing synchronized message updates across connected users and devices. This supports faster communication and coordination among SK officials through both the web and mobile applications.

Agora API (Version 2025)

Agora API provides real-time communication capabilities used to support virtual meetings within the SK 360° system. It enables SK officials to conduct remote meetings, coordination sessions, and discussions without relying on external meeting platforms. Agora was selected for SK 360° because of its reliable real-time communication capabilities, low-latency streaming, scalable infrastructure, and strong support for both web and mobile integration.

Bootstrap (Version 5.3, 2025)

Bootstrap 5.3 is used as a responsive front-end framework that provides pre-designed components, grid-based layouts, and interface elements for the web application. It helps speed up interface development and maintain consistency across pages. In SK 360°, Bootstrap is used together with Tailwind CSS to support responsive design, structured layouts, and a clean professional appearance. This integration enhances the overall user experience by ensuring accessibility and visual consistency across different devices and screen sizes.

Chart.js (Version 4.x, 2025)

Chart.js is used to generate graphical representations of dashboard-based descriptive analytics and monitoring data, including submission compliance, submission trends, completed and pending reports, participation metrics, Point System Ranking summaries, annual budget information, and budget utilization. It enables the Dashboard

Module to present system information through charts and other visual representations, supporting federation-level monitoring and data-driven decision-making.

PHPMailer / Mailgun API (Version 2025)

PHPMailer and Mailgun API are utilized to support email-based system functions, particularly email OTP verification for public feedback and commenting and other configured email alerts. These tools provide reliable email delivery and support secure communication processes within SK 360°. They help ensure that required verification codes and email-based system messages are delivered to their intended recipients.

Figma (2025)

Figma is a cloud-based UI/UX design tool used to create the visual design, layout, and interactive prototypes of web and mobile applications. It was selected for SK 360° due to its real-time collaboration features and accessibility through web browsers without requiring installation. Figma supports wireframes, high-fidelity designs, and user flow prototypes while maintaining a consistent interface through its component-based system. These features improve design efficiency, team communication, and overall user-centered development.

XAMPP (Version 8.2.4, 2025)

XAMPP is a local development server environment utilized to integrate Apache, PHP, and MySQL for web application development. It was used in the development of SK 360° to enable offline testing and debugging of backend functionalities prior to system deployment, ensuring stability and proper functionality. This environment allows developers to simulate real-world server conditions during the development phase. As a result, it supports efficient testing, error detection, and system refinement before deployment to the live hosting environment.

Hostinger (Version 2025)

Hostinger is utilized as the web hosting platform for the live deployment of the SK 360° system. It provides secure hosting services, SSL encryption, and database support to ensure continuous system availability and data protection. The platform enables the system to remain accessible online for users across different barangays while maintaining

the integrity of stored information. Its reliable hosting infrastructure supports stable system performance and ensures consistent access to system features and services.

GitHub (Version 2025)

GitHub is used for version control and collaborative development. It allows the development team to manage source code, track changes, and maintain project documentation efficiently. GitHub helps ensure organized development and proper version tracking throughout the SK 360° project.

Table 3 Hardware Requirements for Systems Development

Hardware Component	Technical Specifications
Developer Workstation (Laptop/Desktop)	Desktop or laptop computer equipped with Intel Core i5 / AMD Ryzen 5 processor, 8 GB RAM, 256 GB SSD storage, and Windows 10 / Windows 11 or macOS 12 operating system.
Mobile Device (Testing Unit)	Android smartphone with Android 11 or later for testing mobile application features such as notifications, chat, meetings, and report submission.
Server	Cloud-based web hosting service (Hostinger) with PHP, MySQL, and SSL support for secure and continuous system access.
Networking Devices	Stable internet connection with a minimum speed of 5 Mbps, Wi-Fi router, and Ethernet connectivity to support real-time communication and data synchronization.

Table 3 indicates that these are the hardware specifications for developing and testing the proposed system.

Developer Workstation (Laptop/Desktop)

The developer workstation is used for system design, coding, debugging, and documentation. It runs the development tools, programming environments, and local servers needed to build and test the web and mobile applications efficiently. It also supports multitasking and performance-intensive processes required during development.

Mobile Device (Testing Unit)

Mobile devices are used to test the Android version of the system, ensuring proper functionality, responsiveness, and compatibility across supported Android platforms. The supported testing environment includes Android 11 and later versions.

Server

The server is hosted on a cloud-based web hosting platform (Hostinger), where the web application and database are deployed. It handles user requests, data processing, and secure storage of system information. The hosting platform ensures system availability, reliable performance, and secure access through SSL encryption for authorized users. It also supports continuous system access and basic data backup mechanisms.

Networking Devices

Networking devices provide internet connectivity required for system development, testing, and deployment. A stable network connection supports real-time data synchronization, document uploads, and communication features of the system. They ensure reliable and continuous access to online services and system resources.

Network Requirements

The SK 360° system relies on a stable and secure internet connection to enable real-time data synchronization between the web application, mobile application, server, and database. The network configuration supports smooth API communication, fast system response, and continuous access to core features such as user authentication, accomplishment report submission, document uploads, announcements, real-time chat, video call, descriptive analytics dashboards and KPI-based monitoring, and notification services. Reliable network infrastructure is essential to ensure system availability, data consistency, and secure access for SK officials across barangays in Lipa City, especially since the platform supports real-time coordination and centralized documentation.

Wi-Fi (Wireless Fidelity) / Wired Connection

During system development, testing, and deployment, a wired or wireless internet connection with a minimum speed of 5–10 Mbps is required for hosting, debugging, and synchronizing system components. This setup enables seamless communication between the web front-end, Laravel backend, and MySQL database in both local (XAMPP) and live hosting environments.

Mobile Data

When Wi-Fi or wired internet is unavailable, a mobile data connection with a minimum speed of 5 Mbps can be used to access and test system features. This allows authorized SK officials to submit reports, view announcements, access monitoring functions, participate in real-time chat, and receive notifications while on the go, with the application optimized for varying mobile data speeds in the Philippines.

Cloud Hosting Server

The production environment of the SK 360° system is deployed on a cloud hosting platform (e.g., Hostinger, 2025) to ensure scalability, security, and continuous availability. The cloud server supports SSL encryption for secure data transmission and enables centralized data storage and synchronization between web and mobile platforms, while managing real-time updates for announcements, notifications, chat messages, and dashboard monitoring data

Peopleware Requirements

Project Manager

The project manager oversees the entire development lifecycle of SK 360°, coordinates team activities, manages timelines, resources, and budgets, and ensures alignment with project objectives and academic standards. This role facilitates communication among team members, the capstone adviser, and the client (SK Federation of Lipa City), monitors progress, mitigates risks, and delivers regular status

reports to maintain project momentum and successful completion within the research timeframe.

System Analyst

The system analyst is responsible for gathering and analyzing system requirements from the SK Federation of Lipa City. This role ensures that user needs are properly translated into system features, workflows, and technical specifications before development begins.

UI/UX Designer

The UI/UX designer is responsible for designing the system's interface and user experience. This includes creating layouts, navigation flows, and visual elements that ensure the system is user-friendly, accessible, and suitable for SK officials and public users.

Front-End Developer

The front-end developer builds the user interface of the web and mobile applications using HTML, CSS, JavaScript, and Flutter. This role ensures that all visual components, dashboards, charts, and interactive monitoring features function correctly and that the system is responsive across different devices.

Back-End Developer

The back-end developer handles server-side logic using PHP and Laravel. This includes implementing authentication, report processing, budget liquidation handling, Point System Ranking computation, descriptive analytics processing, system rules, and API integration between the client interface and the database.

Full Stack Developer

The full stack developer is responsible for developing both the front-end and back-end components of the web and mobile applications using technologies such as HTML, CSS, JavaScript, Flutter, PHP, and Laravel. This role ensures seamless integration between the user interface, server-side logic, and database management. They also

handle debugging, system optimization, and end-to-end feature implementation across the entire application.

Documenter / Technical Writer

The documenter or technical writer is responsible for preparing the Research/Capstone Project documentation, including both the system documentation and the manuscript. This role ensures that all project details, methodologies, and outputs are clearly, accurately, and professionally presented. They also organize and maintain records of system processes, user guides, and technical specifications for future reference.

Database Administrator

The database administrator is responsible for designing, managing, and maintaining the MySQL database. This role ensures data integrity, security, backup, and efficient retrieval of records such as reports, budget records, liquidation data, documents, user accounts, Point System Ranking records, KPI data, and announcements.

Quality Assurance (QA) Tester

The QA tester performs system testing to identify bugs, errors, and performance issues. This role ensures that all system features, dashboards, analytics components, and monitoring functionalities operate correctly and reliably and that the platform meets usability and reliability standards before deployment.

Capstone Adviser

The project adviser provides technical guidance, monitors the progress of the project, and ensures that the development follows academic standards and project requirements. The adviser also provides recommendations and guidance throughout the development process to help ensure that the project remains aligned with its objectives and requirements.

3.2 PROJECT TECHNICAL DESCRIPTION

The proposed system, SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, is a web and mobile-based application designed to support the core operational processes of the Sangguniang Kabataan. It provides a centralized platform for managing accomplishment and budget reports, budget allocation and liquidation records, announcements, schedules, communication, and federation-level monitoring among SK officials across different barangays. The system enables authorized SK officials to submit accomplishment reports through document upload only. For budget reporting, the web application supports direct data entry or document upload, while the mobile application supports document upload only. Budget reports are organized through categorized monthly, quarterly, semi-annual, and annual report navigation for easier monitoring and consolidation. This allows flexible and efficient report management while ensuring organized documentation and record-keeping.

The platform also integrates key features such as real-time communication, an event calendar, leadership profile and term transition management, Point System Ranking, and descriptive analytics to enhance coordination, participation, and engagement. Additionally, the Dashboard Module integrates descriptive analytics and monitoring functions that utilize visualization dashboards, and KPI-based monitoring to present summarized operational and financial data through charts and graphical dashboards to support monitoring and data-driven decision-making. Overall, the system is designed to improve efficiency, transparency, accountability, centralized monitoring, and continuity in youth governance by providing a centralized and accessible digital solution for SK operations in Lipa City.

Special Features of the System

SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City introduces an integrated digital solution that addresses the common challenges faced by the SK Federation and barangay SK councils in managing reports, budgets, coordination, and communication. Unlike traditional paper-based and fragmented digital methods, the system consolidates governance processes into a single, secure, and accessible platform. By leveraging modern web and mobile technologies, SK 360° promotes transparency,

efficiency, accountability, descriptive analytics, and continuity in youth governance operations while supporting real-time collaboration among SK officials and stakeholders. The system is designed to support administrative efficiency, federation-level monitoring, budget transparency, and public transparency by balancing secure internal operations with accessible public information. Furthermore, the platform standardizes governance workflows, enabling consistent implementation of policies, reporting procedures, and communication practices across all barangay SK councils. This integrated approach helps strengthen coordination between the SK Federation and barangay levels while ensuring reliable access to accurate and up-to-date information.

Centralized Accomplishment and Budget Documentation

The system centralizes accomplishment and budget-related records generated or submitted through their respective reporting modules for monitoring, review, and record-keeping. Accomplishment reports are submitted through document upload only. For budget reporting, the web application supports direct data entry or document upload, while the mobile application supports document upload only. Budget reports are organized through monthly, quarterly, semi-annual, and annual report navigation to support monitoring, review, consolidation, and long-term record keeping. This centralized approach helps reduce document loss and redundancy while ensuring that records remain accessible to authorized users.

Role-Based Access Control

SK 360° implements role-based access to ensure data security and accountability. The authenticated user roles are the SK Federation President/Admin, SK Chairman, and SK Secretary, with permissions assigned according to their responsibilities. This prevents unauthorized access to sensitive documents and administrative functions while ensuring that users can access only the features relevant to their roles. Public users are not included in the role-based account structure and may only access selected public information and feedback functions through the web application without account registration.

System Notifications and Announcements

The platform provides system notifications to keep users informed of important updates such as report deadlines, submissions, announcements, upcoming activities, and monitoring reminders. This feature improves coordination and reduces delays in communication between the SK Federation and barangay SK councils. System notifications help users stay informed about reporting and activity requirements.

Integrated Communication and Messaging System

SK 360° includes a built-in messaging feature that allows SK officials to communicate directly within the platform. This eliminates the need to rely on external messaging applications, ensuring that all official discussions, clarifications, and coordination remain documented and centralized. Maintaining communication records within the system also supports transparency and accountability in official interactions.

Calendar and Event Management

The system includes a digital calendar for scheduling SK activities, meetings, deadlines, and youth programs. This feature helps users monitor upcoming events, report schedules, and submission deadlines and avoid scheduling conflicts, promoting better time management and organized program implementation. Shared visibility of schedules improves coordination across barangay and federation levels.

Point System Ranking and Performance Monitoring

The system incorporates a Point System Ranking and performance monitoring mechanism to encourage participation, compliance, and accountability among barangay SK councils. Points are assigned based on defined documentation, compliance, late-submission, and official-meeting participation criteria, allowing the SK Federation to monitor barangay performance through recorded positive and negative points. This feature promotes accountability, sustained participation, and recognition of consistent performance among participating councils.

Leadership Transition Support and Centralized Archiving

SK 360° includes a centralized document archive designed to support smooth leadership transitions within the SK councils. This feature preserves historical records generated or submitted through connected system modules, including accomplishment reports, budget documents, and event records, ensuring continuity of governance and reducing the risk of data loss during leadership transitions. Archived records remain accessible to authorized users for reference, yearly comparison, monitoring, evaluation, and documentation purposes.

Dashboard Analytics and Monitoring

The Dashboard Module presents summarized data such as submitted reports, completed and pending documents, participation records, Point System Ranking summaries, yearly comparisons, and budget utilization statistics. The system uses descriptive analytics, and KPI-based monitoring to organize and visualize operational and financial data entered by SK officials. Graphical representations such as bar charts, pie charts, KPI cards, and yearly comparison dashboards allow the SK Federation to easily monitor submission performance, participation levels, completed and pending reports, Point System Ranking performance, and how allocated budgets are distributed across SK activities and expenses. This allows the SK Federation to monitor the performance of barangay SK councils and supports data-driven decision-making and evaluation. Visual summaries, and KPI-based monitoring enable faster assessment of compliance, participation, budget utilization, and overall barangay performance.

Cloud Hosting and Data Security

The platform is deployed on a cloud-based hosting server with SSL encryption to ensure secure data transmission and storage. Sensitive information such as budget allocation records, liquidation files, reports, OTP verification records, and user credentials is protected through authentication mechanisms and database security controls, ensuring system reliability and data integrity. Regular backups and access monitoring further enhance system security and reliability.

Cross-Platform Accessibility

Developed using standard web technologies for the web application and Flutter for the mobile application, SK 360° provides a consistent and responsive user experience across supported devices. Authorized SK Federation President/Admin, SK Chairman, and SK Secretary accounts may access appropriate system functions through both web and mobile platforms. Public users are limited to selected public information and feedback functions through the web application only. This setup improves accessibility while maintaining the defined role and platform restrictions of the system.

User-Centered Interface Design

The system follows a clean and intuitive UI/UX design using Bootstrap 5.3 and Tailwind CSS to support responsive layouts, reusable components, consistent styling, and a modern user experience. Simple navigation, readable layouts, and clear icons help authorized SK officials use the web and mobile applications effectively, while public users can access the designated public interface through the web application only. This design approach improves usability, reduces the learning curve, and supports consistent system use among its intended users.

Email OTP-Verified Public Feedback and Commenting

SK 360° allows public users to submit anonymous or non-anonymous feedback and comments through the web application without requiring account registration. An email OTP is required before the submission is accepted to help verify authenticity and reduce unauthorized or automated postings. This feature supports organized public participation while maintaining appropriate privacy and verification controls. It also provides the SK Federation with a more structured way of receiving and reviewing community feedback. Verified submissions can help improve the reliability of public input received through the system. Public feedback and comments are available through the web application only and are not accessible through the mobile application.

System Architecture

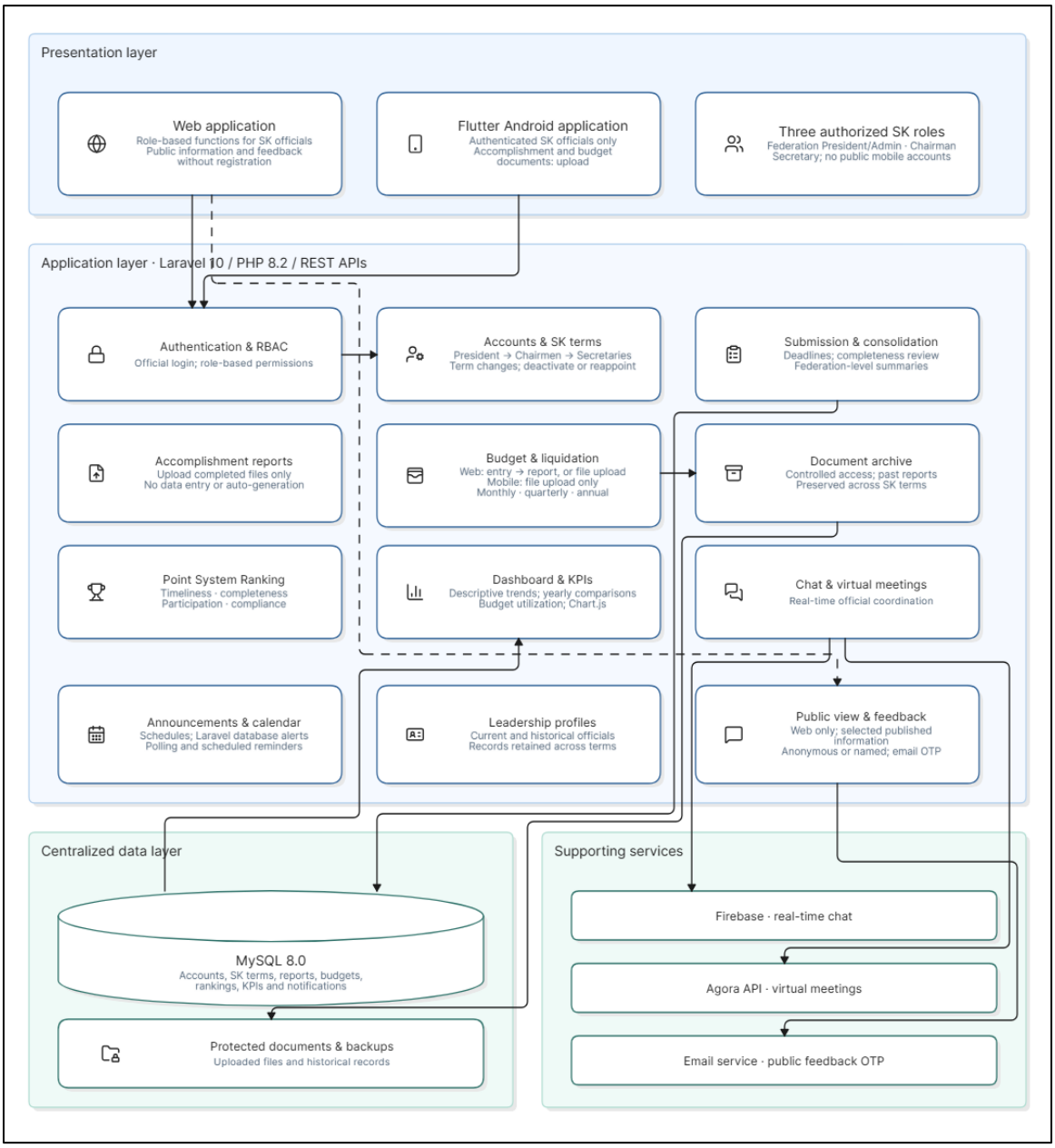


Figure 1 Overall System Architecture

This figure illustrates the system architecture of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, presenting the overall structure and interaction of its core components. It shows how the presentation layer, application layer, and data layer work together to manage user requests, process system operations, and store data efficiently. The architecture follows a structured approach that ensures

centralized data management, real-time communication, and seamless integration between web and mobile platforms, supporting the system's functionality, scalability, and reliability.

As illustrated in Figure 1, the system follows a three-tier client-server architecture that separates concerns for better maintainability, scalability, and security. The presentation layer consists of user interfaces, including the web application and the Flutter-based mobile application. These interfaces handle user interactions, input validation, and visual rendering, providing responsive and user-friendly experiences across devices. Authenticated users with administrator-managed roles, including the SK Federation President/Admin, SK Chairman, and SK Secretary, access authorized system functions through the web and mobile applications. Public users access selected public information and feedback functions through the web application only without requiring account registration.

The application layer is powered by PHP with Laravel, serving as the core business logic of the system. It processes client requests through RESTful APIs, enforces role-based access control, manages authentication, and handles operations such as report consolidation, budget allocation and liquidation management, document uploads, system notifications, communication, Point System Ranking, descriptive analytics, and KPI-based monitoring. This layer ensures secure data processing, system coordination, and compliance with defined workflows.

The data layer utilizes MySQL as the database management system, storing structured data such as user profiles, leadership and term records, reports, budget allocation and liquidation records, documents, chat records, Point System Ranking data, KPI records, and descriptive analytics data. It ensures secure data storage, reliable access, and data integrity through encryption and backup mechanisms.

This architecture supports scalability by allowing independent management of system layers and ensures secure and efficient system operations. It enables centralized coordination, real-time communication, and reliable data management, making it suitable for handling multiple users across different barangays. The analytics component of the system utilizes descriptive analytics, visualization dashboards, and KPI-based monitoring

to present summarized insights through charts and dashboards, supporting monitoring and decision-making processes.

This design also supports efficient handling of concurrent user requests from multiple barangays, ensuring smooth system performance during peak usage periods. The separation of system layers allows for easier maintenance and updates without affecting the overall functionality of the platform. In addition, the architecture enables seamless integration of external services such as notification systems and real-time communication tools, ensuring consistent and reliable system operations. Furthermore, the structured architecture enhances system flexibility, allowing future enhancements and additional features to be incorporated with minimal disruption.

Table 4 System Modules of the Project

Module No.	Module Name	Description
[SYSTEM/APPLICATION] Web		
1	Login Module	Enables secure authentication for authorized SK officials using administrator-issued credentials with input validation and error handling.
2	User Management Module	Allows the SK Federation President/Admin to create and manage SK Chairman accounts, including account activation, deactivation, role-based access, term transition, and reappointment management.
3	Dashboard Module	Displays an overview of reports, announcements, notifications, system updates, descriptive analytics, KPI-based monitoring, submission compliance, budget utilization, and other monitoring summaries through charts and dashboard visualizations in a centralized dashboard interface.
4	Home Page Module	Serves as the main entry point providing navigation, summaries, and access to system features.
5	Accomplishment Reporting Module	Enables authorized users to upload completed accomplishment report files for centralized storage, review, and consolidation.
6	Budget and Financial Documentation Module	Supports direct entry or upload of monthly, quarterly, semi-annual, and annual budget reports, including allocation and liquidation tracking.
7	Document Management and Archiving Module	Preserves and provides authorized access to records generated or submitted through connected system modules, including Accomplishment Reports, Budget Documents, and Event Records.
8	Announcement Module	Allows publishing of announcements, updates, and memos for SK officials and users.
9	Module Management Module	Manages accomplishment and budget submission slots with Upcoming, Open, Past Deadline, and Closed statuses.
10	Report Consolidation Module	Consolidates selected accomplishment and budget reports into PDF files and includes available quality documentation.
11	Notification Module	Delivers system notifications for updates, submissions, and deadlines within the system.
12	Activity Calendar and Scheduling Module	Displays events, meetings, submission schedules, and deadlines in a centralized calendar view.
13	Chat Module	Enables real-time communication among users through a built-in messaging system.

14	Meetings Module	Allows the President/Admin to manage meetings while SK Chairmen and Secretaries may participate, including virtually.
15	Leadership Information Module	Provides verified current and historical leadership information and allows authorized SK Chairmen to manage barangay leadership records, including the SK Secretary, Treasurer, and Council Members. Adding an SK Secretary also establishes the corresponding authorized Secretary system account, while Treasurer and Council Member records do not receive system login access.
16	Point System Ranking	Displays performance-based rankings and point summaries based on defined documentation, compliance, late-submission, and official-meeting participation criteria.
17	Public View Module	Allows public users to access selected information, including announcements, calendar, leadership, annual budget, and the published LYDP, and submit OTP-verified feedback without registration.
18	Profile Settings Module	Enables users to manage account details, passwords, and preferences securely.
[SYSTEM/APPLICATION] Mobile		
1	Mobile Login Module	Provides secure login for authorized SK officials with credential verification and error handling.
2	Mobile Home Page Module	Displays announcements, activities, and quick access features in a mobile-friendly interface.
3	Mobile Accomplishment Reporting Module	Enables authorized users to upload completed accomplishment report files using mobile devices.
4	Mobile Budget Documents Module	Enables authorized users to upload completed budget report documents using mobile devices for centralized storage, monitoring, budget allocation tracking, and liquidation documentation.
5	Mobile Consolidation Module	Allows authorized SK officials to monitor submission statuses, summary counts, reporting periods, and quality documentation through filters. The module is for monitoring and viewing only and does not generate consolidated PDF files.
6	Mobile Module Management Module	Allows the SK Federation President/Admin to create and manage submission slots for accomplishment and budget report submissions through the mobile application.
7	Mobile View Reports Module	Allows the SK Federation President/Admin to view, search, filter, and open available submitted accomplishment and budget report records through the mobile application.
8	Mobile Announcement Module	Allows the President/Admin to manage announcements while authenticated SK officials can view them.
9	Mobile Leadership Profiles Module	Provides access to verified information about SK officials.
10	Mobile Chat Module	Enables real-time messaging and communication among users.
11	Mobile Meetings Module	Allows authorized SK officials to participate in meetings, while meeting management is limited to the SK Federation President/Admin.
12	Mobile Activity Calendar Module	Allows the President/Admin to manage events while authenticated SK officials can view schedules, events, meetings, and deadlines.
13	Mobile Rankings Module	Shows Point System Rankings and point summaries based on defined performance criteria.
14	Mobile Profile Module	Allows users to manage account information and settings.
15	Mobile Notification Module	Provides system notifications for updates, announcements, submission reminders, meeting reminders, and other system activities.

CHAPTER 4
METHODOLOGY, SYSTEM PROCESSES AND DESIGNS

This chapter presents the methodology, requirement analysis and documentation, the design of the system using different models and the development of the proposed system application.

4.1 REQUIREMENTS ANALYSIS AND DOCUMENTATION

The SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City employed a structured requirements analysis and documentation process to address the identified needs of the proposed system. This section presents the methods used by the proponents to identify, analyze, and document the requirements of the project. It covers the research methodology used, project development methodology, functional requirements, and non-functional requirements of the proposed system.

The requirements analysis process involved direct consultation with the SK Federation President of Lipa City through an interview to gather information regarding the existing processes, operational needs, and system requirements of the SK Federation. The information gathered served as the basis for defining and documenting the required functionalities and operational characteristics of SK 360°. Through this process, the proponents established the requirements for the proposed web and mobile applications, including accomplishment reporting, budget and financial documentation, document management and archiving, announcements, submission management, report consolidation, notifications, activity scheduling, communication, leadership information, Point System Ranking, analytics and monitoring, public information access, and other supporting functions within the scope of the project.

Research Methodology Used

The proponents used a descriptive quantitative approach to evaluate the perceived software quality and acceptability of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City. This approach involved collecting numerical ratings through a structured questionnaire and analyzing the responses using descriptive statistics. The evaluation covered functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Requirements identification was supported by an interview with the SK Federation President of Lipa City. The interview provided information about existing processes, operational concerns, and required system functions. These findings informed the development of the web and Android mobile applications, while the questionnaire provided a separate assessment of the developed system from the respondents' perspectives.

The evaluation also includes functional and non-functional technical testing by five (5) IT professionals. Their questionnaire responses form part of the survey evaluation, while their technical test records provide separate evidence concerning the system's behavior, measured performance, and conformity with documented requirements.

Research Design

The survey component used a descriptive quantitative research design to summarize the respondents' assessments of SK 360°. The questionnaire gathered numerical ratings of the software quality characteristics included in the evaluation instrument. These ratings describe the respondents' perceptions and do not, by themselves, establish that all technical requirements have been satisfied.

Thirty-six (36) respondents participated in the survey: one (1) SK Federation President, ten (10) SK Chairmen, ten (10) SK Secretaries, ten (10) public users, and five (5) IT professionals. The respondent groups provided perspectives associated with administrative responsibilities, barangay-level operations, public information access, and technical evaluation.

The respondent distribution will be presented according to user role, while the system evaluation results will be summarized using the combined questionnaire responses of the thirty-six (36) respondents across the eight software quality characteristics. Technical testing findings will be discussed separately to distinguish perceived software quality from observed system behavior and measured technical results.

Respondents and Sampling

The survey included thirty-one (31) intended users and five (5) IT professionals, resulting in thirty-six (36) respondents. The intended users consisted of one (1) SK Federation President, ten (10) SK Chairmen, ten (10) SK Secretaries, and ten (10) public

users. The respondent distribution covered the principal user roles and included a separate group with a technical perspective.

Participant selection was based on the respondents' relevance to the system and the functions being evaluated. The SK officials represented the administrative and barangay-level users of SK 360°, while public users represented users of the publicly accessible web application. The IT professionals participated as both questionnaire respondents and technical evaluators.

The SK Federation President's evaluation concerned administrative, monitoring, approval, consolidation, and management functions. The SK Chairmen and SK Secretaries provided assessments related to reporting, budget and financial documentation, communication, notifications, scheduling, leadership information, and other authorized functions. Public users provided assessments concerning publicly accessible information and features.

The thirty-six (36) respondents constitute the survey sample, not the entire population of potential SK 360° users. The total reflects the combined respondent groups rather than a population-based sample-size calculation. Findings will therefore be interpreted in relation to the participating respondents and the system functions they experienced.

Data Collection Methods and Instrument

The study used an interview for requirements identification and a structured questionnaire for the survey evaluation. Separate software test documents support the functional and non-functional assessment conducted by the five (5) IT professionals. These instruments serve different purposes: the interview identifies operational requirements, the questionnaire gathers perceptions of software quality, and the test documents record observed technical results.

The questionnaire was drawn from the ISO/IEC 25010-based evaluation instrument presented in Management Information System of Barangay Daang Bukid, Bacoor City, Cavite by Santiago et al. (2022). It was used for the evaluation of SK 360° and retained the eight software quality characteristics included in the instrument:

functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The questionnaire contains subcriteria under each characteristic and uses a five-point rating scale. Respondents rated software quality using the following numerical values and verbal interpretations.

Numerical Rating	Verbal Interpretation
5	Excellent
4	Very Good
3	Good
2	Fair
1	Poor

The instrument requests the respondent's position and provides an optional name field. Respondent categories will be used to organize the analysis, while names will not be included in the presentation of individual ratings. The questionnaire does not contain a gender field; therefore, gender distributions will not be reported unless separately collected and documented.

The software test documents record the module or component tested, preconditions, test procedures, expected results, actual results, and relevant observations. Functional test cases will include a pass-or-fail status. Non-functional test documents will also record applicable measurements, scores, testing conditions, and technical remarks.

Technical Testing by IT Professionals

Five (5) IT professionals serve as technical evaluators of SK 360°. These same individuals also answered the research questionnaire and are included in the thirty-six (36) survey respondents. Their participation in both activities does not increase the total respondent count beyond thirty-six.

Functional testing will assess whether implemented features operate according to documented requirements and expected behavior. Where unit testing is conducted, it will examine individual software units or components in isolation. Testing complete modules

or connected workflows will be documented separately as functional or integration testing, as applicable.

Non-functional testing will assess relevant system qualities, including performance, accessibility, compatibility, reliability, and security. The evaluators will record the testing environment, tools used, observed issues, and measured results where applicable. Pass-or-fail judgments will be based on documented acceptance criteria rather than on questionnaire ratings.

Technical test results will remain separate from the questionnaire dataset. The survey provides respondents' assessments of software quality, while technical testing provides evidence from executed test cases and measurements. Both sources will inform the discussion of the system's strengths, limitations, and areas requiring improvement.

Data Collection Procedure

Requirements information was gathered through an interview with the SK Federation President of Lipa City. The information obtained was used to identify existing processes, operational concerns, and required system functions. These requirements served as a reference for system development and the preparation of evaluation materials.

The survey questionnaire was answered by thirty-six (36) respondents from the five identified respondent groups. The collected questionnaires will be reviewed for completeness and organized according to respondent category. Responses will be encoded as recorded, without changing answers to improve the apparent evaluation results.

The five (5) IT professionals will also document technical testing using the prepared test forms. Each record will identify the tested function or quality characteristic, testing conditions, actual observations, and results. Identified defects will be documented and, where corrections are implemented, subjected to retesting.

The survey dataset and technical testing records will be maintained separately. Missing questionnaire responses will not be replaced with invented ratings. The number of valid responses used in each calculation will be recorded to ensure that the reported results correspond to the available data.

Data Analysis

Questionnaire responses will be analyzed using descriptive statistics. Frequency and percentage will summarize the distribution of respondents by category. The weighted mean will describe the average rating for each questionnaire item, while the standard deviation will describe variation among the respondents' ratings.

The survey analysis will cover the eight software quality characteristics included in the evaluation instrument. The weighted mean and standard deviation will be calculated using the combined valid questionnaire responses of the thirty-six (36) respondents. Respondent categories will be presented separately in the demographic profile to describe the composition of the evaluation participants. Blank responses will be excluded from the relevant calculations and will not be treated as zero ratings.

The weighted mean will be interpreted using the five-point evaluation scale presented in Section 4.5, ranging from Poor to Excellent. Higher mean ratings indicate a more favorable perceived assessment of the system. Standard deviation will describe the consistency of ratings; it will not be interpreted as evidence that respondents rated the system positively.

Technical test results will be analyzed independently of the survey ratings. Functional results will be compared with expected behavior, while non-functional measurements will be assessed against the documented testing criteria. Defects, corrective actions, retest outcomes, and unresolved limitations will be reported as supported by the test records.

The survey findings will be interpreted as perceptions of software quality rather than proof of technical compliance. Ratings for characteristics that respondents could not directly observe, such as internal maintainability, will be discussed cautiously and considered alongside the IT professionals' technical findings.

Ethical Considerations

The proponents will observe ethical principles throughout the system evaluation and technical testing of SK 360°. All respondents and IT professionals will be treated with respect, fairness, confidentiality, and transparency.

Informed Consent

Before participation, all respondents and IT professionals will receive an informed consent form explaining the study's purpose, procedures, expected activities, collected information, possible risks and benefits, and use of the results. Participants will be given sufficient time to review the form and ask questions. Participation will begin only after voluntary consent is provided.

Voluntary Participation

Participation will be voluntary. Respondents and IT professionals may refuse to participate, skip questions, or withdraw at any time without penalty or negative consequences. Their decision will not affect their relationship with the proponents, SK Federation, academic institution, or other organizations involved in the study. Participants will be encouraged to provide honest responses based on their actual system experience.

Privacy and Confidentiality

Only information necessary for identifying the respondent group and conducting the evaluation will be collected. Individual names will not be included in the presentation of results. Survey evaluation results will be presented in aggregate, while respondent categories will be reported in the demographic profile. Technical findings will be presented separately without identifying individual evaluators.

Data Protection and Secure Handling

Completed questionnaires, test documents, encoded data, and related records will be stored securely. Digital files will be protected through passwords and access restrictions, while printed records will be kept in a secured location. Access will be limited to the proponents and authorized academic personnel. The proponents will observe the principles of confidentiality and data minimization under the Philippine Data Privacy Act of

2012, or Republic Act No. 10173. Personal information will be processed only for the legitimate purpose of evaluating SK 360°.

Protection from Harm and Possible Risks

The study is expected to involve minimal risk. Participants may experience minor inconvenience, time burden, or fatigue while using the system and completing the questionnaire. These risks will be minimized by providing clear instructions, allowing questions and reasonable breaks, and permitting participants to stop when necessary. Sensitive information unrelated to the study will not be collected.

The IT professionals will test the system using assigned test cases and will not be required to access unrelated personal information. Test accounts or controlled sample records may be used to reduce the exposure of actual confidential data.

Protection of Participants' Rights

Participants will be informed about the study before participation and may ask questions, refuse to answer any item, or withdraw without penalty. They will also be informed that their responses will be used for academic evaluation, system improvement, statistical analysis, and manuscript preparation. The proponents will not alter or misrepresent the collected responses.

Participants Requiring Special Protection

The study will include adult participants capable of providing informed consent. Public users will be at least eighteen years old unless an approved procedure for minors is established by the institution. Individuals unable to provide legal consent will not be included without the required authorization. The proponents will avoid recruiting individuals who may be vulnerable to pressure or exploitation.

Recruitment and Non-Coercion

Participants will be invited according to the established respondent groups and inclusion criteria. Recruitment will be conducted respectfully and transparently. Participants will be informed that participation will not affect their position, access to services, relationship with the SK Federation, or academic institution. No participant will

be promised favorable results, special treatment, or personal benefits in exchange for participation.

Benefits of the Study

The participants may not receive direct financial or material benefits from participating in the study. However, their feedback may help the proponents identify errors, usability concerns, access issues, performance problems, and other areas requiring improvement in SK 360°.

The study may benefit the SK Federation by providing information that can support the refinement of its reporting, documentation, communication, scheduling, monitoring, and public information functions. The results may also contribute to the development of a more organized and centralized digital platform for youth governance.

Compensation and Incentives

Participation will be voluntary and will not require payment from the respondents or IT professionals. No participant will be required to pay any fee to participate in the evaluation. Unless separately approved by the institution, the proponents will not provide financial compensation or incentives that could influence participants to provide favorable responses.

Conflict of Interest

The proponents will declare any actual or potential conflict of interest that may affect the conduct or interpretation of the evaluation. The proponents will not modify the questionnaire responses, test results, or technical findings to make the system appear more effective than its actual performance.

The results will be reported honestly, including failed test cases, negative feedback, technical limitations, and areas that still require improvement. The proponents will remain responsible for the accuracy and integrity of the data analysis and final conclusions.

Data Retention and Disposal

The collected questionnaires, test records, and encoded data will be retained only for the period necessary to complete the academic requirements, analysis, verification, and documentation of the study. After the required retention period, physical and digital records containing identifiable information will be securely disposed of or permanently deleted, subject to the requirements of the academic institution.

Publication and Reporting of Results

The results will be presented in summarized form. The manuscript will not identify individual respondents by name or disclose personal responses that could unnecessarily reveal their identity. Technical issues identified by the IT professionals will be reported as system findings rather than as personal failures of the evaluators.

The proponents will ensure that the final presentation reflects the actual findings of the evaluation. Positive and negative results will be considered in the interpretation of the system's performance and acceptability.

AI Disclosure Statement

Artificial intelligence tools may be used as supplementary assistance during the preparation of the manuscript and system documentation. These tools may assist with grammar checking, sentence organization, proofreading, formatting suggestions, and clarification of technical descriptions.

AI tools will not be used to fabricate respondents, questionnaire responses, test results, statistical values, interview answers, or technical findings. The questionnaire content, data collection, system testing, interpretation of results, and final research decisions will remain under the responsibility of the proponents.

All AI-assisted suggestions will be reviewed, verified, and revised by the proponents before inclusion in the manuscript. The proponents will ensure that the final content accurately reflects the actual scope, development process, evaluation procedures, and findings of SK 360°. The proponents remain fully responsible for the accuracy, originality, ethical compliance, interpretations, conclusions, and final presentation of the study.

Project Development Methodology

The proponents employed the Agile Software Development Methodology in developing SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City. Agile was selected because it supports iterative development, testing, feedback, and revision, allowing the proponents to refine the system as requirements and operational concerns are clarified. This approach is suitable for SK 360° because its interconnected web and Android mobile functions require coordinated development and continuous evaluation to remain aligned with the operational needs of the SK Federation of Lipa City.

The development framework consists of seven stages: Plan, Design, Develop, Test, Deploy, Review, and Launch, as illustrated in Figure 2. The Plan stage establishes the requirements and development priorities, while the Design, Develop, Test, Deploy, and Review stages form an iterative cycle through which system functions are implemented, evaluated, and refined. Feedback obtained during the Review stage informs succeeding design and development activities. The Launch stage marked the operational release of the system following testing, revisions, and implementation preparations.

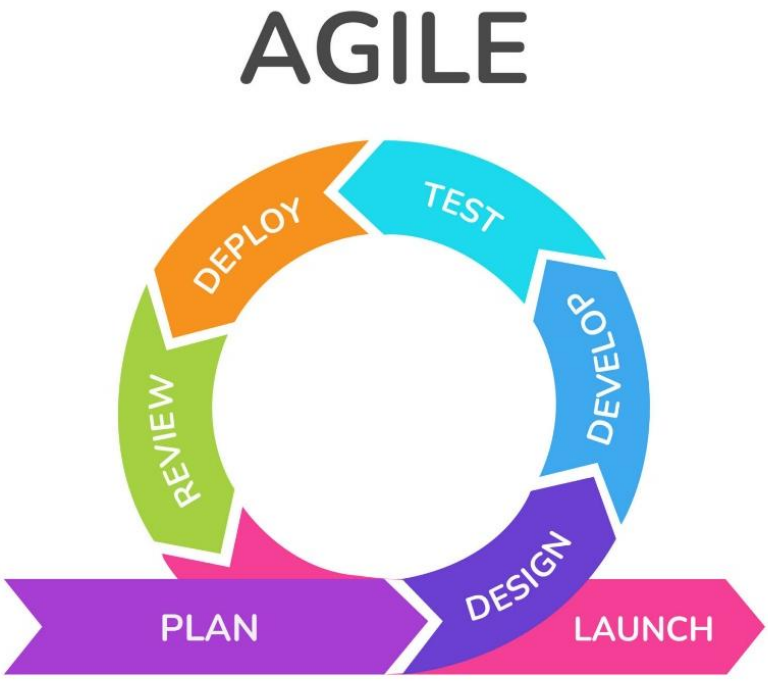


Figure 2 Project Development Methodology

The Plan stage involved identifying the operational needs and system requirements of the SK Federation of Lipa City through an interview with the SK Federation President, examining existing processes and operational concerns, documenting the required system functions, organizing and prioritizing the requirements, determining development tasks, assigning team responsibilities, and establishing the project schedule.

The Design stage focused on translating the identified requirements into the structure, workflows, and interfaces of the system. The proponents prepared the system architecture, database structure, process flows, user interface designs, and prototypes for the web and Android mobile applications while considering user roles, access permissions, and platform-specific functions. Design revisions supported changes identified during subsequent development and review activities.

The Develop stage involved implementing the planned system functions and integrating the user interfaces, application logic, database, and supporting services. Development covered accomplishment reporting, budget and financial documentation, document management and archiving, announcements, submission-slot management, report consolidation, notifications, scheduling, communication, leadership information, Point System Ranking, descriptive analytics, and public information access. Functions were implemented according to the defined scope and platform-specific requirements.

The Test stage focused on evaluating the implemented functions against their documented requirements and expected behavior. The proponents examined valid and invalid inputs, role-based access restrictions, system outputs, and interactions among connected modules. Testing identified errors, inconsistencies, and other issues requiring correction. Findings were documented, and affected functions were revised and retested as development progressed.

The Deploy stage involved preparing and configuring the web and Android applications in the designated deployment environment for testing and evaluation. Activities included configuring the application server, database, application programming interfaces, and supporting services, as well as preparing the Android application for installation on compatible devices. Deployment checks verified that the applications could communicate with the centralized backend and required services. Updated versions were deployed during successive iterations to support further evaluation.

The Review stage involved examining testing findings and available user or client feedback to determine whether the implemented functions addressed the identified requirements. The proponents identified issues, prioritized necessary changes, and documented revisions. These findings guided further design, development, testing, and deployment activities. Repeating this cycle supported improvements in system functionality, usability, reliability, and overall operation.

The Launch stage involved making SK 360° available for operational use following system testing, revisions, and implementation preparations. The SK Federation President reviewed the web and mobile applications on September 28, 2026, as part of the client acceptance process. By October 10, 2026, the system was already live and had been formally turned over to the SK Federation of Lipa City. The proponents also presented the system to the SK Federation President and Barangay SK Chairpersons and conducted the survey for the study. The web application provides access to authorized SK officials and designated public information, while the Android application is limited to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts.

Table 5 Project Development Phases, Activities and Deliverables

Phases	Activities	Deliverables
Plan	Conduct an interview with the SK Federation President; identify existing processes, operational concerns, and system requirements; document and prioritize requirements; assign team responsibilities; and establish development tasks and schedules.	Requirements documentation; prioritized requirements; initial module list; development plan; task assignments; project schedule.
Design	Prepare the system architecture, database structure, workflows, user interfaces, and prototypes; define role-based access and platform-specific functions; and revise designs based on identified requirements and feedback.	System architecture; database design; process flows; interface designs; system prototypes.
Develop	Implement the web and Android mobile functions; integrate interfaces, application logic, database operations, and supporting services; and revise implemented components as needed.	Working application increments; implemented modules; integrated database and services; updated source code.
Test	Execute test cases; compare actual and expected results; check validation, access restrictions, outputs, and module interactions; document issues; and retest corrected functions.	Test records; documented defects and issues; retest results.
Deploy	Prepare the deployment environment; configure the server, database, APIs, and supporting services; prepare the Android application for installation; verify connectivity; and deploy updated versions for evaluation.	Configured deployment environment; deployed evaluation versions; installable Android application; deployment verification records.
Review	Examine testing findings and available user or client feedback; identify and prioritize improvements; document revisions; and return identified changes to the appropriate development activities.	Review findings; feedback records; prioritized revision tasks; updated requirements or designs where necessary.
Launch	Complete final deployment checks; prepare user accounts and access; conduct user orientation; release the evaluated system for operational use; and provide system documentation and initial technical support.	Operational system release; configured user access; user guides; orientation records; initial support arrangements.

Functional Requirements

The functional requirements of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City define the functions and services that the proposed system must provide to its intended users. These requirements were identified based on the operational needs of the SK Federation and the intended functions of the proposed web and mobile applications. The functional requirements cover user authentication and account management, report and financial documentation, communication, scheduling, monitoring, analytics, public information access, and other supporting functions of the system.

Table 6 presents the major functional modules of the SK 360° web and mobile applications and their corresponding descriptions. The table provides an overview of the required system functionalities, while the detailed functional requirements for each module are presented in the Appendix section of the manuscript.

Table 6 Functional Requirements of the Project

Functional Requirement ID	Function Name(s)	Description
WEB-01	Login Module	Enables secure authentication for authorized SK officials using administrator-issued credentials, including input validation and error handling.
WEB-02	User Management Module	Allows the SK Federation President/Admin to create and manage SK Chairman accounts according to the defined role-based access control. The module also supports SK term transition by allowing the President/Admin to initiate a new term, retain or assign the President for the succeeding administration, deactivate outgoing official access, and reappoint eligible officials while preserving previous service records.
WEB-03	Dashboard Module	Displays an overview of reports, announcements, notifications, system updates, descriptive analytics, KPI-based monitoring, submission compliance, budget utilization, and other monitoring summaries through charts and dashboard visualizations in a centralized dashboard.
WEB-04	Home Page Module	Serves as the main entry point of the web application, providing navigation, summaries, and access to system features.
WEB-05	Accomplishment Reporting Module	Enables authorized users to upload completed accomplishment report files for centralized storage, review, and consolidation.

WEB-06	Budget and Financial Documentation Module	Supports budget report submission through direct data entry or document upload. Valid budget information submitted through direct data entry is used to automatically generate a standardized budget report. The module also supports monthly, quarterly, semi-annual, and annual budget organization, budget allocation tracking, liquidation documentation, monitoring, and visual analytics for transparency.
WEB-07	Document Management and Archiving Module	Preserves and provides authorized access to records generated or submitted through connected system modules, including Accomplishment Reports, Budget Documents, and Event Records.
WEB-08	Announcement	Allows the SK Federation President/Admin to create, publish, update, and manage official announcements, updates, and memoranda, while intended users can view published information according to their access level.
WEB-09	Module Management Module	Allows the SK Federation President/Admin to manage accomplishment and budget submission slots with Upcoming, Open, Past Deadline, and Closed statuses.
WEB-10	Report Consolidation Module	Supports the consolidation of selected accomplishment reports and monthly, quarterly, semi-annual, and annual budget reports into consolidated PDF files, including available quality documentation for barangay submissions.
WEB-11	Notification Module	Provides system notifications for updates, submissions, and deadlines within the system.
WEB-12	Activity Calendar and Scheduling Module	Allows the SK Federation President/Admin to create and manage events, meetings, submission schedules, and deadlines through a centralized calendar, while authorized users can view available schedule information according to their access level.
WEB-13	Chat Module	Enables real-time communication among authorized users through an integrated messaging system.
WEB-14	Meetings Module	Allows the President/Admin to manage meetings while SK Chairmen and Secretaries may participate, including virtually.
WEB-15	Leadership Information Module	Provides verified current and historical leadership information and allows authorized SK Chairmen to manage their barangay leadership records, including the SK Secretary, Treasurer, and Council Members. When an SK Secretary is added, the system creates or links the corresponding Secretary account for authenticated system access. Treasurer and Council Member records do not receive login accounts.
WEB-16	Point System Ranking Module	Displays rankings and point summaries based on defined documentation, compliance, late-submission, and official-meeting participation criteria.

WEB-17	Public View Module	Allows public users to access selected announcements, calendar information, leadership information, annual budget information, and the published LYDP without registration, and submit OTP-verified feedback or comments.
WEB-18	Profile Settings Module	Allows users to securely manage account details, passwords, and preferences.
MOB-01	Mobile Login Module	Provides secure authentication for authorized SK officials through mobile devices with credential verification and error handling.
MOB-02	Mobile Home Page Module	Displays announcements, activities, and quick-access features through a mobile-friendly interface.
MOB-03	Mobile Accomplishment Reporting Module	Enables authorized users to upload completed accomplishment report files using mobile devices.
MOB-04	Mobile Budget Documents Module	Enables authorized users to upload completed budget and financial documents using mobile devices for centralized storage, monitoring, budget allocation tracking, and liquidation documentation.
MOB-05	Mobile Consolidation Module	Allows authorized SK officials to monitor submission statuses, summary counts, reporting periods, and quality documentation through filters. It does not generate consolidated PDF files.
MOB-06	Mobile Module Management Module	Allows the SK Federation President/Admin to create, view, search, filter, update, and manage submission slots for accomplishment and budget report submissions through the mobile application.
MOB-07	Mobile View Reports Module	Allows the SK Federation President/Admin to view, search, filter, and open available submitted accomplishment and budget report records through the mobile application.
MOB-08	Mobile Announcement Module	Allows the SK Federation President/Admin to create, publish, update, and manage announcements through the mobile application, while authenticated SK officials can view published announcements through the centralized mobile feed.
MOB-09	Mobile Leadership Profiles Module	Provides access to verified information about SK officials through the mobile application.
MOB-10	Mobile Chat Module	Enables real-time messaging and communication among users through mobile devices.
MOB-11	Mobile Meetings Module	Allows authorized SK officials to participate in meetings, while meeting management is limited to the President/Admin.
MOB-12	Mobile Activity Calendar Module	Allows the SK Federation President/Admin to create and manage events or activities through the mobile application, while authenticated SK officials can view schedules, events, meetings, and deadlines through a mobile-friendly calendar.
MOB-13	Mobile Rankings Module	Displays Point System Rankings and point summaries based on defined performance criteria.

MOB-14	Mobile Profile Module	Allows users to manage their account information and settings through mobile devices.
MOB-15	Mobile Notification Module	Provides system notifications for announcements, updates, submission reminders, meeting reminders, deadlines, and other system activities through the mobile application.

Non-Functional Requirements

The non-functional requirements of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City define the operational characteristics, quality attributes, and constraints that the system must satisfy in addition to its functional capabilities. These requirements describe how the web and mobile applications should perform and operate to provide an effective, reliable, secure, accessible, and maintainable system for its intended users.

The non-functional requirements were established based on the operational needs and intended use of the system and are aligned with the software quality attributes identified for the project. Consideration was given to functionality, performance efficiency, scalability, portability, compatibility, reliability, availability, maintainability, security, and usability. These requirements are intended to ensure that SK 360° can support the centralized management, monitoring, communication, documentation, and public information functions defined within the scope of the project.

Table 7 presents the identified non-functional requirements and the corresponding specifications and constraints of the proposed system for both the web and mobile applications.

Table 7 Non-Functional Requirements of the Project

Non-Functional Criteria	Description
Functionality	The system should: 1. Provide reporting, financial documentation, document management, announcements, submissions, consolidation, communication, scheduling, rankings, analytics, monitoring, and public information functions. 2. Perform according to documented requirements. The system is constrained to: 1. Functions within the approved scope of SK 360°. 2. Role-based access for restricted functions.
Performance	The system should: 1. Process requests, submissions, retrieval, notifications, and dashboard operations efficiently. 2. Provide responsive web and mobile interactions.

	The system is constrained to: 1. Available internet connection and hosting resources. 2. Document sizes and data volumes affecting loading and processing times.
Scalability	The system should: 1. Accommodate increasing users, reports, documents, and records. 2. Support additional users and records without major structural changes. The system is constrained to: 1. Hosting and database capacity. 2. The SK Federation and participating barangay SK councils.
Portability	The system should: 1. Be accessible through supported web browsers and compatible mobile devices. 2. Allow users to access the intended system functions across supported platforms. The system is constrained to: 1. The supported browsers, mobile operating systems, and device specifications. 2. Internet connectivity requirements for online system access.
Compatibility	The system should: 1. Operate with the required database, APIs, and supporting technologies. 2. Support data exchange between web and mobile applications. The system is constrained to: 1. Compatibility with integrated external services. 2. Supported versions of required software components.
Reliability	The system should: 1. Consistently perform its intended functions during normal operation. 2. Preserve the integrity of stored reports, documents, user information, and other system records. The system is constrained to: 1. System reliability may be affected by server, database, network, or other infrastructure failures. 2. Recovery depends on the available system and data management mechanisms.
Availability	The system should: 1. Make authorized system functions available to users when required. 2. Allow public users to access designated public information without requiring account registration. The system is constrained to: 1. System access depends on internet connectivity and the availability of the hosting environment. 2. Restricted functions remain accessible only to authorized users.
Maintainability	The system should: 1. Allow authorized personnel to identify errors and update components. 2. Support maintenance with minimal impact on unrelated functions. The system is constrained to: 1. Maintenance by authorized personnel. 2. Related component and documentation updates following major changes.
Security	The system should: 1. Protect user accounts, credentials, reports, financial documents, and other restricted information through authentication and access controls. 2. Restrict system functions and information according to user roles and permissions. The system is constrained to: 1. Users can only access functions and information permitted by their assigned roles. 2. System security is dependent on proper account management, authentication, access control, and the security of the underlying hosting environment.
Usability	The system should: 1. Provide a clear, understandable, and user-friendly interface for authorized SK officials and public users. 2. Allow users to navigate the system and perform intended tasks efficiently. The system is constrained to: 1. Usability may vary depending on the user's familiarity with digital systems and available device. 2. Users require appropriate access credentials for restricted system functions.

4.2 DESIGN OF SOFTWARE, SYSTEMS, PRODUCTS AND/OR PROCESSES

Use Case Diagram

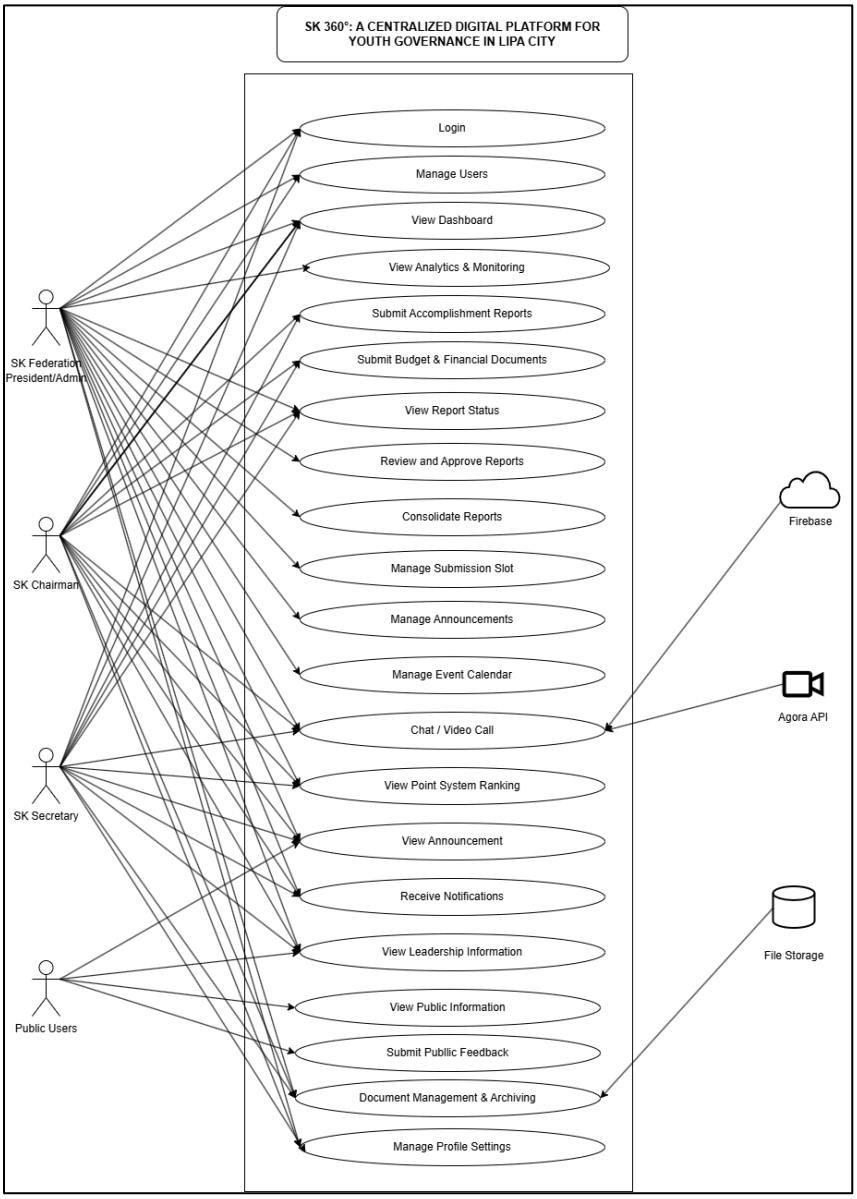


Figure 3 Use Case Diagram

Figure 3 presents the Use Case Diagram of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, illustrating the interactions between the system and its primary actors across the web and mobile applications. The diagram identifies four main actors: the SK Federation President/Admin, SK Chairman, SK Secretary, and Public Users, whose access to system functions is determined by their respective roles and

permissions. The SK Federation President/Admin manages SK Chairman accounts according to the defined hierarchy, monitors submissions, manages announcements and submission slots, reviews applicable submitted reports for administrative completeness, consolidates selected accomplishment and budget reports into consolidated PDF files, checks available quality documentation associated with barangay submissions, and accesses dashboard-based analytics and monitoring functions. The SK Chairman submits accomplishment reports and budget or financial records, manages leadership records for the SK Secretary, Treasurer, and Council Members of the assigned barangay, accesses relevant system information, and participates in communication and meetings. The SK Secretary submits accomplishment reports and budget-related documents and accesses announcements, schedules, communication, notifications, and other authorized functions. Public Users have limited access to selected public information, including announcements, calendar information, leadership information, annual budget information, and the published LYDP, without requiring account registration. The Point System Ranking is not accessible to public users. The use cases represent the major interactions supported by SK 360°, while the detailed functions of the web and mobile applications are further presented through the functional decomposition diagrams and system descriptions. The diagram provides an overview of the role-based interactions within the system and serves as a basis for the succeeding use case narratives and process designs. The diagram also shows the external services used by selected system functions. Firebase supports real-time chat and messaging, while the Agora API supports virtual meetings and video calls. System notifications are handled through the Laravel backend and database notification mechanism, with updates retrieved through JavaScript polling and scheduled meeting notifications supported by Laravel Scheduler.

Use Case Narratives

Table 8 Use Case Narrative of User Authentication and Login

Use Case Name	**Login**
Actor(s)	SK Federation President/Admin, SK Chairman, SK Secretary
Description	This use case describes the process by which authorized SK officials access the SK 360° web application. The user provides valid account credentials, which are verified by the system before access is granted. After successful authentication, the system identifies the user's role and provides access to the appropriate dashboard and system functions. This process helps protect the system from unauthorized access and ensures that users can only access functions appropriate to their assigned responsibilities.
Pre-condition	The user must have an active and authorized SK 360° account created through the defined user hierarchy. The user must have access to the web application and provide the required login credentials.

Post-condition	After successful authentication, the user's identity and role are verified, and the system grants access to the appropriate dashboard and authorized functions. If authentication fails, the user remains outside the protected areas of the system.
Main Scenario	1. The user opens the SK 360° web application. 2. The system displays the login interface. 3. The user enters the registered username or email and password. 4. The system validates the provided credentials. 5. The system identifies the user's assigned role. 6. The system grants access to the appropriate dashboard. 7. The user can access the functions permitted for the assigned role.
Alternative Scenario	1. The user enters an incorrect username, email, or password. 2. The system rejects the authentication attempt. 3. The system displays an appropriate error message. 4. The user may correct the credentials and attempt to log in again. 5. If the account is inactive or unauthorized, the system prevents access to protected functions.

Table 9 Use Case Narrative of User Account Management

Use Case Name	**Manage Users**
Actor(s)	SK Federation President/Admin
Description	This use case allows the SK Federation President/Admin to create and manage SK Chairman accounts according to the organizational hierarchy and role-based access control of the SK Federation. The User Management function also allows the SK Federation President/Admin to initiate a new SK term, manage the transition of the presidency, deactivate outgoing official access, and reappoint eligible officials while preserving their previous service records.
Pre-condition	The SK Federation President/Admin must be successfully logged into the system with the appropriate administrative privileges. For regular account management, the required information of the SK Chairman whose account will be created or managed must be available. For a term transition, an active current term must exist and the SK Federation President/Admin must have permission to initiate the succeeding term.
Post-condition	The appropriate SK Chairman account is created, updated, activated, or deactivated according to the authorized action. During a term transition, the succeeding term is recorded, outgoing official access is deactivated while historical service records are preserved, and eligible officials may be reappointed. The current SK Federation President remains active during the transition and may either continue for the succeeding term or transfer the role to the newly assigned President.
Main Scenario	1. The SK Federation President/Admin logs into the system. 2. The actor accesses the Manage Users function. 3. The system displays the available SK Chairman account-management and term-management options. 4. The President/Admin selects the option to create an SK Chairman account. 5. The President/Admin enters the required Chairman information and account credentials. 6. The system validates the submitted information, verifies the account details, and confirms that the selected role is permitted. 7. The system creates and stores the SK Chairman account. 8. The system assigns the appropriate SK Chairman role and permissions and displays a confirmation that the account was successfully created.
Alternative Scenario	1. If the required information is incomplete or invalid, the system displays an appropriate validation message and requires the actor to correct the information. 2. If an account with the same credentials or user information already exists, the system prevents duplicate account creation. 3. If an unauthorized account-management action is attempted, the system prevents the action. 4. If account creation fails, no account is created, and the system notifies the actor of the unsuccessful transaction. 5. If the SK Federation President/Admin selects Start New Term, the system displays the term transition options. 6. The President/Admin enters the succeeding term information and selects whether the current President will continue or a succeeding President will be assigned. 7. If the succeeding term start year is in the past, the system rejects the term creation and displays an appropriate validation message. 8. The system displays a confirmation that the outgoing official accounts will be deactivated while the current President remains active during the transition.

	9. After confirmation, the system creates the succeeding term and deactivates the outgoing official accounts. 10. If the current President continues, the existing President account remains active for the new term. If a new President is assigned, the current President temporarily remains active until the succeeding President completes the required account setup. 11. Eligible officials from the previous term may be reappointed, and the system records their new term assignments while preserving their previous service history.

Table 10 Use Case Narrative of Dashboard Access and Overview

Use Case Name	**View Dashboard**
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case describes how authorized SK officials access the centralized dashboard after logging into SK 360°. The dashboard provides a summarized view of important information such as report submissions, announcements, notifications, submission compliance, budget utilization, submission trends, and other analytics and monitoring information presented through charts, KPI indicators, and summary cards. It serves as a central starting point where users can quickly monitor important records and navigate to the functions available to them.
Pre-condition	The user must have a valid authorized account and must have successfully logged into the system. Relevant dashboard information must be available in the system.
Post-condition	The system displays the dashboard with the information and functions appropriate to the user's role. The user can review summaries and proceed to the corresponding system modules.
Main Scenario	1. The user successfully logs into SK 360°. 2. The system identifies the user's role. 3. The system retrieves information relevant to the user's account. 4. The system displays report summaries, announcements, notifications, analytics, KPI indicators, monitoring summaries, and other applicable information. 5. The user reviews the displayed information. 6. The user selects a desired function or module from the dashboard.
Alternative Scenario	1. Some dashboard information is unavailable because there are no records for the selected category. 2. The system displays the available information and indicates that no record is currently available for the affected category. 3. The user may proceed to another available function or return to the dashboard later.

Table 11 Use Case Narrative of Accomplishment Report Submission

Use Case Name	**Submit Accomplishment Reports**
Actor(s)	SK Chairman and SK Secretary
Description	This use case describes how authorized SK officials submit a completed accomplishment report file through SK 360°. The user selects the completed report file, the system validates the file, and a valid submission is stored in the centralized system for authorized access, review, monitoring, and consolidation.
Pre-condition	The user must be logged into an authorized account with permission to submit accomplishment reports, and a completed accomplishment report file must be available.
Post-condition	The uploaded accomplishment report file is successfully recorded and stored in the centralized system and becomes available to authorized users for review, monitoring, and consolidation.
Main Scenario	1. The user opens the Accomplishment Reporting Module. 2. The system displays the file upload function. 3. The user selects the completed accomplishment report file. 4. The system validates the selected file. 5. The user submits the report. 6. The system stores the uploaded report and displays a confirmation message.
Alternative Scenario	1. The user does not select a file or selects an invalid or unsupported file. 2. The system displays a validation message. 3. The user selects a valid completed accomplishment report file. 4. The user submits the report again.

Table 12 Use Case Narrative of Budget and Financial Document Submission

Use Case Name	Submit Budget & Financial Documents
Actor(s)	SK Chairman and SK Secretary
Description	This use case describes how authorized SK officials submit budget and financial records through the SK 360° web application. Users may either enter the required budget information directly into the system or upload an externally prepared budget document. For direct data entry, the system automatically generates a standardized budget report. Budget reports are organized according to monthly, quarterly, semi-annual, and annual reporting categories to support monitoring, budget allocation tracking, liquidation documentation, and centralized record management.
Pre-condition	The user must be logged into an authorized account with permission to submit budget and financial records. The required budget information or completed supporting document must be available.
Post-condition	The submitted budget report or financial document is recorded and stored in the centralized system and becomes available to authorized users for monitoring and documentation.
Main Scenario	1. The user opens the Budget and Financial Documentation Module. 2. The system displays the available monthly, quarterly, semi-annual, and annual budget reporting categories. 3. The user selects the applicable budget reporting category. 4. The user chooses either direct data entry or document upload. 5. For direct data entry, the user enters the required budget information and the system validates the entries and generates the standardized budget report. For document upload, the user selects a completed budget or financial document and the system validates the file. 6. The user reviews and submits the budget report or document. 7. The system records and stores the submission and displays a confirmation message.
Alternative Scenario	1. The entered information is incomplete or invalid, or the uploaded document is invalid or unsupported. 2. The system displays an appropriate validation message. 3. The user corrects the information or selects a valid document. 4. The user submits the budget report or document again.

Table 13 Use Case Narrative of Report Status Monitoring

Use Case Name	View Report Status
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to monitor the current status of submitted accomplishment reports and budget or financial records. The system provides information that helps users determine whether a submission has been received, remains pending, has been reviewed, or requires further action.
Pre-condition	The user must be successfully logged into the system and must have permission to view report records. At least one report or submission record must be available in the system.
Post-condition	The system displays the current status and relevant information of the selected report or financial record.
Main Scenario	1. The user opens the Report Status function. 2. The user selects the applicable report or barangay and, when applicable, the relevant budget reporting category. 3. The system retrieves the corresponding submission record. 4. The system displays the report details and current status. 5. The user reviews the status and determines whether additional action is needed.
Alternative Scenario	1. The requested report or record cannot be found. 2. The system displays a message indicating that no matching record is available. 3. The user modifies the search criteria and performs the search again.

Table 14 Use Case Narrative of Report Review and Approval

Use Case Name	Review and Approve Reports
Actor(s)	SK Federation President/Admin
Description	This use case allows the SK Federation President/Admin to review submitted accomplishment reports and budget or financial documents for completeness and compliance with submission requirements. The administrator may approve a submission or return it for correction. Financial validation remains the responsibility of the barangay SK Chairman before submission; administrative approval does not constitute financial auditing.
Pre-condition	The SK Federation President/Admin must be logged into an authorized account. A report or document submitted by an authorized SK Chairman or SK Secretary must be available for review.
Post-condition	The submission is approved or returned for correction, and the resulting status is recorded in the system.
Main Scenario	1. The President/Admin opens the submitted reports section. 2. The system displays available accomplishment reports and budget or financial documents. 3. The President/Admin selects a submission. 4. The system displays the submitted document and its supporting information. 5. The President/Admin checks its completeness and compliance with submission requirements. 6. If the requirements are satisfied, the President/Admin selects Approve. 7. The system records the approval and updates the submission status.
Alternative Scenario	1. If the submission is incomplete or requires correction, the President/Admin selects Return for Correction, and the system records the corresponding status. 2. If the document cannot be accessed, the system displays an error message, and the review cannot be completed. 3. If saving the review action fails, the system informs the President/Admin and retains the previously recorded status. 4. If an unauthorized user attempts to approve or return a submission, the system denies the action.

Table 15 Use Case Narrative of Report Consolidation

Use Case Name	Consolidate Reports
Actor(s)	SK Federation President/Admin
Description	This use case allows the SK Federation President/Admin to consolidate selected accomplishment reports and monthly, quarterly, semi-annual, and annual budget reports into PDF files, including available quality documentation for barangay submissions. The consolidation supports federation-level monitoring, reporting, and documentation without performing financial auditing.
Pre-condition	The SK Federation President/Admin must be logged into an authorized account. Applicable accomplishment reports or budget records must be available in the system.
Post-condition	The selected records and available quality documentation are compiled into a consolidated PDF file for federation-level monitoring and documentation.
Main Scenario	1. The SK Federation President/Admin opens the Report Consolidation Module. 2. The administrator selects the applicable accomplishment report or budget-report category. 3. The administrator selects the available source records for consolidation. 4. The system retrieves the selected records and available quality documentation. 5. The system compiles the selected records into a consolidated PDF file. 6. The system displays the generated consolidated PDF file. 7. The administrator reviews the consolidated PDF file.
Alternative Scenario	1. If some records or quality documentation are unavailable, the system proceeds with the available selected records. 2. If no applicable records are available, the system displays an appropriate no-data message. 3. If PDF generation fails, the system displays an error message and allows the administrator to retry.

Table 16 Use Case Narrative of Announcement Management

Use Case Name	Manage Announcements
Actor(s)	SK Federation President/Admin
Description	This use case allows the SK Federation President/Admin to create, edit, publish, and manage official announcements within SK 360°. The function can be used to communicate important updates, reminders, deadlines, meetings, activities, and other information relevant to authorized SK officials and, when designated for public access, public users through the web application.
Pre-condition	The administrator must be logged in and authorized to manage announcements. The administrator must have the information necessary to prepare the announcement.
Post-condition	The announcement is successfully saved and, when published, becomes available to authorized users and, when designated as public, through the web-based Public View.
Main Scenario	1. The administrator opens the Announcement Module. 2. The administrator selects the option to create or manage an announcement. 3. The administrator enters the announcement title, content, and other required details. 4. The system validates the information. 5. The administrator saves or publishes the announcement. 6. The system stores the announcement and makes it available to the intended users.
Alternative Scenario	1. The announcement contains incomplete or invalid information. 2. The system displays a validation message. 3. The administrator corrects the information. 4. The administrator saves or publishes the announcement again.

Table 17 Use Case Narrative of Web Module Management

Use Case Name	Manage Submission Slots and Published LYDP
Actor(s)	SK Federation President/Admin
Description	This use case allows the SK Federation President/Admin to create, update, and manually close submission slots for accomplishment reports and budget documents. It also allows the administrator to publish or replace the Local Youth Development Plan (LYDP) PDF available through the web application's public Annual Budget & LYDP section.
Pre-condition	The SK Federation President/Admin must be logged into an authorized account. Required submission-slot information or a valid LYDP PDF must be available. An existing slot or published document must be available when updating or replacing a record.
Post-condition	The submission slot is created, updated, or closed, and its status is reflected in the system. For LYDP publication, the selected PDF becomes accessible through the public Annual Budget & LYDP section without requiring login.
Main Scenario	1. The President/Admin opens Module Management. 2. The system displays existing submission slots and their statuses. 3. The President/Admin creates a new slot or selects an existing slot for modification. 4. The President/Admin enters or updates the title, description, opening schedule, deadline, report type, intended role, and applicable submission requirements. 5. The system validates the information. 6. The President/Admin saves the changes. 7. The system records the slot and displays its applicable status: Upcoming, Open, Past Deadline, or Closed.
Alternative Scenario	1. If required information is missing or invalid, the system displays validation messages and allows correction. 2. To close an existing slot, the President/Admin selects the close action, and the system prevents further submissions to that slot. 3. If the deadline has passed but the slot has not been manually closed, the system displays Past Deadline and continues accepting late submissions. 4. To publish or replace the LYDP, the President/Admin selects the LYDP management option, uploads a valid PDF, and saves it. The system makes the document available in the public section. 5. If the uploaded file is invalid or saving fails, the system displays an appropriate message and allows another attempt. 6. If an unauthorized user attempts a management action, the system denies access.

Table 18 Use Case Narrative of Event Calendar Management

Use Case Name	Manage Event Calendar
Actor(s)	SK Federation President/Admin
Description	This use case allows the SK Federation President/Admin to create, update, and manage calendar events, including activities, meetings, programs, and submission deadlines. Authenticated SK officials may view available schedules, while public users may access events designated for public viewing through the web application only.
Pre-condition	The administrator must be logged into the system and authorized to manage calendar information. The event or schedule details must be available for entry or modification.
Post-condition	The event information is saved in the centralized calendar and becomes available to authorized users and, when designated as public, to public users through the web application.
Main Scenario	1. The administrator opens the Event Calendar. 2. The administrator creates a new event or selects an existing event. 3. The administrator enters or modifies the event title, date, time, location, audience, and related event settings. 4. The system validates the schedule information. 5. The administrator saves the event. 6. The system records and displays the updated schedule.
Alternative Scenario	1. Required event information is incomplete or a scheduling conflict is detected. 2. The system displays an appropriate warning or validation message. 3. The administrator reviews and modifies the schedule information. 4. The administrator saves the updated event again.

Table 19 Use Case Narrative of Real-Time Chat

Use Case Name	Real-Time Chat
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to communicate through real-time messaging within SK 360°. Users can exchange messages for coordination, discussions, and other organizational activities through the integrated chat function.
Pre-condition	The user must be logged in and authorized to use the chat function. A stable internet connection and available messaging service are required.
Post-condition	The message is transmitted to the intended recipient or conversation and becomes available according to the system's communication functions and access controls.
Main Scenario	1. The user opens the Real-Time Chat function. 2. The user selects an available recipient or conversation. 3. The user enters a message. 4. The user sends the message. 5. The system transmits and synchronizes the message to the intended recipient or conversation. 6. The participants continue communication through the chat function.
Alternative Scenario	1. The user loses internet connectivity or the messaging service becomes unavailable. 2. The system indicates that the message cannot be transmitted. 3. The user may retry once the connection or service becomes available.

Table 20 Use Case Narrative of Meetings

Use Case Name	Meetings
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows the SK Federation President/Admin to schedule, manage, and finalize meetings, while authorized SK Chairmen and SK Secretaries may participate in face-to-face or virtual meetings. The system supports attendance recording and applies applicable participation points based on validated meeting attendance.
Pre-condition	The user must be logged in and authorized to access the Meetings function. For virtual meetings, a stable internet connection and available meeting service are required.

Post-condition	The meeting is successfully scheduled or completed, and applicable attendance records are stored in the system. Upon meeting finalization, validated participation points are recorded according to the defined Point System Ranking criteria.
Main Scenario	1. The user opens the Meetings Module. 2. The system displays available meeting information. 3. The SK Federation President/Admin creates or manages a meeting. 4. The system saves and displays the meeting details. 5. Authorized SK Chairmen and SK Secretaries access the scheduled meeting. 6. Participants attend the meeting either face-to-face or virtually. 7. The system records applicable meeting attendance. 8. The President/Admin ends and finalizes the meeting. 9. The system finalizes validated attendance records and applies the corresponding participation points.
Alternative Scenario	1. The meeting service or internet connection becomes unavailable. 2. The system displays an appropriate connection or service error. 3. The user may attempt to reconnect to the meeting.

Table 21 Use Case Narrative of Point System Ranking

Use Case Name	**View Point System Ranking**
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to view the Point System Ranking and related performance summaries based on defined documentation, compliance, late-submission, and official-meeting participation criteria.
Pre-condition	Point and ranking information must have been recorded or generated by the system. The authorized SK official must have appropriate access to the Point System Ranking function.
Post-condition	The system displays the available ranking information, point summaries, and relevant performance indicators according to the user's access level.
Main Scenario	1. The user opens the Point System Ranking function. 2. The system retrieves the available ranking data. 3. The system organizes the ranking according to the defined criteria. 4. The system displays the rankings and corresponding point summaries. 5. The user reviews the displayed ranking information.
Alternative Scenario	1. No ranking information is currently available. 2. The system informs the user that ranking data is unavailable. 3. The user may return to the previous interface or check again later.

Table 22 Use Case Narrative of Announcement Viewing

Use Case Name	**View Announcement**
Actor(s)	SK Federation President/Admin, SK Chairman, SK Secretary, and Public Users
Description	This use case allows users to view official announcements, notices, updates, reminders, and other information published through SK 360°. Authorized SK officials can access announcements relevant to their responsibilities, while public users may access announcements specifically designated for public viewing through the web application only.
Pre-condition	At least one announcement must have been created and published in the system. The user must have permission to access the selected announcement. Public users may access only announcements designated for public viewing through the web application.
Post-condition	The selected announcement and its relevant details are displayed according to the user's access level.
Main Scenario	1. The user opens the Announcement section. 2. The system retrieves published announcements available to the user's access level. 3. The system displays the list of available announcements. 4. The user selects an announcement. 5. The system displays the complete announcement information.
Alternative Scenario	1. No published announcement is available. 2. The system displays a message indicating that there are currently no announcements. 3. The user may return to the main interface or check again later.

Table 23 Use Case Narrative of Notification Management

Use Case Name	Receive Notifications
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	Notifications may inform users about announcements, report submissions, deadlines, scheduled activities, reminders, and other system events.
Pre-condition	The user must have an active authorized account and access to the notification service. A system event that requires or triggers a notification must occur.
Post-condition	The appropriate notification is generated and delivered or made available to the intended user. The user can open the notification and access the related information when applicable.
Main Scenario	1. A relevant event occurs within the system. 2. The system determines the users who should be notified. 3. The system generates the appropriate notification. 4. The notification is delivered through the available notification mechanism. 5. The user receives and opens the notification. 6. The user views the related information or proceeds to the corresponding system function.
Alternative Scenario	1. The user's device or connection is temporarily unavailable. 2. The notification cannot be delivered immediately. 3. The system retains or makes the notification available according to the notification mechanism. 4. The user can access the notification when connectivity or service becomes available.

Table 24 Use Case Narrative of Leadership Information Management and Viewing

Use Case Name	Manage and View Leadership Information
Actor(s)	SK Federation President/Admin, SK Chairman, SK Secretary, and Public Users
Description	This use case allows authorized users to view verified current and historical leadership information. Authorized SK Chairmen may also manage leadership records for their assigned barangay, including the SK Secretary, Treasurer, and Council Members. Adding an SK Secretary creates or links the corresponding Secretary system account, while Treasurer and Council Member records do not receive system login access. Public users may view only leadership information designated for public access through the web application.
Pre-condition	Leadership records must be available for viewing. For leadership management, the SK Chairman must be successfully logged in and authorized to manage records for the assigned barangay.
Post-condition	The selected leadership information is displayed according to the user's access level. When an authorized SK Chairman adds or updates a leadership record, the validated information is saved. If the added official is an SK Secretary, the corresponding Secretary system account is created or linked.
Main Scenario	1. The user opens the Leadership Information function. 2. The system retrieves available current and historical leadership records. 3. The user selects a leadership record to view. 4. The system displays the verified leadership information. 5. If the user is an authorized SK Chairman, the user may add or update the SK Secretary, Treasurer, or Council Member record. 6. The SK Chairman enters or updates the required official information. 7. The system validates and saves the leadership record. 8. If the official is an SK Secretary, the system creates or links the corresponding Secretary account and assigns the appropriate role and barangay access. 9. If the official is a Treasurer or Council Member, the system stores the information as a leadership record only.
Alternative Scenario	1. The requested leadership record is unavailable or has not yet been entered. 2. If submitted leadership information is incomplete or invalid, the system displays a validation message and requires correction. 3. If an unauthorized user attempts to modify leadership records, the system denies the action.

Table 25 Use Case Narrative of Public Information Access

Use Case Name	View Public Information
Actor(s)	Public Users
Description	Publicly accessible information includes announcements, calendar information, leadership information, annual budget information, and the published LYDP through the SK 360° Public Information Portal. The Point System Ranking is not accessible to public users. Public users may also submit feedback or comments anonymously or non-anonymously through the Public Information Portal, subject to email OTP verification before the submission is recorded.
Pre-condition	The information must have been designated and published for public access. The public user must have access to the SK 360° Public Information Portal through the web application.
Post-condition	The selected public information is displayed through the web application without requiring account registration or login. If the public user submits feedback or comments, the system records the submission only after successful email OTP verification and displays a confirmation message.
Main Scenario	1. The public user opens the web-based Public View of SK 360°. 2. The system displays the available public information categories. 3. The user selects the desired category. 4. The system retrieves the information designated for public access. 5. The system displays the selected information. 6. If the public user selects the feedback or commenting function, the user chooses whether to submit anonymously or non-anonymously and enters the required information, including an email address. 7. The system sends an OTP to the provided email address. 8. The user enters the OTP and the system verifies it. 9. If the OTP is valid, the system records the feedback or comment and displays a confirmation message.
Alternative Scenario	1. The requested information is restricted or not designated for public viewing. 2. The system prevents access to the restricted information. 3. The system displays an appropriate access or availability message. 4. If the OTP is invalid, the system displays an error message and allows the user to request or enter another OTP.

Table 26 Use Case Narrative of Document Management and Archiving

Use Case Name	Document Management & Archiving
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to access and retrieve records preserved from connected system modules, including accomplishment reports, budget documents, and event records.
Pre-condition	The user must be logged into an authorized account with permission to access archived records generated or submitted through connected system modules.
Post-condition	The selected archived record is successfully retrieved and made available to authorized users according to their access permissions.
Main Scenario	1. The user opens Document Management and Archiving. 2. The user searches for or selects an available archived record. 3. The system verifies the user's access permission. 4. The system retrieves the selected record from the connected module data. 5. The system displays the archived record to the authorized user.
Alternative Scenario	1. The user attempts to access a document without the required permission. 2. The system denies the unauthorized request. 3. The system displays an appropriate access message. 4. If an archived record is unavailable or cannot be retrieved, the system informs the user that the record is unavailable.

Table 27 Use Case Narrative of Profile Settings Management

Use Case Name	Manage Profile Settings
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to manage information associated with their own SK 360° account. Users can update applicable profile details, manage account credentials such as passwords, and adjust available account preferences. The system validates the changes and securely stores the updated information to maintain accurate user records and support secure account management.
Pre-condition	The user must have an active SK 360° account and must be successfully logged into the system.
Post-condition	The updated profile information, credentials, or preferences are successfully validated and stored in the system. The user's account reflects the changes the next time the information is accessed.
Main Scenario	1. The user opens Profile Settings. 2. The system displays the user's current account information and available settings. 3. The user selects the information or setting to be modified. 4. The user enters the new information. 5. The system validates the changes. 6. The user confirms the update. 7. The system securely saves the updated information.
Alternative Scenario	1. The user enters incomplete or invalid information. 2. The system identifies the invalid information. 3. The system displays a validation message. 4. The user corrects the information. 5. The user submits the updated information again.

Table 28 Use Case Narrative of Mobile Login

Use Case Name	Login
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to securely access SK 360° through their mobile devices. The user enters valid account credentials, and the mobile application communicates with the system to verify the user's identity. After successful authentication, the user is granted access to the mobile functions corresponding to their assigned role.
Pre-condition	The user must have an authorized SK 360° account, have the mobile application available on a supported device, and have sufficient connectivity for authentication.
Post-condition	The user's credentials are successfully verified and the mobile application provides access to the authorized functions and mobile home page.
Main Scenario	1. The user opens the SK 360° mobile application. 2. The mobile login screen is displayed. 3. The user enters the registered credentials. 4. The system validates the credentials. 5. The system identifies the user's role. 6. The user is granted access to the mobile application.
Alternative Scenario	1. The user enters invalid credentials. 2. The system rejects the login attempt. 3. The system displays an error message. 4. The user corrects the credentials and attempts to log in again.

Table 29 Use Case Narrative of Mobile Accomplishment Reporting

Use Case Name	Submit Accomplishment Reports
Actor(s)	SK Chairman and SK Secretary
Description	This use case allows authorized SK officials to upload completed accomplishment report files through the SK 360° mobile application. The uploaded file is validated and transmitted to the centralized system for storage, monitoring, review, and consolidation.
Pre-condition	The user must be logged into an authorized account with permission to submit accomplishment reports through the mobile application. A completed accomplishment report file must be available.
Post-condition	The uploaded accomplishment report file is successfully recorded and stored in the centralized system.
Main Scenario	1. The user opens Mobile Accomplishment Reporting. 2. The system displays the file upload function. 3. The user selects the completed accomplishment report file. 4. The system validates the selected file. 5. The user confirms the submission. 6. The system stores the uploaded accomplishment report and displays a confirmation message.
Alternative Scenario	1. The user does not select a file or selects an invalid or unsupported file. 2. The system displays a validation message. 3. The user selects a valid completed accomplishment report file. 4. The user submits the report again.

Table 30 Use Case Narrative of Mobile Budget Documents

Use Case Name	Submit Budget & Financial Documents
Actor(s)	SK Chairman and SK Secretary
Description	This use case allows authorized SK officials to upload completed budget and financial documents through the SK 360° mobile application. Uploaded documents are validated and stored in the centralized system to support monitoring, budget allocation tracking, liquidation documentation, and centralized financial record management.
Pre-condition	The user must be logged into an authorized account with permission to access the mobile budget and financial functions. A completed budget or financial document must be available for upload.
Post-condition	The uploaded budget or financial document is successfully recorded and stored in the centralized system for authorized monitoring and documentation.
Main Scenario	1. The user opens Mobile Budget Documents. 2. The system displays the document upload function. 3. The user selects the completed budget or financial document. 4. The system validates the selected file. 5. The user confirms the submission. 6. The system stores the uploaded document and displays a confirmation message.
Alternative Scenario	1. The user does not select a file or selects an invalid or unsupported document. 2. The system displays a validation message. 3. The user selects a valid completed budget or financial document. 4. The user submits the document again.

Table 31 Use Case Narrative of Mobile Consolidation

Use Case Name	Monitor Report Submissions
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to monitor barangay report submissions through the mobile application. Users can view submission statuses, summary counts, reporting periods, and available quality documentation using search and filtering functions. Unlike the web-based Report Consolidation Module, the mobile function is limited to monitoring and viewing submission records and does not generate consolidated PDF files.

Pre-condition	The user must be logged into an authorized account with permission to access Mobile Consolidation. Applicable barangay submission records must be available in the system.
Post-condition	The system displays the applicable barangay submission overview, filtered submission information, and available quality documentation according to the selected criteria and the user's access permissions.
Main Scenario	1. The user opens Mobile Consolidation. 2. The system displays the total, submitted, and pending summary counts. 3. The user selects an available reporting year. 4. The system updates the displayed information according to the selected year. 5. The user selects an available reporting period. 6. The system updates the displayed barangay submission information according to the selected period. 7. The user searches for an available barangay. 8. The system displays the matching barangay submission information. 9. The user checks the available quality documentation for an applicable barangay submission. 10. The system displays the available quality documentation information.
Alternative Scenario	1. No barangay submission matches the selected year, reporting period, or search criteria. 2. The system displays an appropriate no-record or empty-state message. 3. If no quality documentation is available for the selected submission, the system displays an appropriate availability message.

Table 32 Use Case Narrative of Mobile Module Management

Use Case Name	**Manage Submission Slots**
Actor(s)	SK Federation President/Admin
Description	This use case allows the SK Federation President/Admin to create and manage accomplishment and budget submission slots through the mobile application using Upcoming, Open, Past Deadline, and Closed statuses.
Pre-condition	The SK Federation President/Admin must be logged into an authorized account with permission to access Mobile Module Management.
Post-condition	The created or updated submission slot information is successfully stored and displayed with the applicable report type, schedule, intended role, and status.
Main Scenario	1. The SK Federation President/Admin opens Mobile Module Management. 2. The system displays the available submission slots and submission summary information. 3. The user searches, filters, or selects an available submission slot, or selects Create New Slot. 4. The user enters or updates the required submission slot information. 5. The user specifies the applicable report type, schedule, deadline, and intended role. 6. The system validates the submitted information. 7. The user saves the submission slot. 8. The system stores the information and displays the created or updated submission slot and its current status. 9. The system displays each submission slot using the applicable Upcoming, Open, Past Deadline, or Closed status. 10. When the deadline passes, the slot becomes Past Deadline and may continue accepting late submissions until the SK Federation President/Admin manually closes it. 11. Once the slot is Closed, the system prevents new submissions.
Alternative Scenario	1. The user submits incomplete or invalid submission slot information. 2. The system displays an appropriate validation message. 3. The user corrects the required information and submits it again. 4. If an SK Chairman or SK Secretary attempts to access Mobile Module Management, the system denies access to the restricted function.

Table 33 Use Case Narrative of Mobile View Reports

Use Case Name	**View Submitted Reports**
Actor(s)	SK Federation President/Admin
Description	This use case allows the SK Federation President/Admin to view available submitted accomplishment and budget report records through the mobile application. The user can

	review report summary counts, search for submitted records, apply report-type filters, and open an available report record.
Pre-condition	The SK Federation President/Admin must be logged into an authorized account with permission to access Mobile View Reports. Submitted accomplishment reports or budget report records must be available in the system.
Post-condition	The selected submitted report record and its available information or document are displayed to the SK Federation President/Admin.
Main Scenario	1. The SK Federation President/Admin opens Mobile View Reports. 2. The system displays the available report summary counts. 3. The user searches for an available submitted report record. 4. The user applies an available report-type filter. 5. The system displays the submitted reports that match the selected criteria. 6. The user selects an available accomplishment or budget report record. 7. The system displays the selected report information and available submitted document.
Alternative Scenario	1. No submitted report matches the selected search or filter criteria. 2. The system displays an appropriate no-record or empty-state message. 3. The user may modify the search or filter criteria. 4. If an SK Chairman or SK Secretary attempts to access Mobile View Reports, the system denies access to the restricted function.

Table 34 Use Case Narrative of Mobile Announcement

Use Case Name	**Manage and View Mobile Announcements**
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows the SK Federation President/Admin to create, publish, and update announcements through the Android application. Authenticated SK officials can view published announcements according to their access permissions and the announcement's audience settings.
Pre-condition	The user must be logged into an authorized mobile account. Creation and modification require SK Federation President/Admin privileges. An existing announcement must be available for viewing or updating.
Post-condition	The selected announcement is displayed. For successful administrative actions, the announcement is saved or updated and made available to its intended audience.
Main Scenario	1. The user opens Mobile Announcement. 2. The system displays published announcements available to the user. 3. The user selects an announcement. 4. The system displays its details. 5. To create or update an announcement, the President/Admin selects the corresponding management option. 6. The President/Admin enters or modifies the announcement information and audience settings. 7. The system validates the information. 8. The President/Admin confirms the action. 9. The system saves the announcement and updates the displayed information.
Alternative Scenario	1. If no announcements are available, the system displays an appropriate message; the President/Admin may still create an announcement. 2. If required information is missing or invalid, the system displays validation messages and allows correction. 3. If a Chairman or Secretary attempts a management action, the system denies access. 4. If saving fails, the system informs the President/Admin that the action was unsuccessful and allows another attempt.

Table 35 Use Case Narrative of Mobile Receive Notifications

Use Case Name	**Receive Notifications**
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to access system notifications through the mobile application. Notifications may inform users about system updates, announcements, report-related reminders, meeting reminders, deadlines, and other applicable system activities. This function helps users remain updated on relevant system information.

Pre-condition	The user must have an active and authorized account, and a system event that generates a notification must occur.
Post-condition	The notification is recorded by the system and made available to the user through the mobile application, allowing the user to review the related information.
Main Scenario	1. A relevant event occurs in the system. 2. The system generates a notification. 3. The system stores the notification and makes it available to the intended user. 4. The user accesses the notification through the mobile application. 5. The user opens the notification. 6. The application displays the related information when applicable.
Alternative Scenario	1. The mobile device is offline or temporarily unable to receive notifications. 2. The notification is not immediately displayed. 3. Once connectivity becomes available, the user can access the notification through the mobile application.

Table 36 Use Case Narrative of Mobile Chat

Use Case Name	**Mobile Chat**
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to communicate through real-time messaging using the mobile application. Users can exchange messages for coordination, discussions, and other organizational activities through the integrated chat function.
Pre-condition	The user must be logged into an authorized account and have a stable internet connection. The intended conversation or recipient must be available.
Post-condition	The message is sent and synchronized to the intended conversation or recipient.
Main Scenario	1. The user opens Mobile Chat. 2. The user selects a conversation or recipient. 3. The user enters a message. 4. The user sends the message. 5. The system transmits and synchronizes the message. 6. The participants continue communication through the chat function.
Alternative Scenario	1. The user's internet connection becomes unstable or unavailable. 2. The system displays an appropriate connection or service message. 3. The user may retry once connectivity becomes available.

Table 37 Use Case Narrative of Mobile Meetings

Use Case Name	**Mobile Meetings**
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authenticated SK officials to access and participate in SK meetings through the mobile application. Meeting management is limited to the SK Federation President/Admin, while SK Chairmen and SK Secretaries may access and participate in available meetings, including virtually.
Pre-condition	The user must be logged into an authorized account, have a stable internet connection, and have access to an available meeting.
Post-condition	The SK Federation President/Admin can manage applicable meeting information, while authorized participants can successfully access or join the selected meeting. Valid virtual participation may be reflected in the applicable attendance record.
Main Scenario	1. The user opens Mobile Meetings. 2. The system displays the meeting information and functions available to the user's role. 3. If the user is the SK Federation President/Admin, the user may create, update, or manage an applicable meeting. Otherwise, the authorized participant selects an available meeting. 4. The system records or retrieves the applicable meeting details. 5. The authorized participant joins the available virtual meeting. 6. The system records valid virtual participation for attendance when applicable. 7. The participants conduct the meeting through the available meeting function.

Alternative Scenario	1. The internet connection or meeting service becomes unavailable. 2. The system displays an appropriate connection or service error. 3. The user may attempt to reconnect to the meeting.

Table 38 Use Case Narrative of Mobile Activity Calendar

Use Case Name	**Manage and View Mobile Calendar Events**
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authenticated SK officials to view calendar events and their details through the Android application. The SK Federation President/Admin can create and update events, including their schedules, locations, and intended audiences.
Pre-condition	The user must be logged into an authorized mobile account. Creating or updating events requires SK Federation President/Admin privileges. An existing event must be available for viewing or modification.
Post-condition	The selected event's details are displayed. For successful administrative actions, the event is saved or updated and becomes available according to its audience settings.
Main Scenario	1. The user opens Mobile Activity Calendar. 2. The system displays calendar events available to the user. 3. The user selects an event to view its details. 4. To manage an event, the President/Admin selects the option to create a new event or update an existing one. 5. The President/Admin enters or modifies the title, date, time, location, audience, and other applicable details. 6. The system validates the information, and the President/Admin saves the event. 7. The system records the changes and displays the updated calendar.
Alternative Scenario	1. If no events are available, the system displays an appropriate message; the President/Admin may still create an event. 2. If required information is missing or invalid, the system displays validation messages and allows correction. 3. If a Chairman or Secretary attempts to create or modify an event, the system denies the action. 4. If saving fails, the system displays an unsuccessful-action message and allows the President/Admin to try again.

Table 39 Use Case Narrative of Mobile View Point System Ranking

Use Case Name	**View Point System Ranking**
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to view the Point System Ranking and point summaries through the mobile application. Ranking information is generated based on defined documentation, compliance, late-submission, and official-meeting participation criteria.
Pre-condition	The user must be logged into an authorized account, and ranking and point information must be available in the system.
Post-condition	The system displays the available rankings, point summaries, and performance indicators on the user's mobile device.
Main Scenario	1. The user opens Mobile Rankings. 2. The system retrieves available ranking records. 3. The system calculates and organizes the ranking according to the defined point criteria from highest to lowest total points. 4. The system displays the Top 10 councils and their corresponding point summaries. 5. The user reviews the displayed ranking information.
Alternative Scenario	1. Ranking data is unavailable. 2. The system informs the user that no ranking information is currently available. 3. The user may return to the previous screen or check again later.

Table 40 Use Case Narrative of Mobile View Leadership Information

Use Case Name	View Leadership Information
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to access verified current and historical leadership information through the mobile application. Users can view available leadership profiles and official information according to their authorized access.
Pre-condition	The user must be logged into an authorized account, and leadership information must be available in the system.
Post-condition	The selected leadership profile or organizational information is displayed on the user's mobile device.
Main Scenario	1. The user opens Mobile Leadership Profiles. 2. The system retrieves available leadership profiles. 3. The user selects a profile. 4. The system retrieves the selected profile information. 5. The system displays the available verified leadership details.
Alternative Scenario	1. The selected leadership profile is unavailable. 2. The system displays an appropriate message. 3. The user may select another available leadership profile.

Table 41 Use Case Narrative of Mobile Manage Profile Settings

Use Case Name	Manage Profile Settings
Actor(s)	SK Federation President/Admin, SK Chairman, and SK Secretary
Description	This use case allows authorized SK officials to manage their account information and available profile settings through the mobile application. Users can review and update applicable account information and settings while maintaining their authenticated access to the system. The updated information is synchronized with the user's account in the centralized platform.
Pre-condition	The user must be logged into an active authorized account through the mobile application.
Post-condition	The updated account information or settings are validated and stored in the system, and the mobile profile reflects the changes.
Main Scenario	1. The user opens Mobile Profile. 2. The system displays the user's current profile information and settings. 3. The user selects the information to update. 4. The user enters the new information. 5. The system validates the changes. 6. The user confirms the update. 7. The system saves the updated information.
Alternative Scenario	1. The user enters invalid or incomplete information. 2. The system displays a validation message. 3. The user corrects the information. 4. The user submits the updated information again.

Functional Decomposition Diagrams

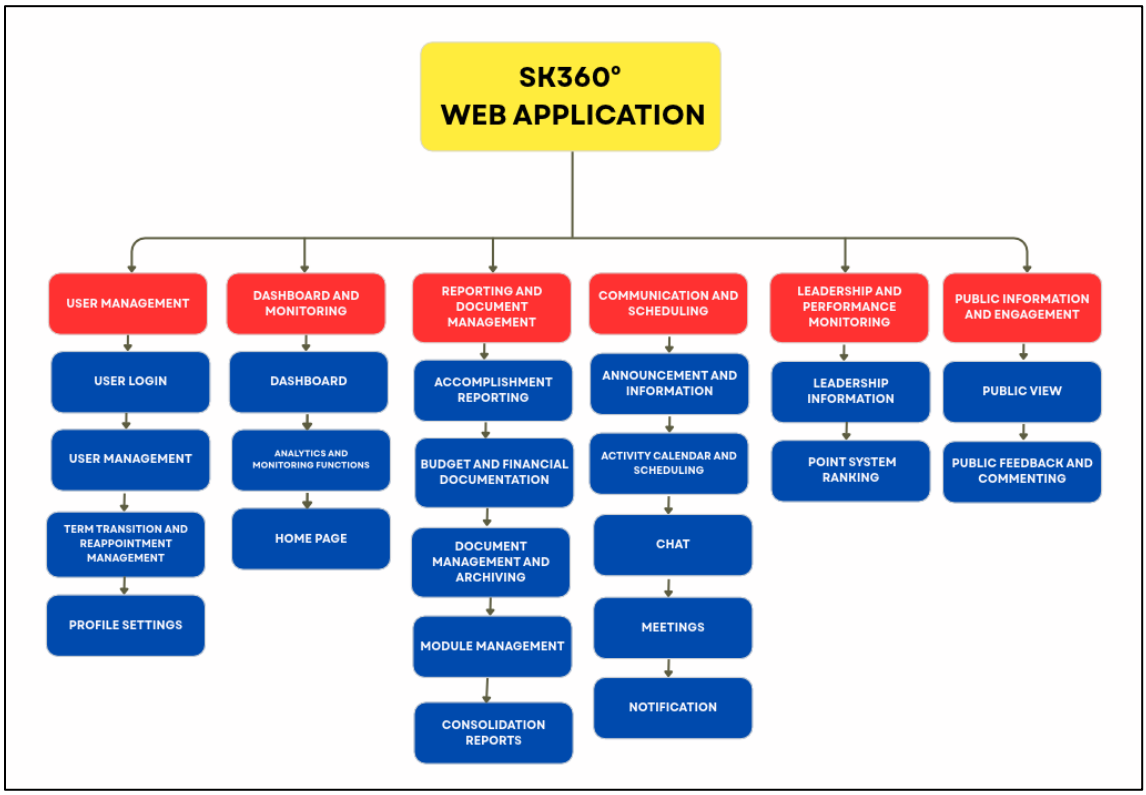


Figure 4 Functional Decomposition Diagram of the Web Application

This figure illustrates the functional structure of the SK 360° Web Application, which is designed to provide authorized SK officials with a centralized platform for managing youth governance activities in Lipa City. The web application provides different modules that support authentication, reporting, documentation, communication, monitoring, scheduling, and public information access. After successfully logging in, users are provided with features and functions based on their assigned roles and access permissions.

The Login module provides secure authentication for authorized SK officials using their authorized account credentials, including input validation and error handling. The User Management module allows the SK Federation President/Admin to create, update, activate, deactivate, and manage SK Chairman accounts according to the defined user hierarchy and role-based access control, including applicable term-transition functions. The Dashboard module provides an overview of reports, announcements, notifications, system updates, descriptive analytics, KPI-based monitoring, submission compliance,

budget utilization, and other monitoring summaries through charts and dashboard visualizations, while the Home Page module serves as the main entry point of the web application and provides navigation and access to system features.

The Accomplishment Reporting module enables authorized users to upload completed accomplishment report files for centralized storage, review, monitoring, and consolidation. The Budget and Financial Documentation module supports budget report submission through direct data entry or document upload. Budget reports are organized according to monthly, quarterly, semi-annual, and annual reporting categories to support budget monitoring, allocation tracking, liquidation documentation, and centralized financial record management. The Document Management and Archiving module preserves and provides authorized access to records generated or submitted through connected system modules.

The Announcement module allows the SK Federation President/Admin to create, update, publish, and manage official announcements, updates, and other applicable information, while authorized SK officials can view announcements available to them. The Module Management module allows the SK Federation President/Admin to create and manage accomplishment and budget submission slots with Upcoming, Open, Past Deadline, and Closed statuses. Past Deadline slots may continue accepting late submissions until manually closed, while Closed slots no longer accept new submissions. The Report Consolidation module compiles selected accomplishment and monthly, quarterly, semi-annual, and annual budget reports into consolidated PDF files and includes available quality documentation. The Notification module provides system notifications for announcements, updates, report submissions, deadlines, and other system activities.

The Activity Calendar and Scheduling module allows the SK Federation President/Admin to create and manage calendar events, including the title, date, time, location, audience, and related event settings, while authenticated SK officials can view available schedules and event details. The Chat module provides real-time communication among authorized users, while the Meetings module allows the President/Admin to manage meetings while SK Chairmen and Secretaries may participate, including virtually. The Leadership Information module provides verified leadership information and directory records of SK officials across barangays.

The Point System Ranking module displays rankings and point summaries based on defined documentation, compliance, late-submission, and official-meeting participation criteria. The Public View module allows public users to access selected announcements, calendar information, leadership information, annual budget information, and the published LYDP, and submit OTP-verified feedback or comments without registration. Lastly, the Profile Settings module allows users to manage their account details, passwords, and preferences.

Together, these modules provide the web application with the functions necessary to support centralized reporting, documentation, communication, monitoring, coordination, and transparency for SK governance.

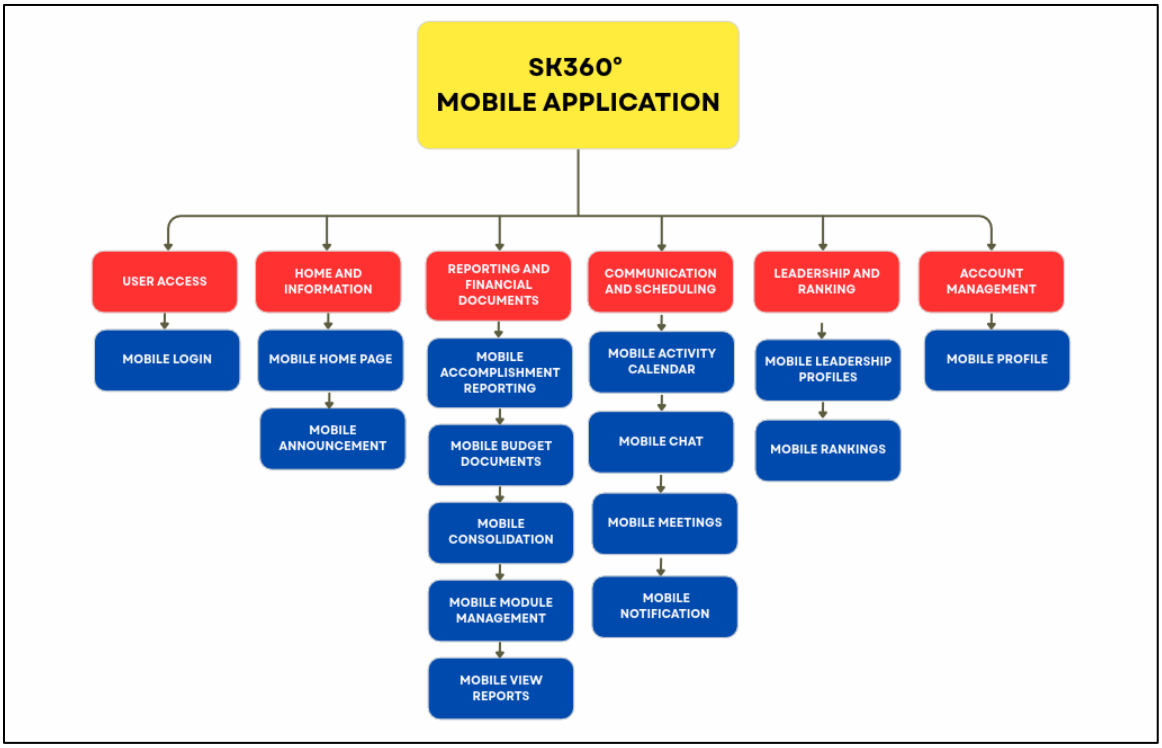


Figure 5 Functional Decomposition Diagram for Mobile Application

This figure illustrates the functional structure of the SK 360° Mobile Application, which provides convenient mobile access to essential SK 360° services for authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts. The mobile application is designed to support reporting, communication, scheduling, notifications, leadership information, rankings, monitoring, and account-related functions according to assigned roles and permissions.

The Mobile Login module provides secure authentication for authorized SK officials accessing the SK 360° system through mobile devices. The Mobile Home Page module provides a mobile-friendly interface where users can view announcements, activities, and quick-access features. The Mobile Accomplishment Reporting module enables authorized users to upload completed accomplishment report files using their mobile devices. The Mobile Budget Documents module enables authorized users to upload completed budget and financial documents for centralized storage, monitoring, budget allocation tracking, and liquidation documentation. The Mobile Consolidation module allows authorized SK officials to monitor submission statuses, summary counts, reporting periods, and quality documentation through filters. It does not generate consolidated PDF files. The Mobile Module Management module allows the SK Federation President/Admin to create and manage accomplishment and budget submission slots with Upcoming, Open, Past Deadline, and Closed statuses through the mobile application. The Mobile View Reports module allows the SK Federation President/Admin to view, search, filter, and open available submitted accomplishment and budget report records.

The Mobile Announcement module allows the President/Admin to manage announcements while authenticated SK officials can view them. The Mobile Leadership Profiles module allows users to view verified information about SK officials through the mobile application. The Mobile Chat module enables messaging and communication among authorized users, while the Mobile Meetings module allows authorized SK officials to participate in meetings, with management limited to the President/Admin.

The Mobile Activity Calendar module allows the President/Admin to manage events while authenticated SK officials can view schedules and deadlines. The Mobile Rankings module displays Point System Rankings and summaries based on performance criteria. The Mobile Profile module allows users to manage their information and settings. The Mobile Notification module provides notifications for announcements, updates, submission reminders, meeting reminders, deadlines, and system activities.

Together, these modules provide authenticated SK officials mobile access to SK 360° services while supporting accomplishment reporting, financial documentation, communication, scheduling, leadership information, rankings, notifications, monitoring, and account management.

Process Flow Diagrams

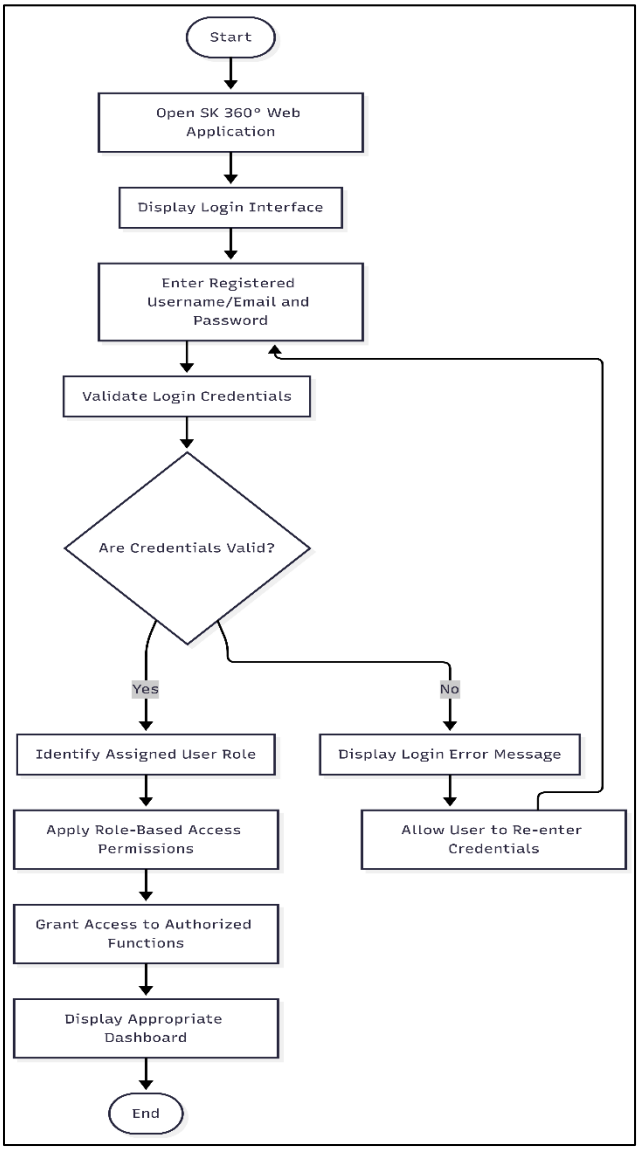


Figure 6 User Login and Authentication

Figure 6 illustrates the process flow of user login and authentication in the SK 360° system. The process begins when an authorized SK official opens the application and enters the registered username or email and password. The system validates the provided credentials and, if valid, identifies the user's assigned role and applies the corresponding role-based access permissions. The user is then granted access to the functions available for the assigned role and directed to the appropriate authenticated web dashboard or mobile home page. If the credentials are invalid, the system displays an error message and allows the user to re-enter the required credentials.

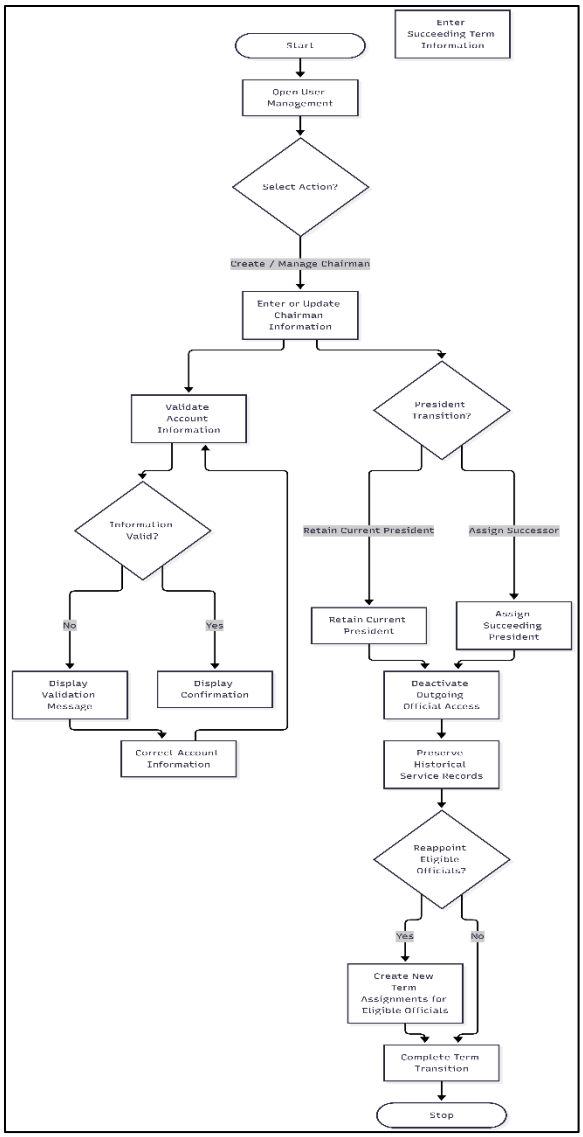


Figure 7 User Account Management

Figure 7 illustrates the process flow for user account and term management. The SK Federation President/Admin can create and manage SK Chairman accounts through the User Management module. The system validates the submitted account information, creates or updates the Chairman account, and applies the appropriate role-based permissions and account status. If the succeeding term start year is in the past, the system rejects the term creation and displays an appropriate validation message. During the transition, the President/Admin may retain the current President or assign a succeeding President, while outgoing official access is deactivated and previous service records are preserved. Eligible officials may also be reappointed for the succeeding term without overwriting their previous service history.

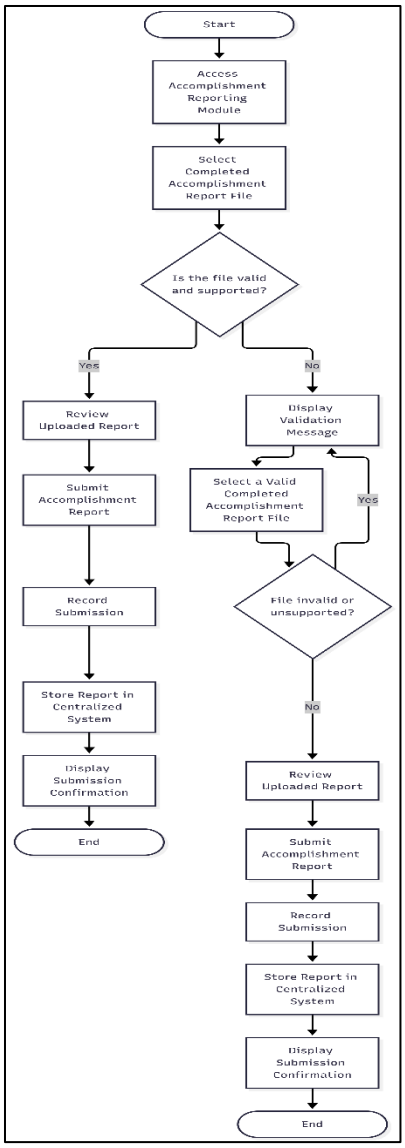


Figure 8 Accomplishment Report Submission

Figure 8 illustrates the process flow for accomplishment report submission. The process begins when an authorized SK Chairman or SK Secretary accesses the Accomplishment Reporting module and selects a completed accomplishment report file. The system validates the uploaded file. If the file is valid, the user reviews and submits the report, and the system records and stores the submission in the centralized system. If the applicable submission slot is already Past Deadline, the valid report may still be accepted and recorded as a late submission until the SK Federation President/Admin manually closes the slot. Once the slot is Closed, the system prevents new submissions. If the uploaded file is invalid or unsupported, the system displays an appropriate validation message and requires the user to select a valid completed accomplishment report file.

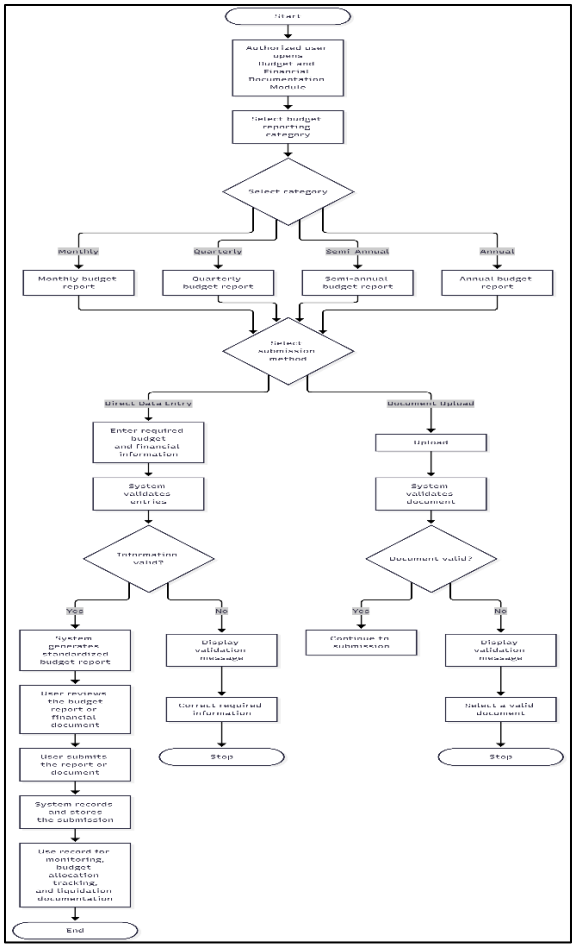


Figure 9 Budget and Financial Documentation

Figure 9 illustrates the process flow for budget and financial documentation through the web application. The process begins when an authorized user accesses the Budget and Financial Documentation Module and selects the applicable monthly, quarterly, semi-annual, or annual budget reporting category and submission method. For direct data entry, the user enters the required budget and financial information, and the system validates the entries before automatically generating the standardized budget report. For document upload, the user selects a completed budget or financial document, which is then validated by the system. The user reviews and submits the budget report or document, after which the system records and stores the submission for monitoring, budget allocation tracking, and liquidation documentation. Financial verification remains the responsibility of the barangay SK Chairman prior to submission. Direct data entry is available through the web application only, while the mobile application supports budget and financial document upload only.

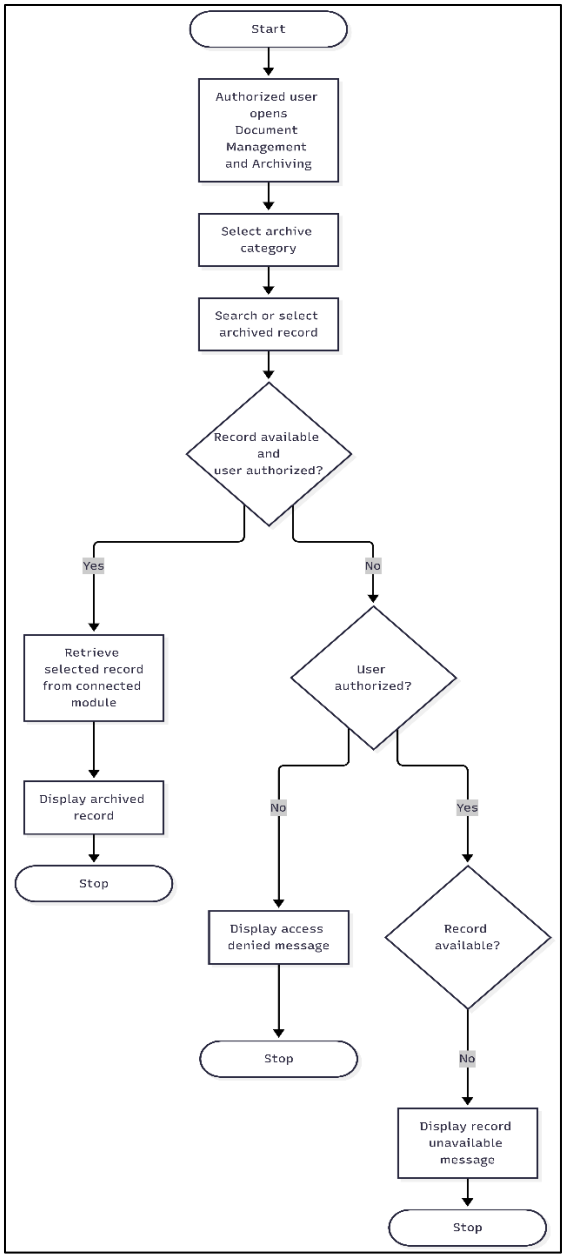


Figure 10 Document Management and Archiving

Figure 10 illustrates the process flow for Document Management and Archiving. Authorized users search for or select records preserved from connected system modules, including Accomplishment Reports, Budget Documents, and Event Records. The system verifies the user's access permission before retrieving and displaying the selected record. Unauthorized requests are denied, while unavailable records result in an appropriate message. The function preserves and retrieves connected-module records and does not operate as a separate generic document-upload repository.

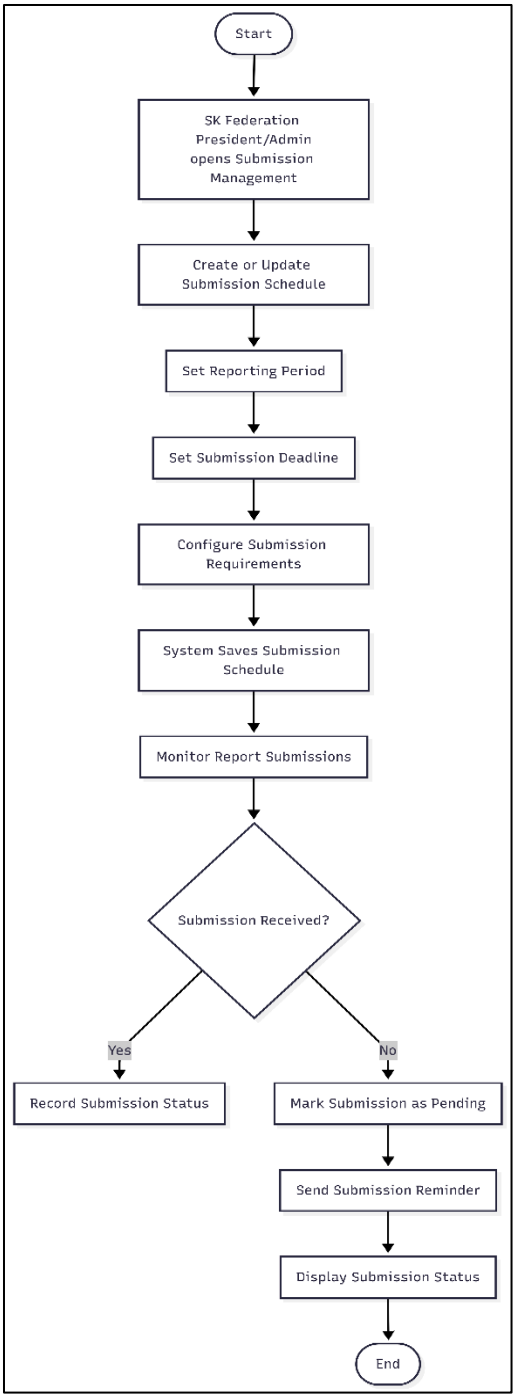


Figure 11 Module Management

Figure 11 illustrates the process flow for Module Management. The SK Federation President/Admin creates or updates accomplishment and budget submission slots. The system displays each slot as Upcoming, Open, Past Deadline, or Closed. After the deadline, the slot becomes Past Deadline and may still accept late submissions until the President/Admin manually closes it. Once Closed, no new submissions are accepted.

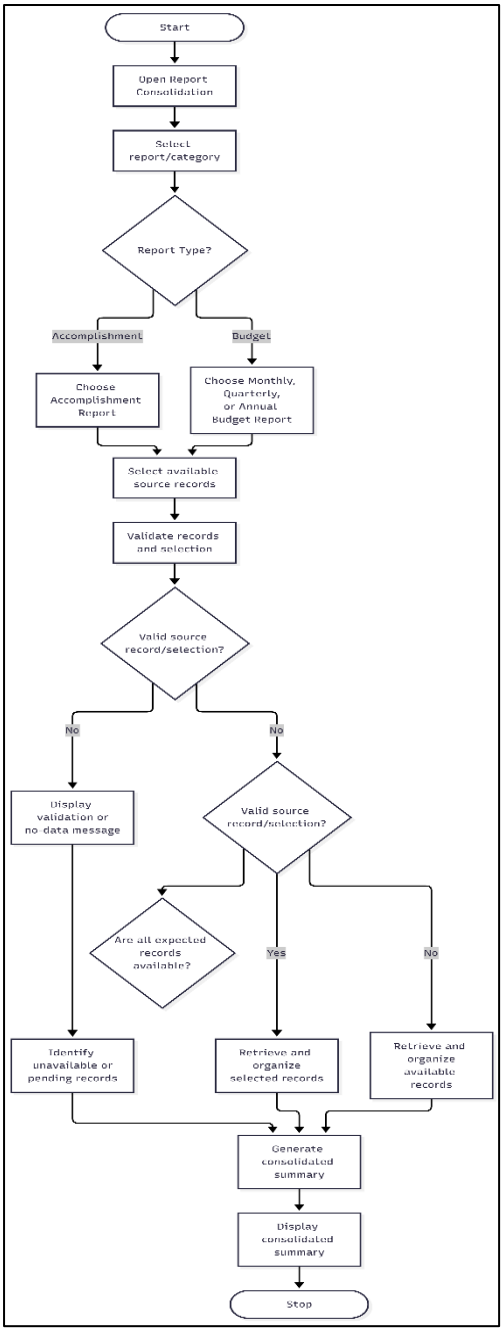


Figure 12 Report Consolidation

Figure 12 illustrates the process flow for report consolidation. The SK Federation President/Admin selects applicable submitted accomplishment reports or monthly, quarterly, semi-annual, and annual budget reports. The system retrieves and compiles the selected records into a consolidated PDF file for federation-level monitoring and documentation, including available quality documentation. If no valid record is available, the system displays an appropriate no-data message.

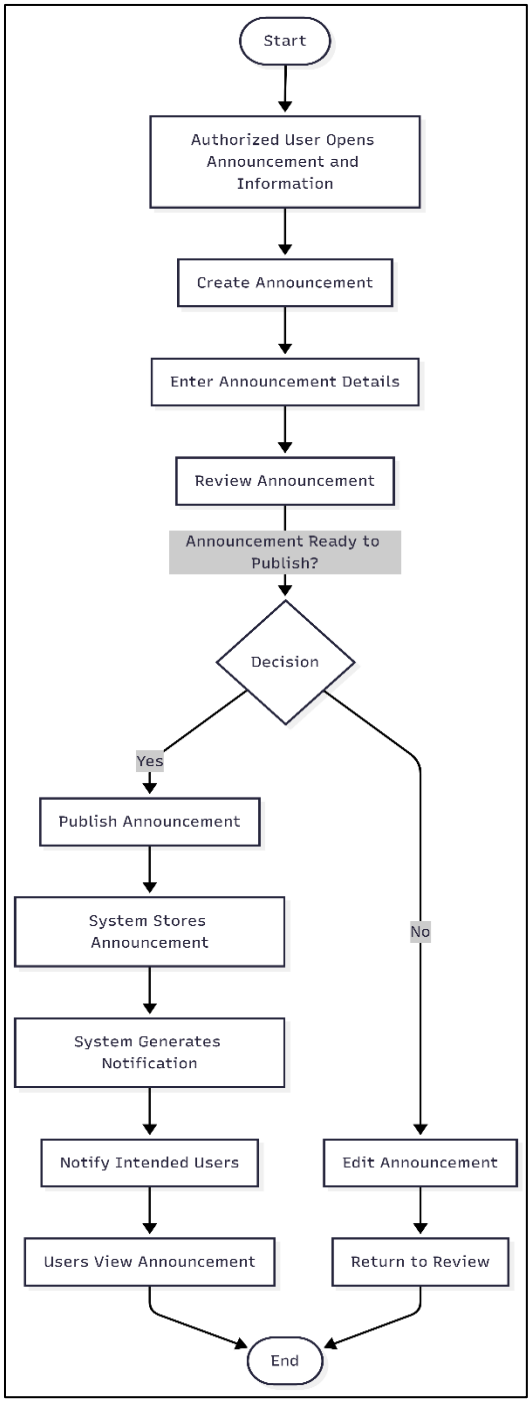


Figure 13 Announcement and Notification

Figure 13 illustrates the process flow for announcements and notifications. The President/Admin creates or updates an announcement and enters the required details. Once confirmed, the system publishes the announcement and generates a system notification for the intended users. Authenticated SK officials can then view the published announcement.

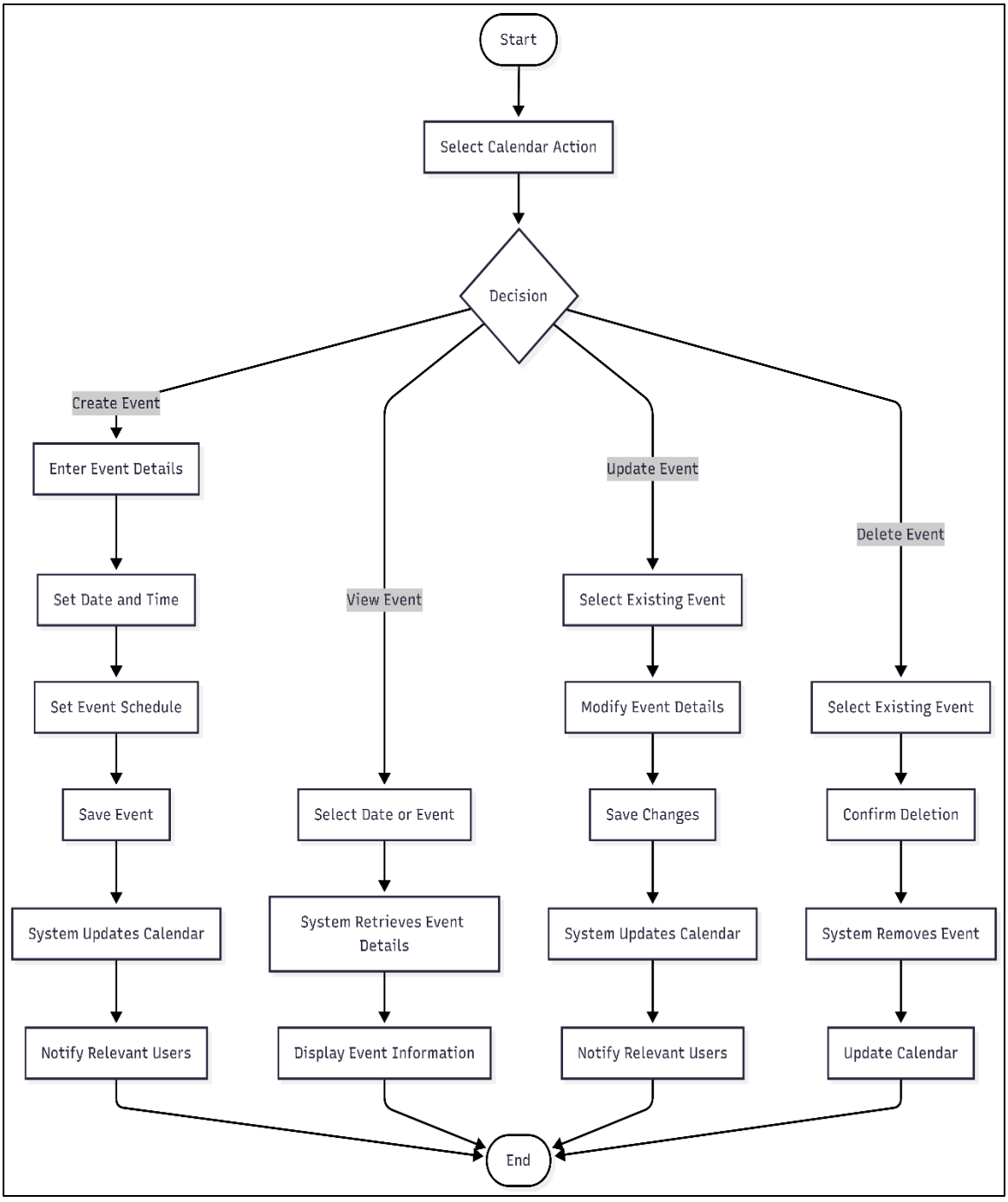


Figure 14 Activity Calendar and Scheduling

Figure 14 illustrates the process flow for activity calendar and scheduling. The President/Admin can create, edit, and delete calendar events by managing the title, date, time, location, audience, and related event settings. Authenticated SK officials can select an event to view its available details.

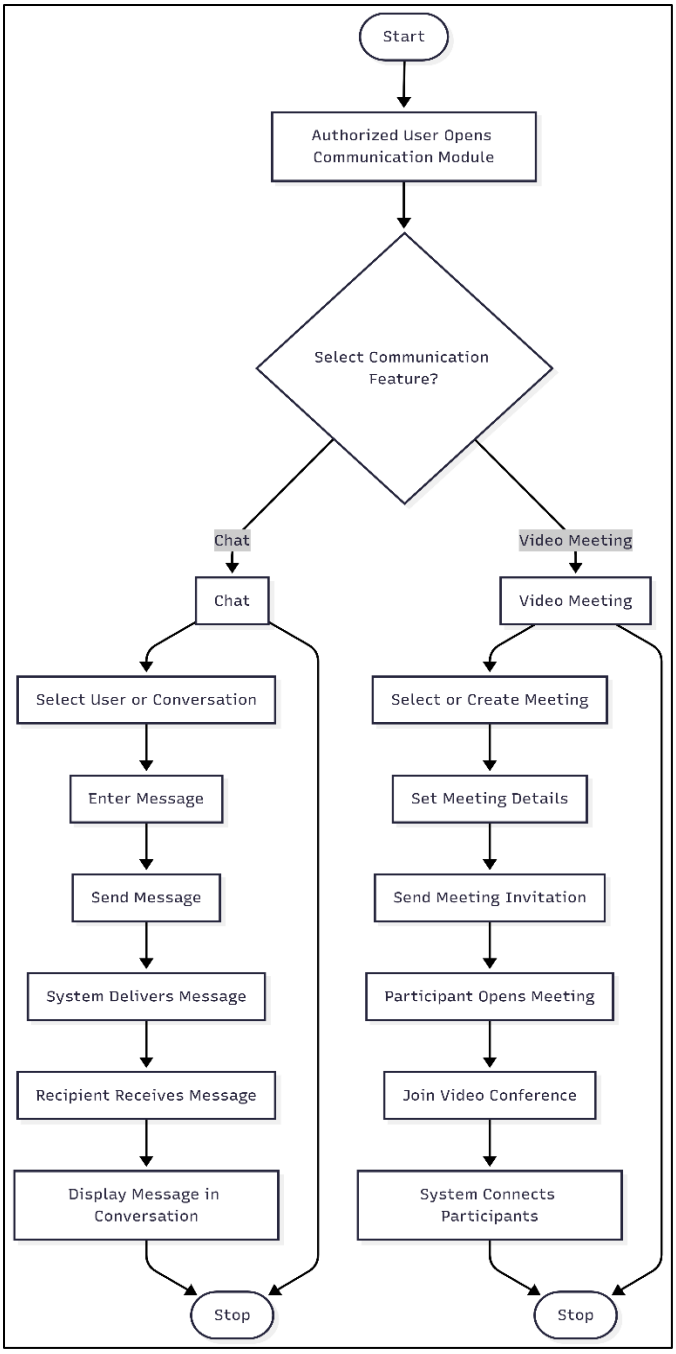


Figure 15 Chat and Meetings

Figure 15 illustrates the process flow for chat and meetings. The process begins when an authorized user accesses the communication functions and selects either the chat or meeting function. For chat, the user selects a conversation, enters a message, and sends it through the system, which delivers and synchronizes the message to the intended recipient. For meetings, the President/Admin schedules and manages the meeting, while SK Chairmen and Secretaries may join and participate, including virtually.

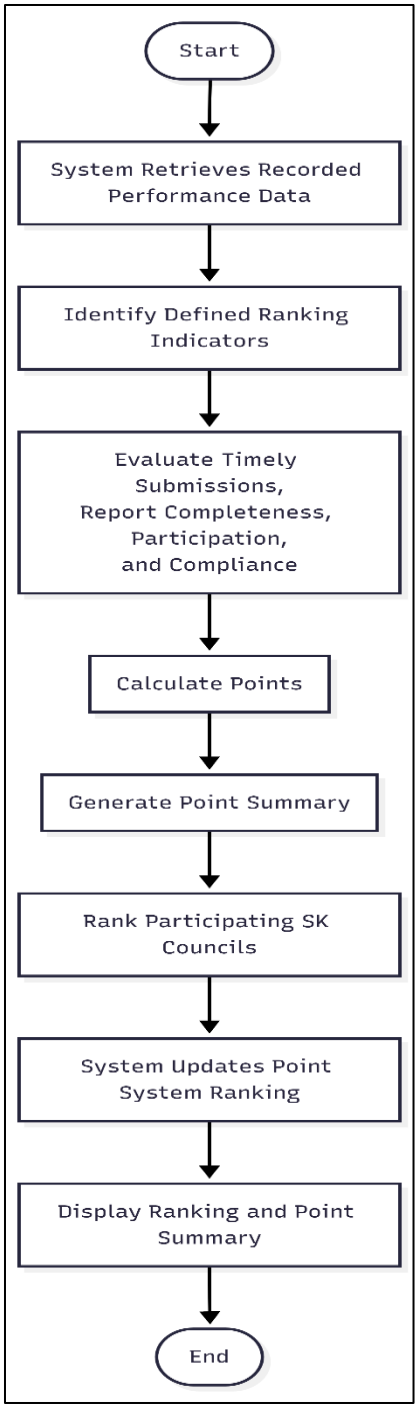


Figure 16 Point System Ranking

Figure 16 illustrates the process flow for the Point System Ranking module. The system retrieves recorded data and calculates points based on defined documentation, compliance, late-submission, and official-meeting participation criteria. The accumulated points are used to generate point summaries and rank participating SK councils, which are then displayed to authorized users.

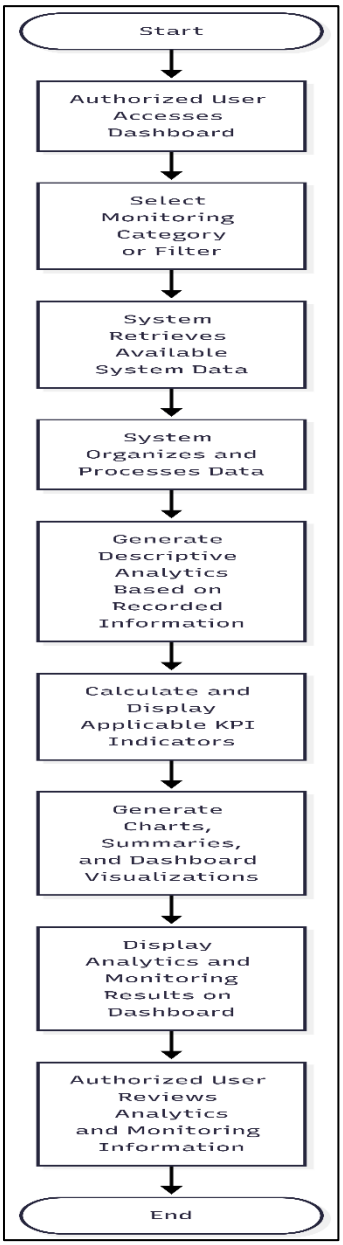


Figure 17 Analytics and Monitoring

Figure 17 illustrates the process flow for analytics and monitoring functions integrated into the Dashboard Module. The process begins when an authorized user accesses the Dashboard and selects the desired monitoring category or filter. The system retrieves and organizes the available operational and financial data and generates descriptive analytics based on the recorded information. The system then calculates and displays applicable KPI indicators, charts, summaries, and dashboard visualizations. The resulting information is presented to authorized users to support performance tracking, monitoring, and data-driven decision-making.

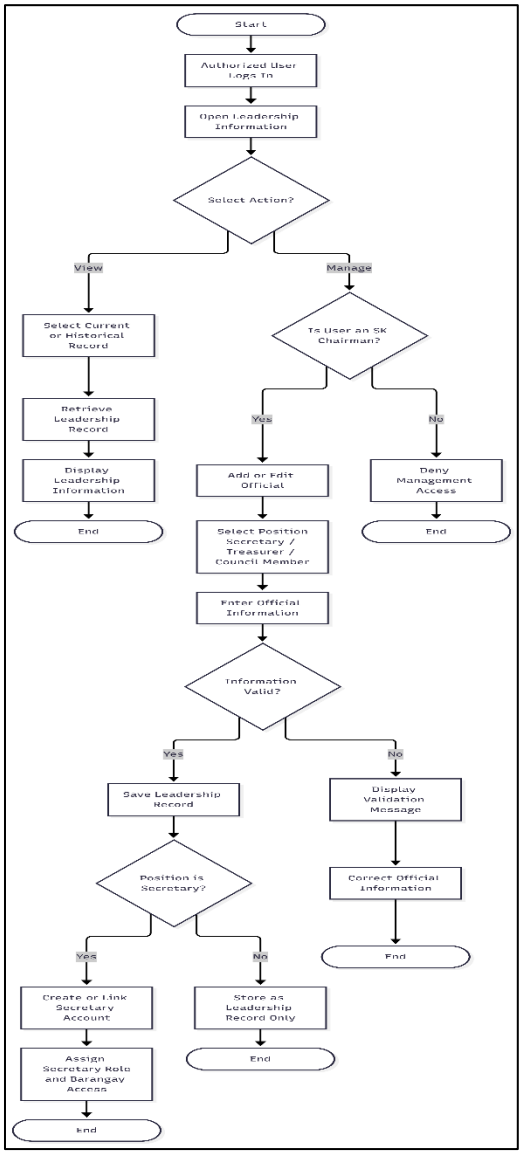


Figure 18 Leadership Information

Figure 18 illustrates the process flow for leadership information management and viewing. Authorized users may access verified current and historical leadership records through the Leadership Information module. The SK Chairman may also manage leadership records for the assigned barangay by adding or updating the SK Secretary, Treasurer, and Council Members. The system validates the submitted official information before saving the leadership record. When an SK Secretary is added, the system creates or links the corresponding Secretary system account and assigns the appropriate role and barangay access. Treasurer and Council Member records are stored as leadership information only and do not receive system login accounts. Historical leadership records remain available for reference and continuity across SK terms.

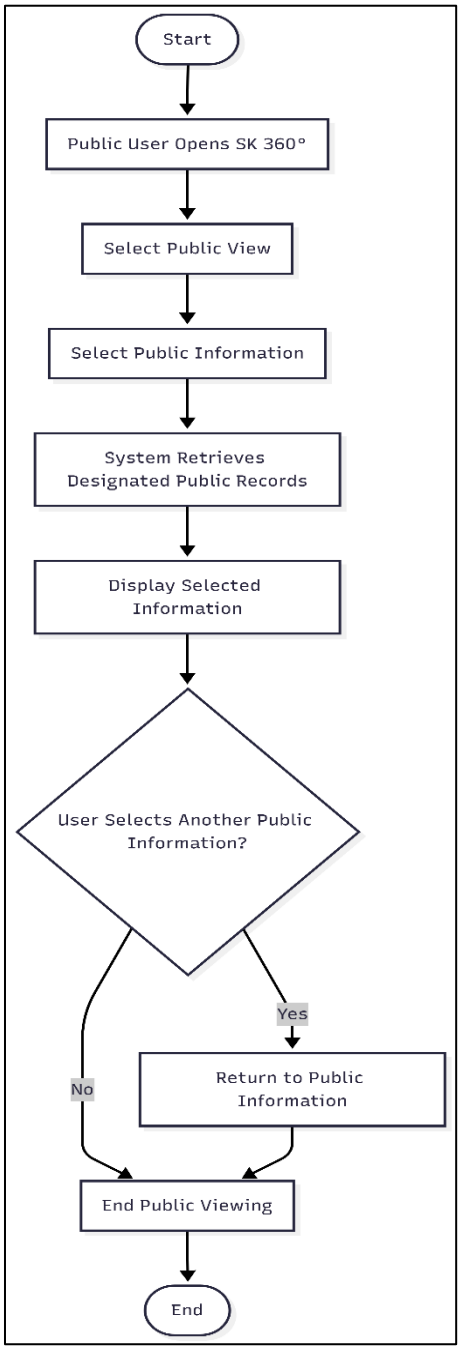


Figure 19 Public View and Information Access

Figure 19 illustrates the process flow for public information access through the web application. Public users may access selected announcements, calendar information, leadership information, annual budget information, and the published LYDP without creating an account or logging in. The system retrieves and displays the selected public information. The Point System Ranking is not available to public users.

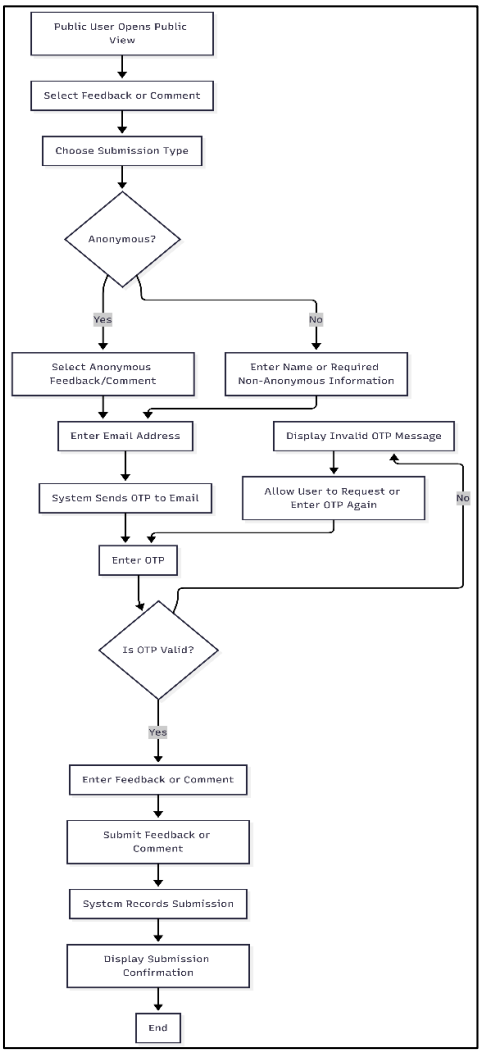


Figure 20 Public Feedback and Commenting with Email OTP

Figure 20 illustrates the process flow for public feedback and commenting with email OTP verification through the web application. The process begins when a public user accesses the web-based Public View and selects the feedback or commenting function. The user chooses whether to submit the feedback or comment anonymously or non-anonymously and provides the required information, including an email address for OTP verification. The system sends an OTP to the provided email address, which the user enters for verification. If the OTP is valid, the user may submit the feedback or comment, and the system records the submission and displays a confirmation message. If the OTP is invalid, the system displays an error message and allows the user to request or enter another OTP.

4.3 DESCRIPTION OF THE SYSTEM

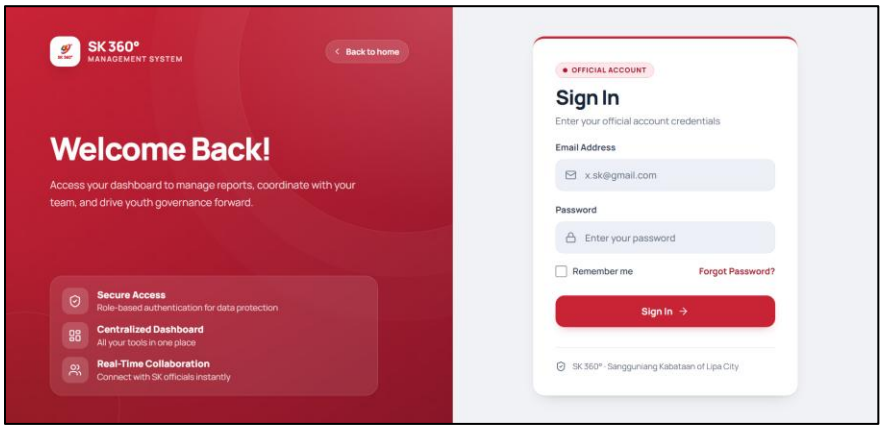


Figure 21 Login Page

Figure 21 presents the Login Page of the SK 360° web application, where users securely access their accounts using valid credentials. The system verifies user identity and prevents unauthorized access through authentication and validation. This ensures secure system entry and directs users to their respective dashboards.

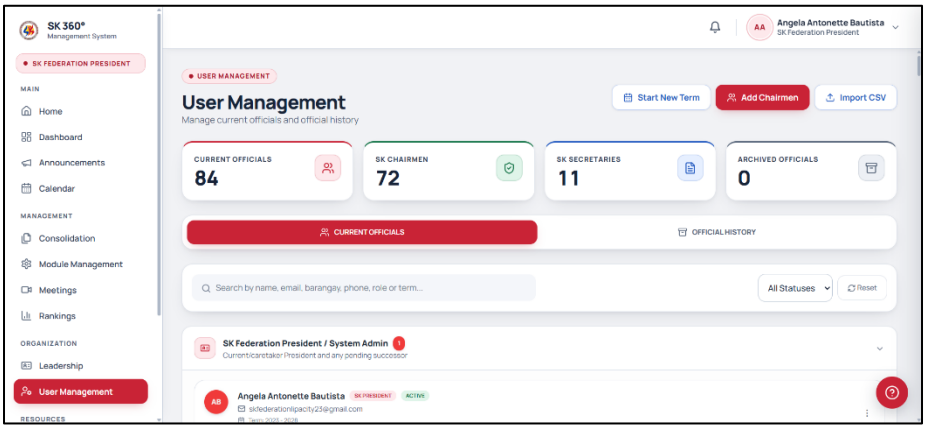


Figure 22 User Management

Figure 22 presents the User Management Module, which allows the SK Federation President/Admin to manage SK Chairman accounts and term transitions. It also supports succeeding President assignment, reappointment of eligible officials, and preservation of previous service records.

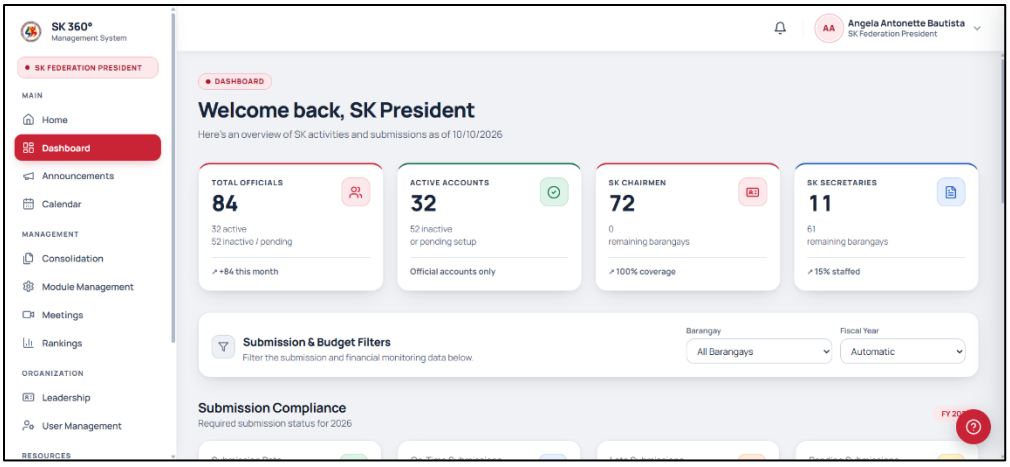


Figure 23 Dashboard Module

Figure 23 presents the Dashboard Module, which serves as the main monitoring interface after user login. It displays report summaries, announcements, notifications, submission compliance, budget utilization, submission trends, descriptive analytics, KPI indicators, charts, and summary cards according to the user's authorized access. This allows authenticated SK officials to monitor relevant system information and access applicable functions from a centralized interface.

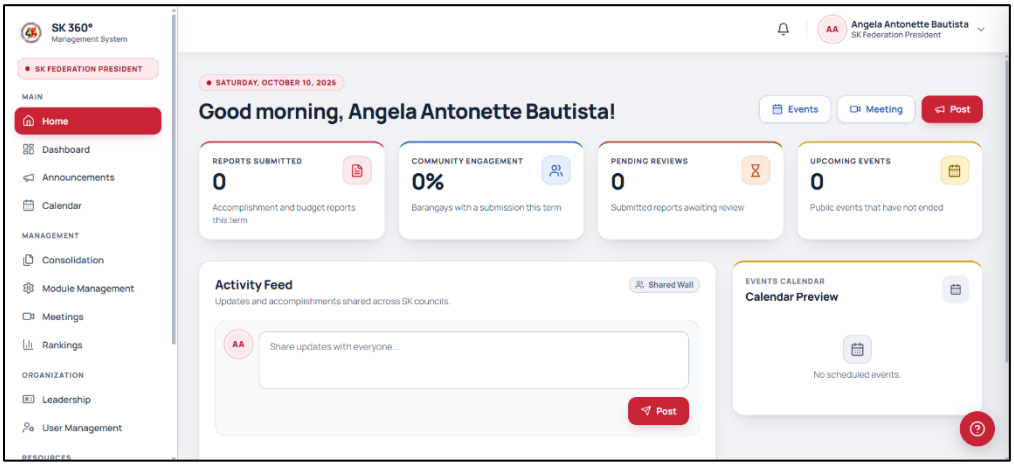


Figure 24 Home Page Module

Figure 24 presents the authenticated Home Page of the SK 360° web application, which provides logged-in SK officials with access to announcements, upcoming activities, and available system functions according to their assigned role. Public users access publicly available information separately through the Public Information Portal.

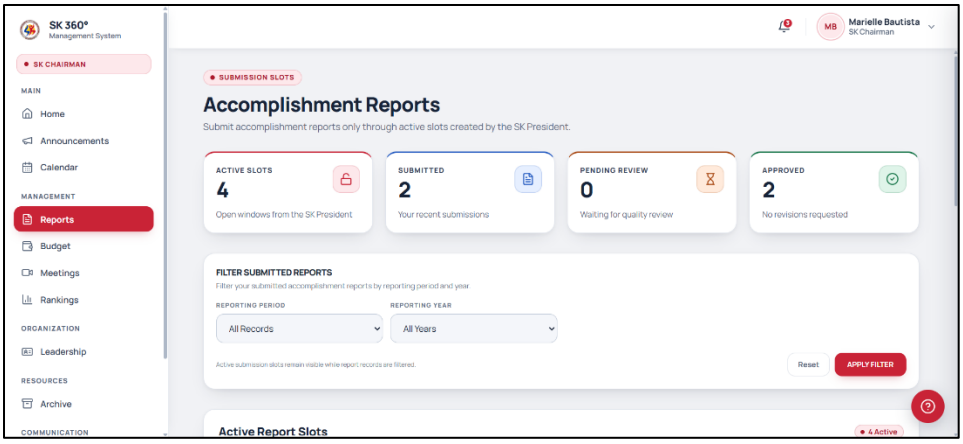


Figure 25 Accomplishment Reporting Module

Figure 25 presents the Accomplishment Reporting Module, which allows authorized SK Chairmen and SK Secretaries to upload completed accomplishment report files for centralized storage, administrative review, monitoring, and consolidation. Accomplishment reporting uses completed-file upload and does not require direct report data entry, report-category selection, reporting-period selection, or automatic report generation.

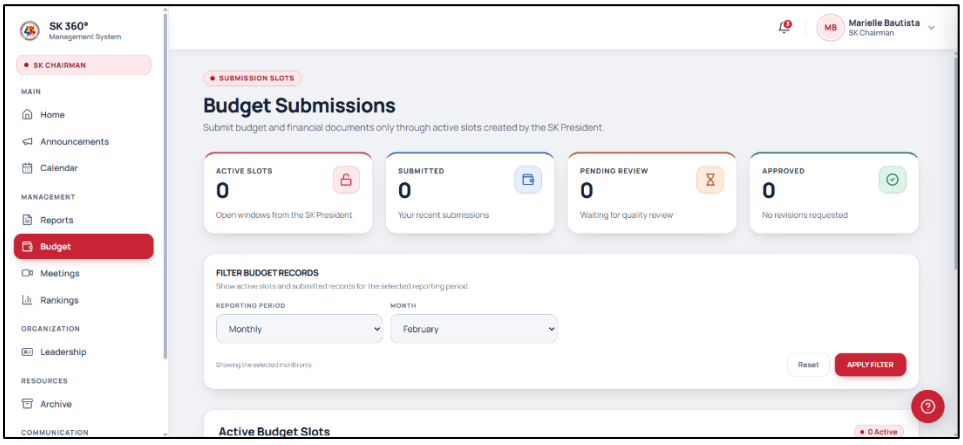


Figure 26 Budget and Financial Documentation Module

Figure 26 presents the Budget and Financial Documentation Module, which supports direct data entry or document upload for monthly, quarterly, semi-annual, and annual budget reports. Direct data entry generates a standardized budget report, while financial verification remains the responsibility of the barangay SK Chairman before submission.

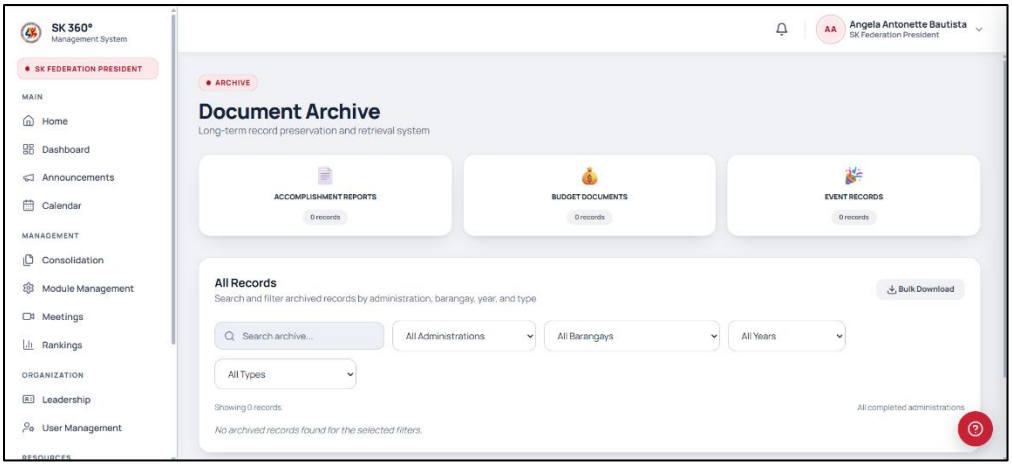


Figure 27 Document Management and Archiving Module

Figure 27 presents the Document Management and Archiving Module, which preserves and provides authorized access to records generated or submitted through connected system modules, including accomplishment reports, budget documents, and event records.

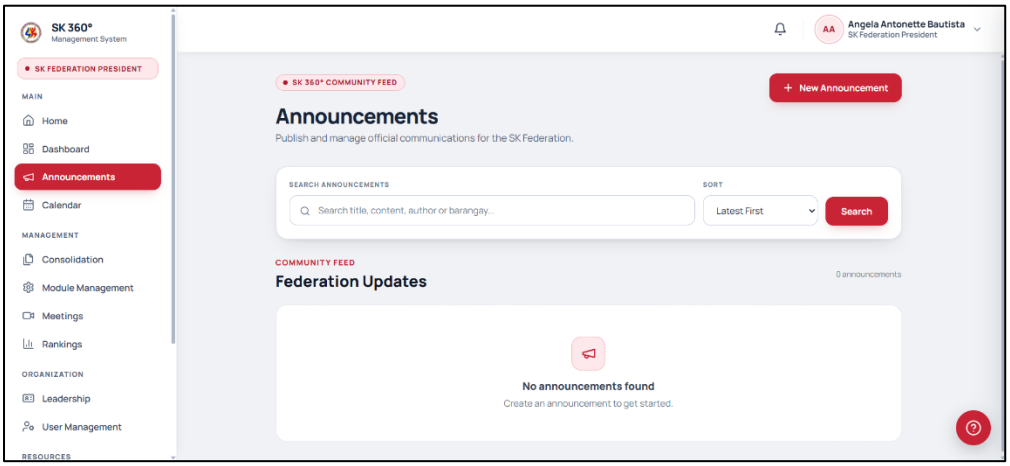


Figure 28 Announcement

Figure 28 presents the Announcement Module, which allows the SK Federation President/Admin to create, update, publish, and manage official announcements and notices. Authenticated SK officials can view applicable announcements, while announcements designated for public access may also be viewed through the web-based Public Information Portal.

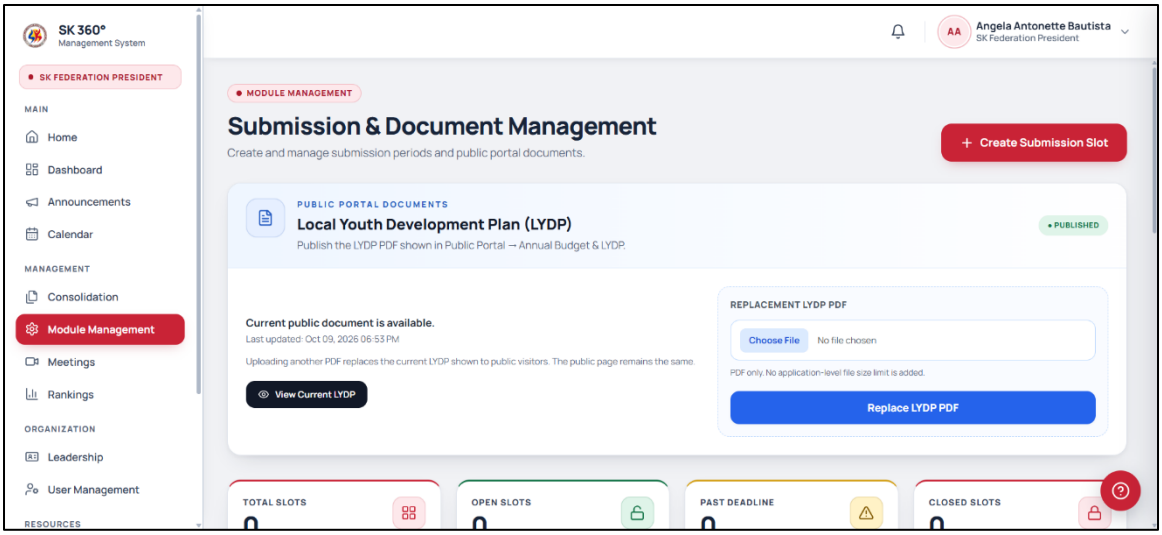


Figure 29 Module Management Module

Figure 29 presents the Module Management feature, which allows the SK Federation President/Admin to create and manage accomplishment and budget submission slots with Upcoming, Open, Past Deadline, and Closed statuses. Past Deadline slots may accept late submissions until manually closed.

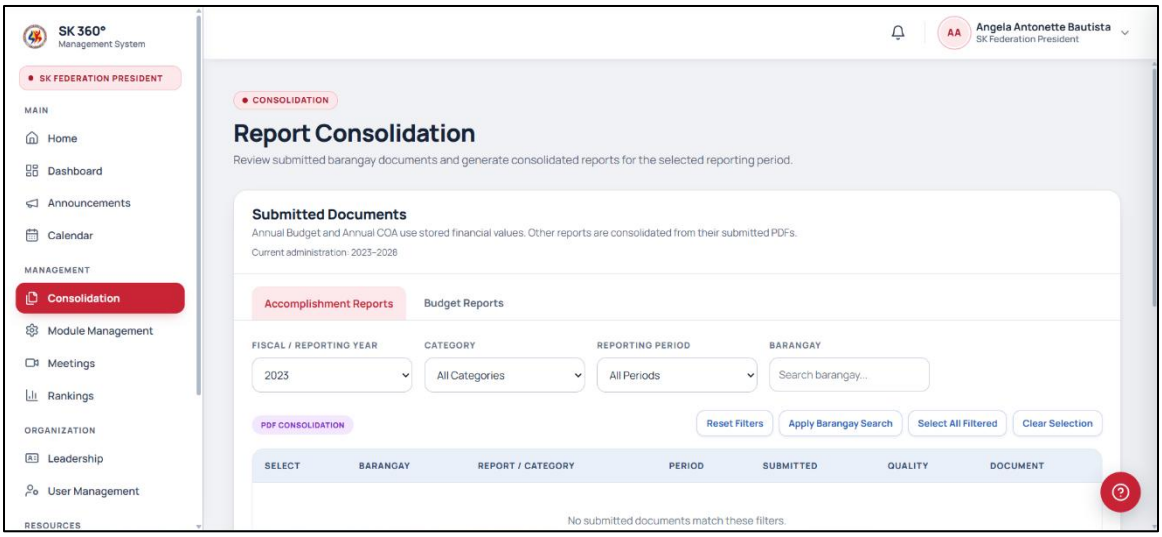


Figure 30 Report Consolidation Module

Figure 30 presents the Report Consolidation Module, which compiles selected accomplishment reports and monthly, quarterly, semi-annual, and annual budget reports into consolidated PDF files, including available quality documentation for barangay submissions.

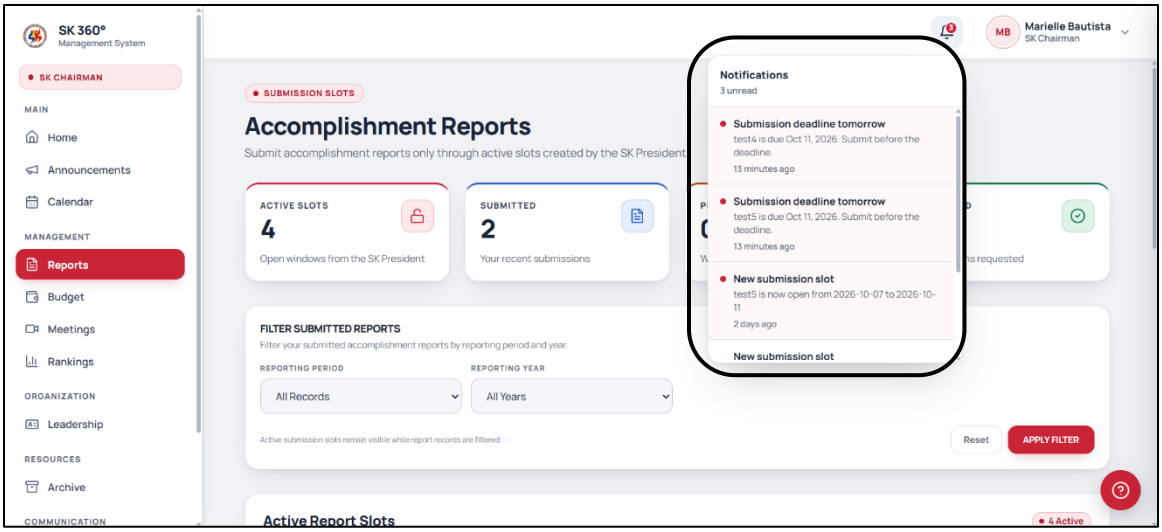


Figure 31 Notification Module

Figure 31 presents the Notification Module, which provides system notifications for updates, announcements, deadlines, submission reminders, meeting reminders, and other system activities. Notifications are stored and displayed within the system for authorized users.

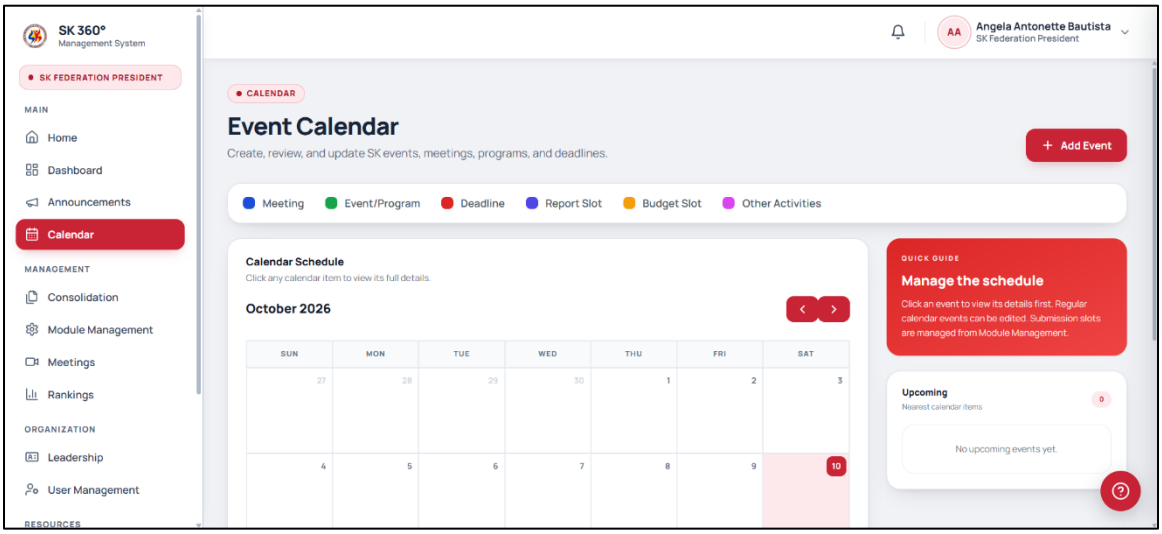


Figure 32 Activity Calendar and Scheduling Module

Figure 32 presents the Activity Calendar Module, which allows the President/Admin to manage calendar events while authenticated SK officials can view schedules, deadlines, and event details.

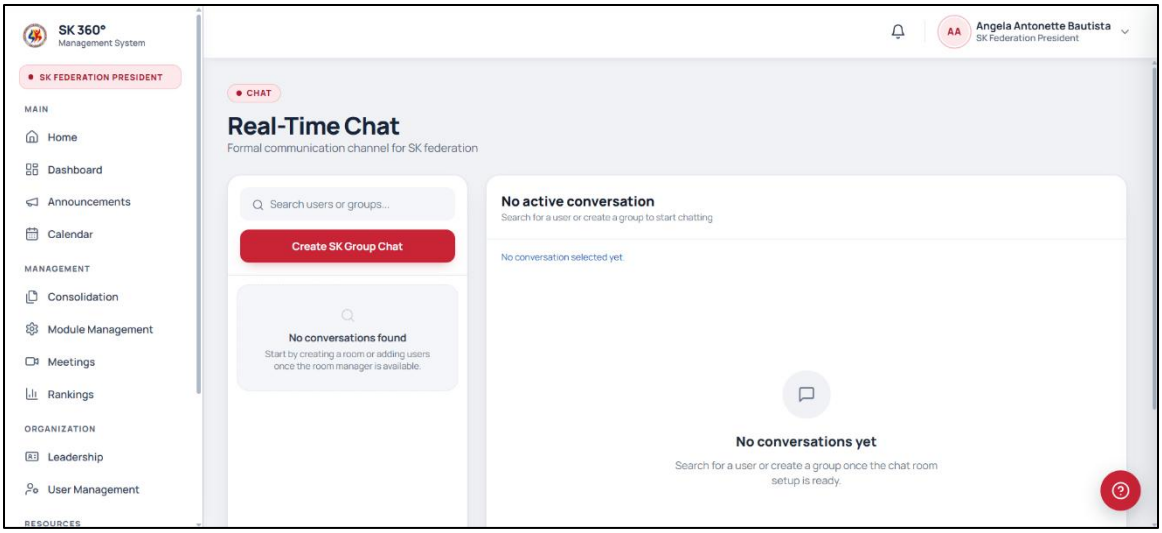


Figure 33 Chat Module

Figure 33 presents the Chat Module, which enables real-time communication among SK officials within the system. Users can send messages and participate in discussions without relying on external platforms. This improves coordination and communication efficiency.

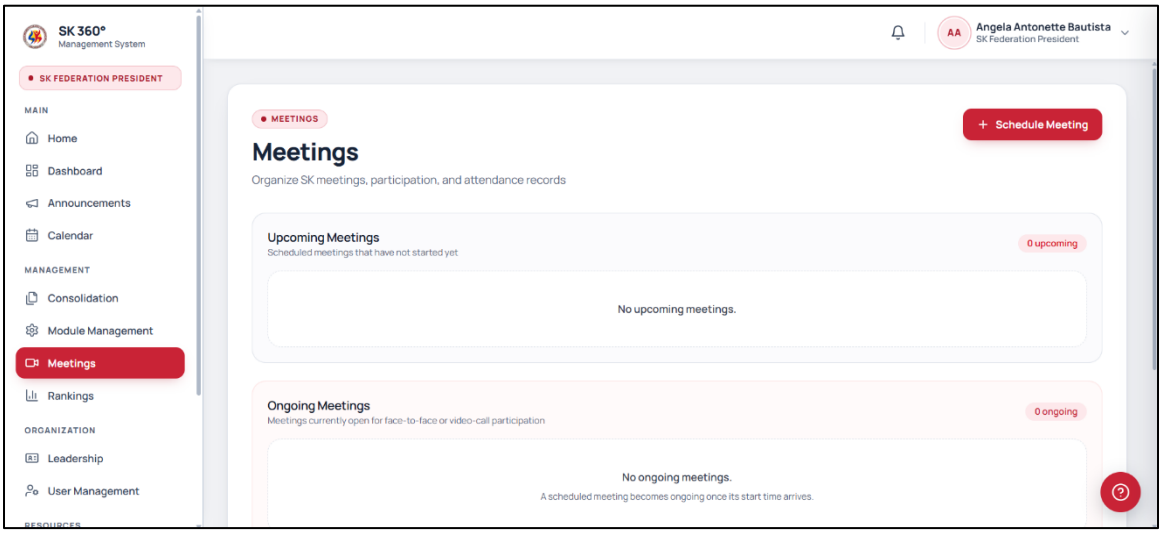


Figure 34 Meetings Module

Figure 34 presents the Meetings Module, which allows the President/Admin to schedule and manage meetings while SK Chairmen and Secretaries may participate, including virtually.

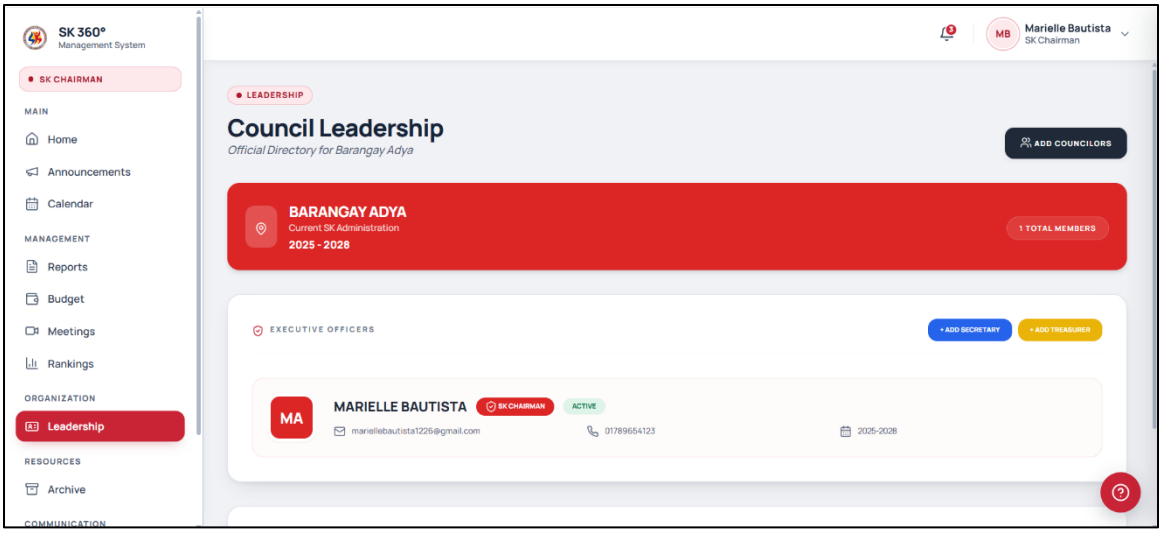


Figure 35 Leadership Information Module

Figure 35 presents the Leadership Information Module, which allows authorized SK Chairmen to manage current leadership records for the Secretary, Treasurer, and Council Members while preserving historical records across SK terms. Adding an SK Secretary also creates or links the corresponding system account, while Treasurer and Council Member records remain for leadership information only.

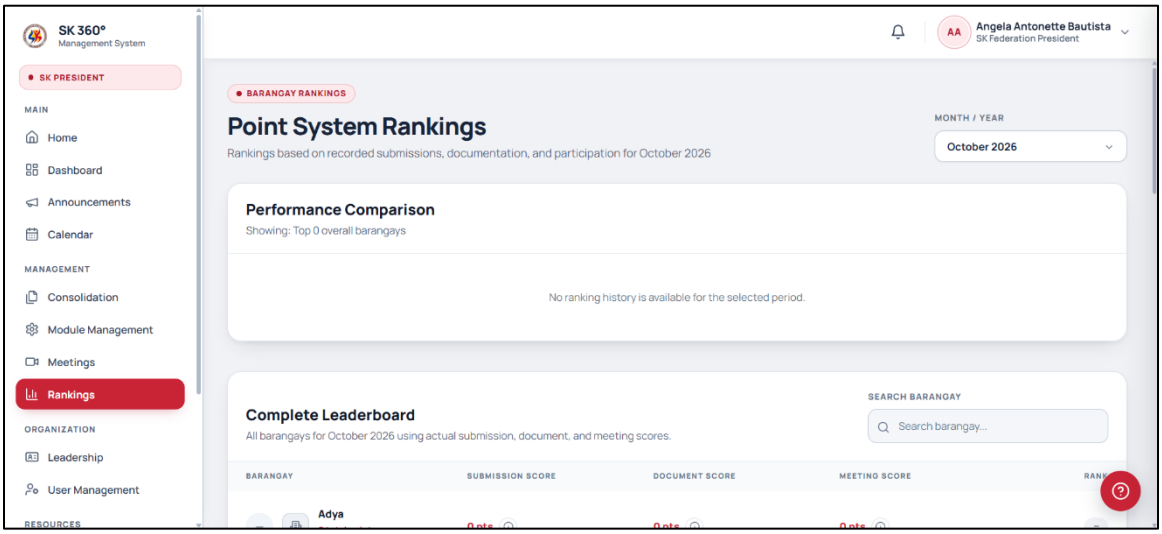


Figure 36 Point System Ranking

Figure 36 presents the Point System Ranking Module, which generates rankings and point summaries based on defined documentation, compliance, late-submission, and official-meeting participation criteria.

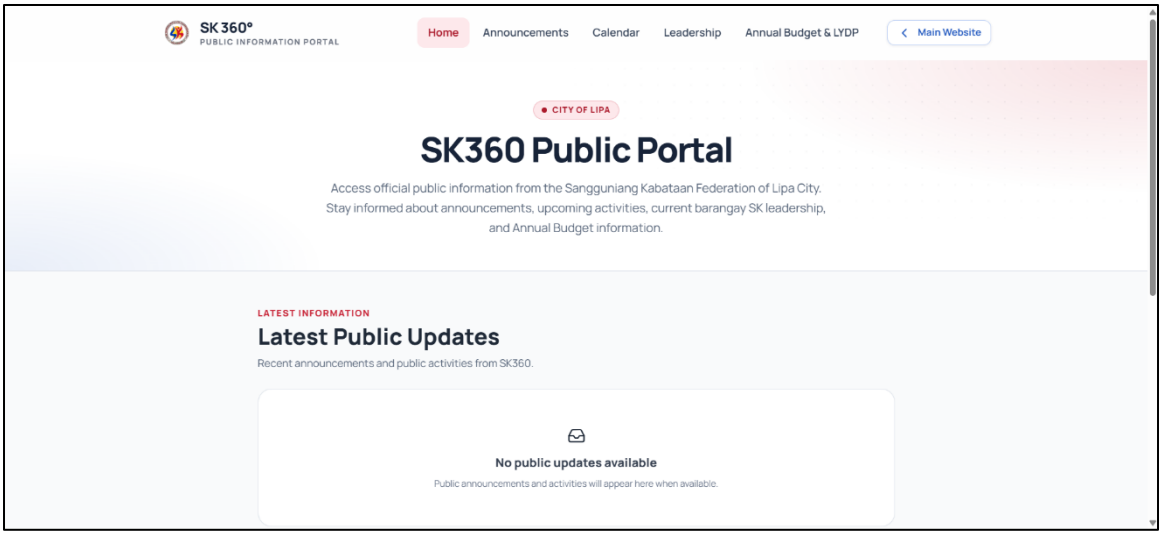


Figure 37 Public View Module

Figure 37 presents the Public View Module, which allows public users to access announcements, calendar information, leadership information, annual budget information, and the published LYDP without registration. The Point System Ranking is not accessible to public users.

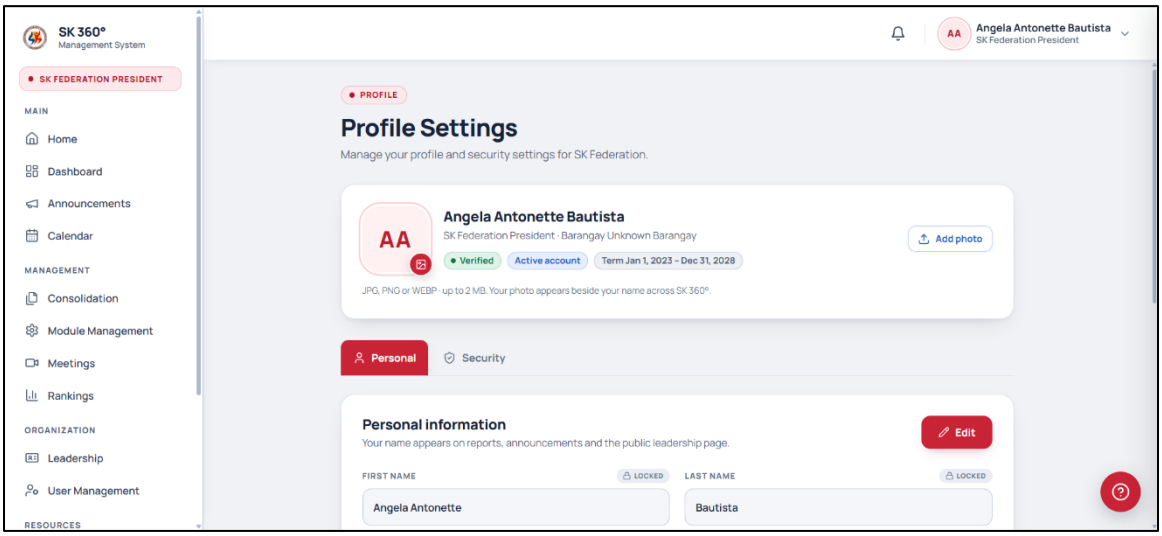


Figure 38 Profile Settings Module

Figure 38 presents the Profile Settings Module, which allows users to manage their account information. It includes updating personal details and passwords securely. This ensures user control and data privacy.

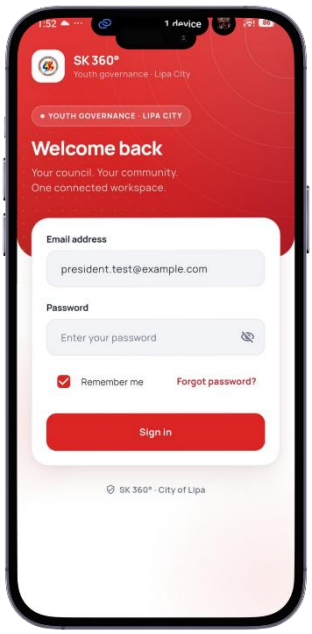


Figure 39 Mobile Login Page

Figure 39 presents the Mobile Login Page, which authenticates authorized SK officials and provides role-based access to the mobile application.

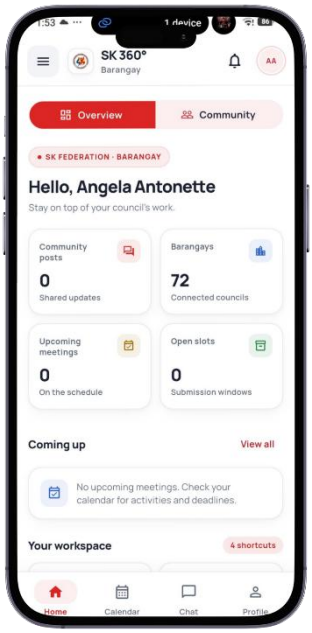


Figure 40 Home Page Module

Figure 40 presents the Mobile Home Page, which provides authenticated SK officials with announcements and quick access to available mobile functions.

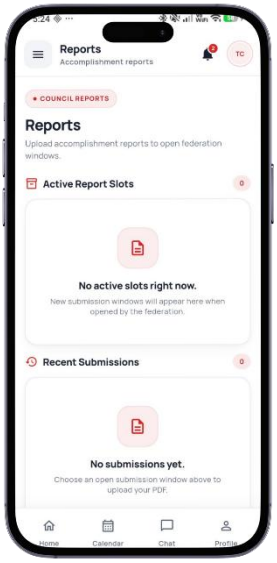


Figure 41 Mobile Accomplishment Reporting

Figure 41 presents the Mobile Accomplishment Reporting Module, which allows SK Chairmen and SK Secretaries to upload completed accomplishment report files through the mobile application.

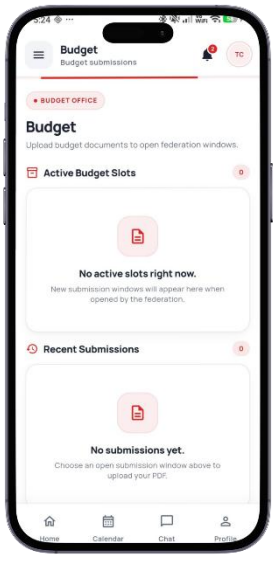


Figure 42 Mobile Budget Documents Module

Figure 42 presents the Mobile Budget Documents Module, which allows SK Chairmen and SK Secretaries to upload completed budget and financial documents through the mobile application. Direct data entry and automatic report generation are not available on mobile.

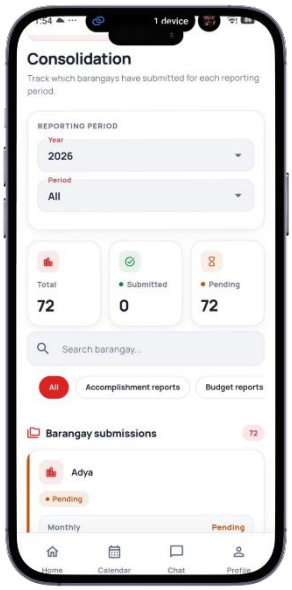


Figure 43 Mobile Consolidation Module

Figure 43 presents the Mobile Consolidation Module, which allows authorized SK officials to monitor submission statuses, summary counts, reporting periods, and available quality documentation through filters. It does not generate consolidated PDF files.

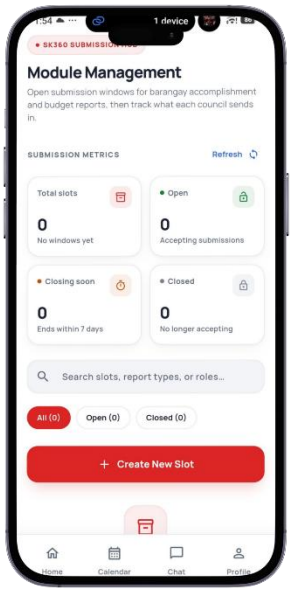


Figure 44 Mobile Module Management

Figure 44 presents the Mobile Module Management feature, which allows the SK Federation President/Admin to create and manage accomplishment and budget submission slots with Upcoming, Open, Past Deadline, and Closed statuses.

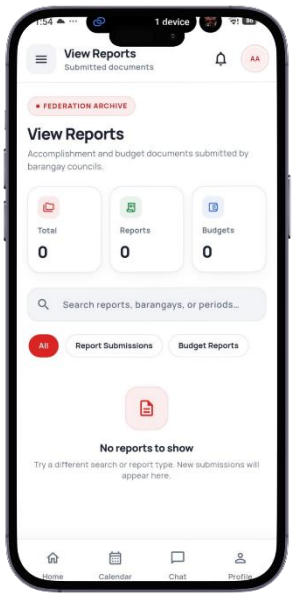


Figure 45 Mobile View Reports

Figure 45 presents the Mobile View Reports feature, which allows the SK Federation President/Admin to view, search, filter, and open submitted accomplishment and budget reports through the mobile application. The feature is view-only.

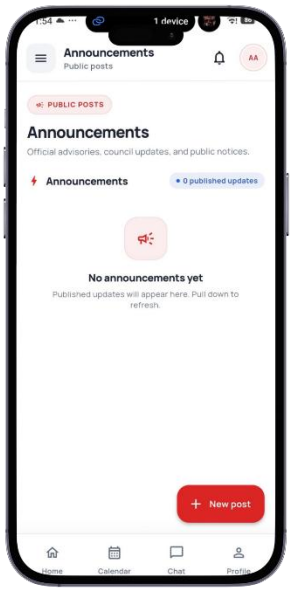


Figure 46 Mobile Announcement

Figure 46 presents the Mobile Announcement feature, which allows the President/Admin to manage announcements while authenticated SK officials can view published announcements.

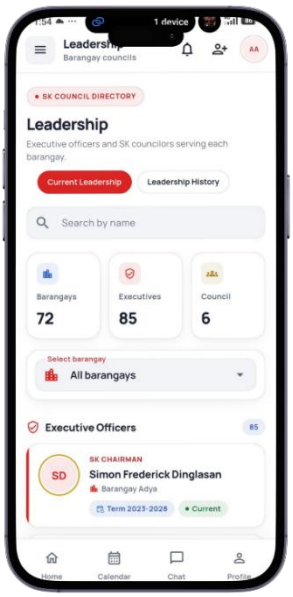


Figure 47 Mobile Leadership Profiles

Figure 47 presents the Mobile Leadership Profiles Module, which displays verified current and historical leadership information of SK officials. The module allows authorized users to view leadership profiles and official details through the Android mobile application, supporting convenient access to leadership information.

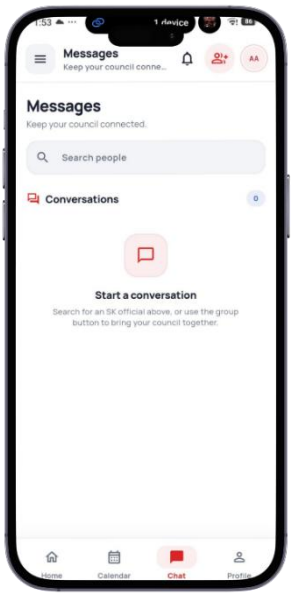


Figure 48 Mobile Chat

Figure 48 presents the mobile chat feature, which enables real-time messaging. It allows users to communicate easily. This improves coordination.

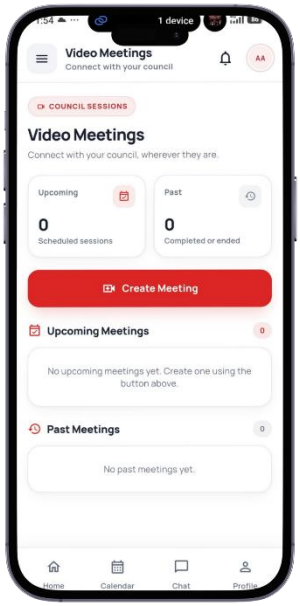


Figure 49 Mobile Meetings

Figure 49 presents the Mobile Meetings feature, which allows authorized SK officials to participate in meetings, while meeting management is limited to the President/Admin.



Figure 50 Mobile Activity Calendar

Figure 50 presents the Mobile Activity Calendar, which allows the President/Admin to manage events while authenticated SK officials can view schedules and deadlines.

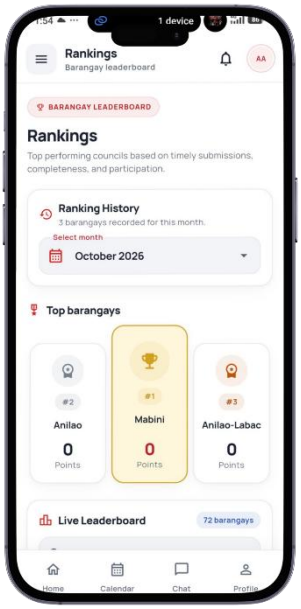


Figure 51 Mobile Rankings

Figure 51 presents the Mobile Point System Ranking feature, which displays rankings and point summaries based on defined performance criteria.

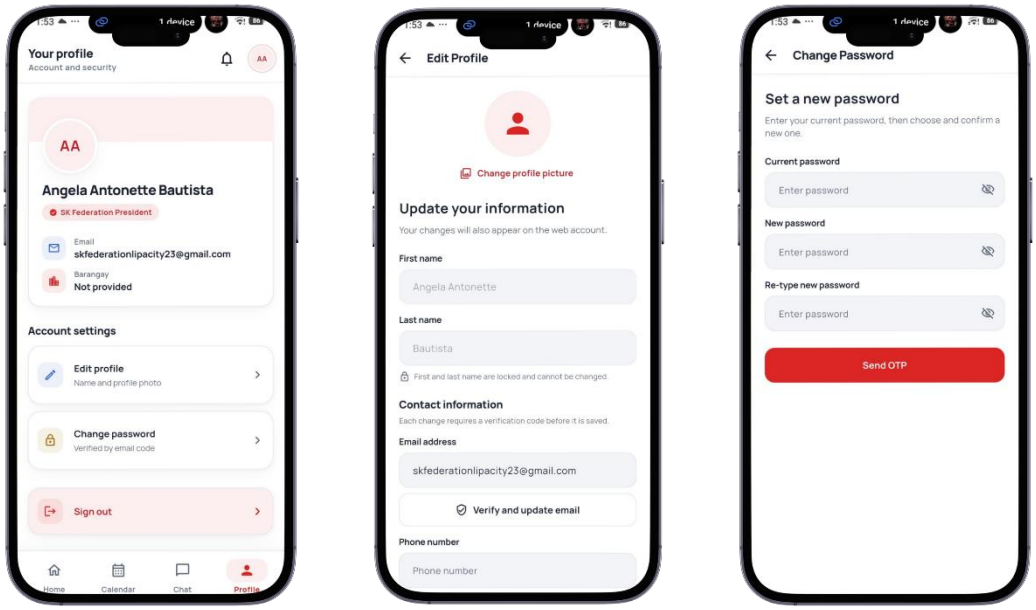


Figure 52 Mobile Profile

Figure 52 presents the Mobile Profile Module, which allows users to manage their account information and settings through the mobile application.

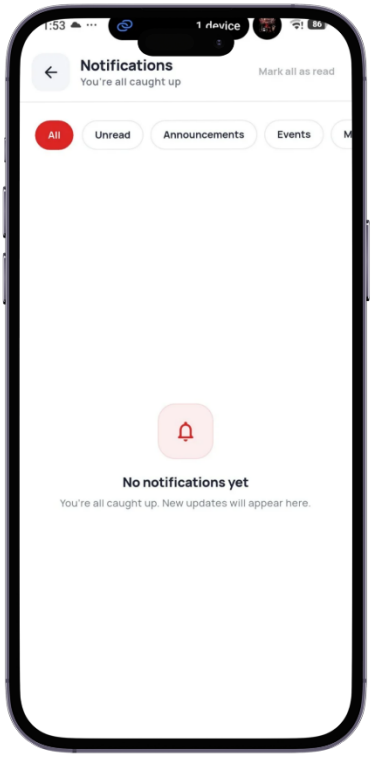


Figure 53 Mobile Notification

Figure 53 presents the Mobile Notification Module, which provides system notifications for updates, announcements, submission reminders, meeting reminders, deadlines, and other system activities.

4.4 DEPLOYMENT AND IMPLEMENTATION PLAN

Deployment Plan

The SK 360° system will be deployed as a centralized digital platform consisting of a web application and an Android mobile application. The web application will be accessed through supported web browsers using computers or laptops, while the mobile application will be deployed on compatible Android smartphones and tablets. Both applications will communicate with the centralized server through an internet connection to provide authorized users with access to reporting, documentation, communication, monitoring, scheduling, notifications, and other system functions.

The web application will be hosted on a cloud-based application server that provides the required runtime environment for the system's backend services. The server will host the SK 360° web application, Laravel backend, and application programming interfaces (APIs), while the MySQL database will serve as the centralized repository for user accounts, accomplishment reports, budget and financial records, documents, announcements, schedules, leadership information, rankings, notifications, and other system-related data. A stable internet connection is required to allow users to access the centralized system and synchronize information between the web and Android mobile applications.

The Android mobile application will be developed using Flutter and deployed on compatible Android devices. The application will communicate with the centralized backend through the internet to access and update system information. Android users will require a compatible device, sufficient storage, and a stable internet connection for functions that require communication with the server and external services.

The deployment will also require supporting software and third-party services for specific system functions. Firebase is used to support real-time chat and messaging, while the Agora API supports virtual meetings. The SK 360° notification system uses custom Laravel database-based notifications, with notification updates retrieved through JavaScript polling and scheduled meeting notifications supported through Laravel Scheduler. An email service is used to deliver OTP verification codes required for public feedback and commenting. These services and system components communicate with the centralized backend through the internet.

The deployment environment will require proper configuration of the server, database, network connectivity, and application dependencies before system release. The Laravel backend and APIs must be configured to communicate securely with the MySQL database and external services. Required environment variables, database credentials, API keys, and service configurations must be properly established, while the deployed system will undergo functional, connectivity, security, and compatibility testing before implementation.

For the web environment, users will require a computer or laptop with a supported modern web browser and a stable internet connection. For the Android environment, users will require a compatible Android smartphone or tablet with sufficient storage and an internet connection. The application server must support the technologies and dependencies required by the Laravel-based web application, backend services, database, and APIs.

The deployment architecture will allow the web and Android mobile applications to access the same centralized backend and database. This provides synchronized information across platforms while maintaining centralized data management. Role-based access control will restrict administrative, reporting, financial, and other system functions to authorized users based on their assigned roles and permissions. Public users may access designated public information through the web-based Public View without creating an account, while public feedback and commenting through the web application will require email OTP verification and may be submitted anonymously or non-anonymously. The Android mobile application is restricted to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary accounts.

During implementation, the system will be configured and tested in a controlled deployment environment before being made available to its intended users. The development team will verify database connectivity, application functionality, user authentication, role-based access, document archiving and retrieval, notifications, communication services, and email OTP verification. Once the system passes the required testing and validation, the web application will be made accessible to authorized SK officials and public users according to their respective access permissions, while the Android mobile application will be distributed only to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary users.

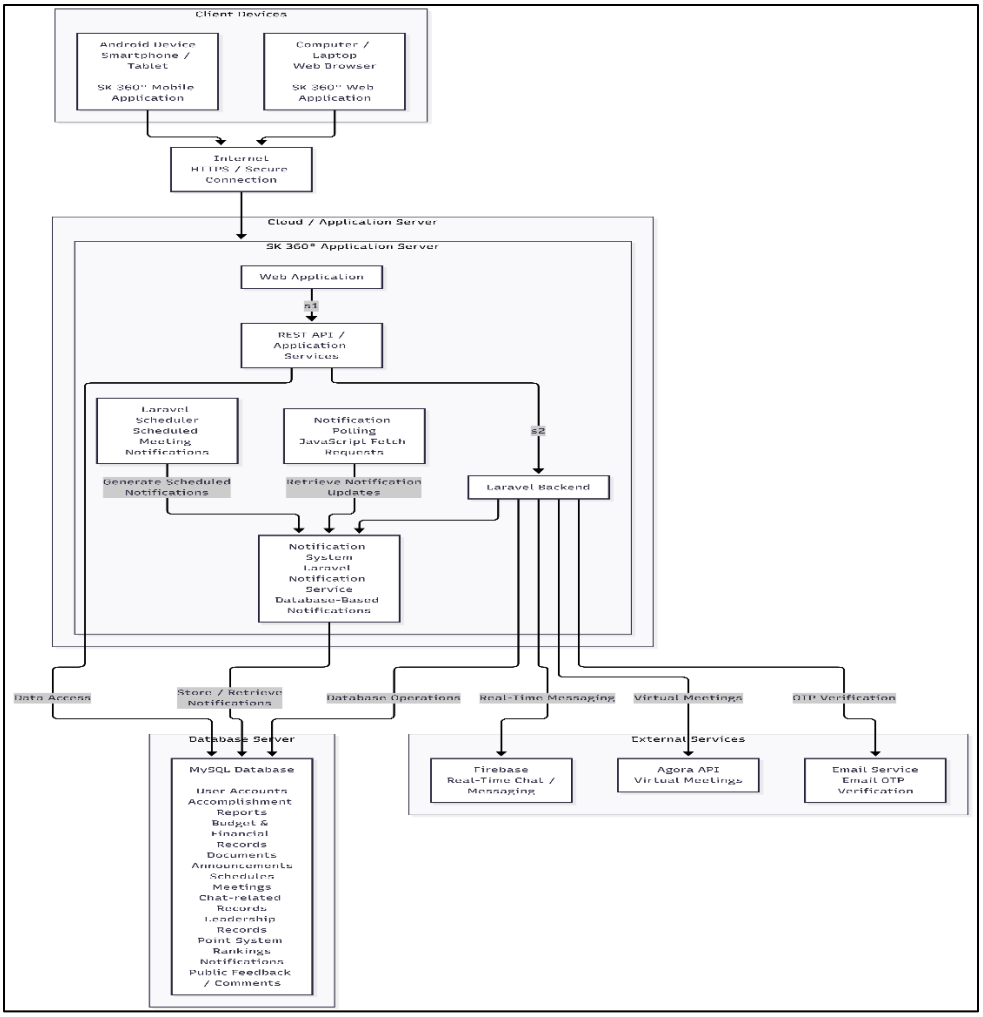


Figure 54 Deployment Diagram

Figure 54 illustrates the deployment architecture of the SK 360° system and shows how the developed software components are distributed across the different deployment targets. The system consists of the web application accessed through a computer or laptop, the Android mobile application accessed through compatible Android devices, the centralized application server, the MySQL database server, and the external services required by selected system functions. Both the web and Android applications require an internet connection to communicate with the centralized application server.

The deployment process begins by preparing the cloud-based application server and configuring the required environment for the SK 360° system. The Laravel-based web application, backend services, and APIs are deployed to the application server, while the MySQL database is configured to store and manage the system's centralized data. The

required database structure and application dependencies are then configured to allow the backend to communicate with the database securely.

After the server environment is configured, the SK 360° web application is made accessible through supported web browsers on computers or laptops. The Flutter-based mobile application is compiled and deployed to compatible Android devices. Both applications are configured to communicate with the centralized backend through the appropriate application programming interfaces and secure internet connections.

The application server is also configured to communicate with the required external services. Firebase is configured to support real-time chat and messaging, the Agora API is configured to support virtual meetings, and the email service is configured to deliver OTP verification codes for public feedback and commenting. System notifications are managed through the Laravel backend and database. These services allow the system to provide its required communication, notification, and verification functions.

The deployment also includes backup and recovery mechanisms to protect the centralized system data and support service continuity. Regular database backups and copies of essential system records are maintained to provide a recovery source in case of data loss, system failure, or other deployment-related issues. Access to backup resources is limited to authorized administrators to help maintain the security and integrity of the stored data. These measures ensure that important system information can be restored when necessary and support the reliability and continuous operation of the SK 360° system.

Once the components are deployed and configured, the system is tested to verify connectivity between the web application, Android application, application server, database, and external services. Authentication, role-based access control, report submission, document archiving and retrieval, notifications, communication, web-based public viewing, and email OTP verification are tested to ensure that the deployed system operates as intended. After successful testing and validation, the web application is made available to authorized SK officials and public users according to their respective access permissions, while the Android mobile application is limited to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary users.

Implementation Plan

The proposed implementation of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City will follow an “instant-on” approach, wherein the complete system will be deployed and made available for use after the development, testing, and necessary preparations have been completed. This approach will allow the SK Federation of Lipa City to access and utilize the web and Android mobile application functionalities of SK 360° upon implementation. Prior to deployment, the proponents will conduct system testing, user orientation, account and access preparation, data configuration, and other necessary implementation activities to ensure that the system is ready for operational use. The implementation activities will be coordinated among the proponents and the designated personnel of the SK Federation to facilitate the proper deployment and initial utilization of the system.

Points-of-Contact

This section contains the names of responsible personnel, roles, and contact information. These points of contact are the representatives with responsibilities relating to system implementation.

Table 42 Implementation Points-of-Contact

Role	Name(s)	Contact Details
CAPSTONE PROJECT TEAM		
Full Stack Developer /UI/UX Designer	Bautista, Marielle O.	bautistamo@students.nu-lipa.edu.ph
System Analyst/ Documentation Assistant	Janda, Giorgia Schen C.	jandagc@students.nu-lipa.edu.ph
Full Stack Developer/QA Tester/ Database Administrator	Monje, Paul Vincent M.	monjepm@students.nu-lipa.edu.ph
Project Manager/ Documenter / Technical Writer/ Front-End Developer	Sumadsad, Alexa E.	sumadsadae@students.nu-lipa.edu.ph
Capstone Adviser	Natividad, Reymark R.	rrnatividad@nu-lipa.edu.ph
CLIENT ORGANIZATION TEAM REPRESENTATIVES		
Client	Sangguniang Kabataan Federation of Lipa City	

The table presents the roles, responsibilities, and contact information of the capstone team and client representatives to ensure clear accountability and coordination throughout the implementation of SK 360°.

Major Implementation Tasks

This phase includes the major activities required for system implementation. It contains the tasks that must be completed from the beginning of the implementation process until the deployment of the system to its intended client and end-users.

Table 43 Lists of Major Implementation Tasks

Major Activities	Description	Responsible Person/Team
User's Training	Conduct orientation and training for the SK Federation representatives on the proper use of the web and Android mobile applications.	Capstone Project Team
User's Acceptance	Allow the client representatives to evaluate the system and confirm that the implemented functionalities meet the identified requirements.	SK Federation / Capstone Project Team
Go Live	Deploy SK 360° and make the web and Android mobile applications available for operational use.	Capstone Project Team
Support	Provide initial technical assistance, troubleshooting, and system support after deployment.	Capstone Project Team

Hardware, Software, Facilities, and Materials Requirement

The implementation of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City requires appropriate software, hardware, networking resources, facilities, and materials to support the deployment, testing, and operation of its web and Android mobile applications. For software requirements, the system requires a modern web browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox for accessing the web application. The development and deployment environment utilizes Visual Studio Code as the primary Integrated Development Environment (IDE), while the system's database operations require the configured database server and management tools used by the proponents. Additional software and supporting libraries required by the system should be installed and properly configured before implementation to ensure that all system functions operate as intended.

On the hardware side, the client requires computers or laptops capable of running the web application, accessing reports and archived records, and performing administrative tasks efficiently. A minimum of an Intel Core i3 processor or equivalent, 8 GB of RAM, and at least 256 GB of storage is recommended for administrative and management devices. For mobile users, compatible Android smartphones or tablets capable of running the Android mobile application are required. The devices should have sufficient storage, processing capability, and battery capacity to support regular system use. These specifications provide adequate resources for accessing system modules, uploading documents, viewing reports, receiving notifications, and performing other supported functions.

For networking requirements, a stable internet connection is necessary to allow users to access the centralized system and its online services. The client should provide reliable Wi-Fi or wired internet connectivity within the designated implementation area. Network access is required for activities such as accomplishment report submission, document uploading, announcements, notifications, communication, scheduling, monitoring, and access to other centralized system functions. Authorized SK officials using the mobile application may also use a reliable mobile data connection when accessing the system outside the organization's designated workspace.

In terms of facilities, the client should provide an appropriate workspace where system orientation, user training, acceptance activities, and initial implementation can be conducted. The facility should have sufficient space, electrical outlets, internet access, and suitable seating arrangements for the participating SK Federation representatives. A projector, display, or other presentation equipment may also be used during the training and orientation sessions when available.

For materials and implementation support, the proponents will provide the necessary system documentation, user guides, training materials, and other instructional resources required to assist the SK Federation representatives during implementation. These materials will support user familiarization, system acceptance, and the initial operation of SK 360°. The identified hardware, software,

networking resources, facilities, and materials will help ensure that the system can be implemented and utilized effectively by the client organization.

Knowledge Transfer and Training Plan

The knowledge transfer and training plan for SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City will be conducted to ensure that the designated representatives of the Sangguniang Kabataan (SK) Federation of Lipa City are properly equipped with the knowledge and skills required to use and operate the web and Android mobile applications. The training will be conducted by the Capstone Project Team during the implementation phase and will follow a structured approach consisting of system orientation, feature demonstration, guided practice, user evaluation, and technical support.

The Project Manager will be responsible for coordinating the overall training activities, communicating with the client representatives, preparing the training schedule, and ensuring that the necessary materials and resources are available. The Project Manager will also monitor the progress of the training and address concerns raised by the client during the knowledge transfer activities. The Back End Developer/UI/UX Designer will demonstrate the relevant system functions and explain the user interface and workflows to help users understand how to navigate and perform tasks within the system. The Full Stack Developer/QA Tester will assist in demonstrating system operations, provide hands-on technical guidance, and address technical concerns encountered during the training. The Database Administrator/System Analyst will explain relevant data management processes, system workflows, and administrative functions, particularly those involving records, reports, and information stored within the system. The Documenter/Technical Writer will prepare and organize the user guides, training materials, and other documentation that the client representatives can use as references after the training.

The training sessions will begin with an overview of SK 360°, including its purpose, major modules, and the roles of the intended users. The proponents will then demonstrate the proper use of the web and Android mobile applications, including functions related to accomplishment reporting, budget and financial

documentation, document archiving and retrieval, announcements, report consolidation, notifications, activity scheduling, communication, leadership information, Point System Ranking, dashboard-based analytics and monitoring, and web-based public information access. The client representatives will be guided through the appropriate workflows and procedures for each function relevant to their responsibilities. The mobile application functions demonstrated during training are limited to authenticated SK Federation President/Admin, SK Chairman, and SK Secretary users.

After the demonstrations, the client representatives will perform guided hands-on activities using the system. These activities will allow the users to practice common tasks such as submitting and reviewing reports, accessing archived records, viewing announcements, monitoring activities, and using other applicable system features. The Capstone Project Team will provide assistance during the hands-on sessions and clarify any questions or difficulties encountered by the users. Feedback gathered during the training will also be used to identify areas that may require additional explanation or clarification.

To support continuous knowledge transfer, the proponents will provide user documentation and training materials containing instructions on system navigation, major functionalities, and basic procedures. These materials will serve as references for the SK Federation representatives after the implementation and will help users independently perform their assigned tasks. Technical concerns that cannot be resolved through the user documentation will be addressed by the Capstone Project Team through the support process established during the implementation period.

The training will conclude with an evaluation and user acceptance activity in which the client representatives will assess their understanding and ability to use the system. Any identified concerns, usability issues, or areas requiring further clarification will be documented and addressed by the proponents when applicable. Through this knowledge transfer and training plan, the SK Federation representatives will be prepared to use SK 360° effectively and integrate the system into their existing operational processes.

4.5 SYSTEM TESTING AND EVALUATION

Unit Testing

Unit testing was conducted to verify that the individual components and functions of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City operate correctly and according to their intended specifications. The testing involved isolating individual system functions and examining their inputs, processes, and expected outputs to identify errors, defects, or unexpected behavior at an early stage of development. The proponents performed unit tests on the developed web and Android mobile application components using the testing and debugging tools available within the development environment. Test cases were prepared based on the expected behavior of each function, and the actual results were compared with the expected results. Identified errors were documented, corrected, and retested to ensure that the affected components performed as intended before being integrated with other system components.

Table 44 Lists of System Components for Unit Testing

Module No.	Module Name	Component	Description
[SYSTEM/APPLICATION] Web			
1	Login	User Login	Verifies user authentication using valid and invalid credentials and applies to the appropriate role-based access.
2	User Management	Account Management	Verifies that the SK Federation President/Admin can create and manage SK Chairman accounts according to the defined role-based access control.
		Leadership Term Management	Verifies that the SK Federation President/Admin can initiate a new SK term using a valid succeeding start year, reject a past start year, retain or assign the President for the succeeding administration, deactivate outgoing official access, and reappoint eligible SK officials while preserving previous service records.
3	Dashboard	Dashboard Summary Viewing	Verifies that authorized users can view applicable summaries of reports, announcements, notifications, system updates, and monitoring information through the centralized dashboard.
		Dashboard Analytics and Monitoring	Verifies that authorized users can view submission compliance, budget utilization, submission trends, applicable KPI indicators, charts, and other monitoring information through the Dashboard Module.
4	Home Page	Home Page Navigation	Verifies that authenticated SK officials can access the web Home Page and navigate to available system functions and information according to their assigned roles.
5	Accomplishment Reporting	Report Submission	Verifies that authorized users can upload completed accomplishment report files to the system.
		Report Review	Verifies that submitted accomplishment reports can be viewed and reviewed by authorized users.

6	Budget and Financial Documentation	Budget Submission	Verifies that an authorized SK Chairman or SK Secretary can enter and submit required budget and financial information under the applicable monthly, quarterly, semi-annual, or annual reporting category and that the system automatically generates the standardized budget report after successful validation.
		Financial Document Upload	Verifies that an authorized SK Chairman or SK Secretary can upload completed budget and financial documents under the applicable monthly, quarterly, semi-annual, or annual reporting category through the web application.
		Submitted Financial Record Viewing	Verifies that authorized SK officials can view and access permitted generated budget reports and uploaded financial documents organized under the applicable monthly, quarterly, semi-annual, or annual reporting categories.
7	Document Management and Archiving	Archive Record Availability	Verifies that applicable Accomplishment Reports, Budget Documents, and Event Records generated or stored through connected system modules are available under their corresponding Archive categories without requiring a separate generic archive upload.
		Archived Record Retrieval	Verifies that authorized users can locate, view, and retrieve available archived Accomplishment Reports, Budget Documents, and Event Records according to their access permissions.
		Archive Record Preservation	Verifies that applicable current and historical records produced or stored through connected system modules remain preserved and accessible under their correct Archive categories according to role-based access and record availability.
8	Announcements	Announcement Creation	Verifies that the SK Federation President/Admin can create and publish announcements using valid announcement information and applicable audience or visibility settings.
		Announcement Management	Verifies that the SK Federation President/Admin can view, update, and manage published announcements.
		Announcement Viewing	Verifies that authenticated SK officials can view published announcements available to their access level.
9	Module Management	Submission Slot Status Tracking	Verifies that submission slots correctly display the Upcoming, Open, Past Deadline, and Closed states based on their configured schedule and manual close status, including the acceptance of late submissions while a slot remains Past Deadline.
		Submission Slot Creation	Verifies that the SK Federation President/Admin can create submission slots for accomplishment and budget report submissions using valid title, description, opening schedule, deadline, report type, intended role, and other applicable submission requirements.
		Submission Slot Management	Verifies that the SK Federation President/Admin can view, update, manually close, and manage created submission slots and that SK Chairman and SK Secretary accounts are prevented from performing submission-slot management actions.
		LYDP Public Document Management	Verifies that the SK Federation President/Admin can publish or replace the current LYDP PDF through Module Management and that the published LYDP is available to public users through the web-based Annual Budget & LYDP section without requiring account registration or authentication.
10	Report Consolidation	Report Consolidation Monitoring and Quality Documentation	Verifies that the SK Federation President/Admin can monitor available and pending barangay accomplishment and budget submissions, check

			available quality documentation, and generate and download the applicable consolidated PDF through the Report Consolidation page.
		Barangay Submission Detail Viewing	Verifies that the SK Federation President/Admin can select a barangay and view its available accomplishment reports, budget records, submission status, and quality documentation through the Report Consolidation page.
11	Notifications	Notification Generation	Verifies that system notifications are generated for applicable activities such as announcements, report-related updates, submission reminders, deadlines, meetings, and other relevant system events.
		Notification Viewing	Verifies that authenticated SK officials can view received notifications and access the related system information when applicable.
12	Activity Calendar and Scheduling	Schedule Creation	Verifies that the SK Federation President/Admin can create calendar events using valid title, date, time, location, audience, and other applicable event settings.
		Schedule Management	Verifies that the SK Federation President/Admin can select an existing event, update valid event information or applicable audience settings, and save the changes successfully.
		Schedule Viewing	Verifies that authorized SK officials can view available calendar events and open a selected event to display its event details.
13	Chat	Message/Communication Function	Verifies that authorized SK officials can send, receive, and view messages through the web application's real-time communication function.
14	Meetings	Virtual Meeting Participation	Verifies that authorized SK officials can access and participate in an available virtual meeting through the web application and that valid virtual participation can be recorded for meeting attendance.
		Meeting Management	Verifies that the SK Federation President/Admin can create, schedule, update, and manage meetings through the web application while SK Chairman and SK Secretary accounts are restricted from meeting-management actions.
		Meeting Attendance and Conference Finalization	Verifies that valid virtual and face-to-face attendance can be recorded, that leaving a virtual meeting does not end the conference for all participants, and that the SK Federation President/Admin can end the meeting for everyone, finalize attendance, and apply the applicable meeting participation points without duplicate point application.
15	Leadership Information	Leadership Information Management	Verifies that an authorized SK Chairman can create and manage leadership records for the SK Secretary, Treasurer, and Council Members of the assigned barangay, including the creation or linking of the corresponding system account for the SK Secretary.
		Leadership Information Viewing	Verifies that authorized SK officials can view available current and historical leadership information according to their access permissions.
16	Point System Ranking	Point Assignment	Verifies that the system applies the configured positive points for qualifying documentation or category criteria, applies a -5 deduction for late required submissions, applies +5 for official meeting attendance and -5 for a missed official meeting, and prevents duplicate point application for the same qualifying event.
		Ranking Generation	Verifies that barangay rankings are generated and updated according to the recorded Point System totals and applicable ranking rules.
		Ranking Viewing	Verifies that authenticated SK officials can view the current Point System Ranking, point summaries,

			and related performance information and that the ranking is not available to public users.
17	Public View	Public Information Viewing	Verifies that public users can access designated announcements, calendar information, leadership information, annual budget information, and the published LYDP through the web application without requiring account registration or authentication, while restricted administrative functions and the Point System Ranking remain inaccessible.
		Comment and Feedback	Verifies that public users can submit anonymous or non-anonymous comments and feedback through the web application after successful email OTP verification.
		Feedback Management	Verifies that authorized system users can view and manage public comments and feedback according to their assigned access permissions.
18	Profile Settings	Profile and Account Settings Management	Verifies that authenticated SK officials can view and update permitted profile information, account settings, passwords, and preferences with appropriate validation.
[SYSTEM/APPLICATION] Mobile			
1	Mobile Login	Mobile User Login	Verifies that the SK Federation President/Admin, SK Chairman, and SK Secretary can log in to the Android mobile application using valid credentials and that invalid credentials are properly rejected.
2	Mobile Home Page	Mobile Home Page Display	Verifies that authenticated SK officials can access the mobile Home Page and view available announcements, activities, and quick-access features according to their assigned role.
3	Mobile Accomplishment Reporting	Mobile Report Submission	Verifies that an authorized SK Chairman or SK Secretary can upload completed accomplishment report files through the mobile application.
4	Mobile Budget Documents	Mobile Financial Document Upload	Verifies that an authorized SK Chairman or SK Secretary can upload completed budget and financial documents through the mobile application.
		Mobile Financial Record Viewing	Verifies that an authorized SK Chairman or SK Secretary can view available submitted budget and financial records within the permitted barangay scope through the mobile application.
5	Mobile Consolidation	Mobile Consolidation and Quality Documentation Monitoring	Verifies that authorized SK officials can view total, submitted, and pending barangay submission counts, apply available year and reporting-period filters, search for barangay records, and check available quality documentation through the mobile application without generating a consolidated report.
6	Mobile Module Management	Mobile Submission Slot Creation	Verifies that the SK Federation President/Admin can create submission slots for accomplishment and budget report submissions using valid scheduling, report-type, intended-role, and other applicable submission information through the mobile application.
		Mobile Submission Slot Management	Verifies that the SK Federation President/Admin can view, search, filter, update, manually close, and manage submission slots through the mobile application, including the applicable Upcoming, Open, Past Deadline, and Closed states, while SK Chairman and SK Secretary accounts are prevented from performing submission-slot management actions.
7	Mobile View Reports	Mobile Submitted Report Viewing	Verifies that the SK Federation President/Admin can view, search, filter, and open available submitted accomplishment and budget report records through the mobile application and that SK Chairman and SK Secretary accounts cannot access Mobile View Reports.

8	Mobile Announcement	Mobile Announcement Creation	Verifies that the SK Federation President/Admin can create and publish announcements using valid announcement information and applicable audience or visibility settings through the mobile application.
		Mobile Announcement Management	Verifies that the SK Federation President/Admin can view, update, and manage published announcements through the mobile application and that unauthorized roles cannot perform announcement-management actions.
		Mobile Announcement Viewing	Verifies that authenticated SK officials can view published announcements available to their access level through the mobile application.
9	Mobile Leadership Profiles	Mobile Leadership Information Viewing	Verifies that authenticated SK officials can view available current and historical leadership information through the mobile application according to their access permissions.
10	Mobile Chat	Mobile Chat/Messaging	Verifies that authenticated SK officials can send, receive, and view messages with authorized users through the mobile application's real-time messaging function.
11	Mobile Meetings	Mobile Meeting Participation	Verifies that authenticated SK officials can access and participate in available virtual meetings through the mobile application and that valid virtual participation can be recorded for meeting attendance.
		Mobile Meeting Management	Verifies that the SK Federation President/Admin can create, schedule, update, and manage meetings through the mobile application while SK Chairman and SK Secretary accounts are limited to permitted meeting-participation functions.
12	Mobile Activity Calendar	Mobile Event Creation	Verifies that the SK Federation President/Admin can create calendar events using valid title, date, time, location, audience, and other applicable event settings through the mobile application.
		Mobile Event Management	Verifies that the SK Federation President/Admin can select an existing calendar event, update valid event information or applicable audience settings, and save the changes through the mobile application.
		Mobile Schedule Viewing	Verifies that authenticated SK officials can view available events, meetings, submission schedules, and deadlines and open an available calendar event to view its details through the mobile application.
13	Mobile Rankings	Mobile Ranking Viewing	Verifies that authenticated SK officials can view the current Point System Ranking, point summaries, and applicable performance information through the mobile application.
14	Mobile Profile	Mobile Profile Management	Verifies that authenticated SK officials can view and update permitted profile information, account settings, passwords, and available preferences through the mobile application with appropriate validation.
15	Mobile Notification	Mobile Notification Viewing	Verifies that authenticated SK officials can receive and view notifications for announcements, report-related updates, submission reminders, deadlines, meetings, and other relevant system events through the mobile application.

Table 45 Summary Results of Unit Testing

Module No.	Module Name	Component	Overall Remarks
[SYSTEM/APPLICATION] Web			
1	Login	User Login	PASSED
2	User Management	Account Management	PASSED
		Leadership Term Management	PASSED
3	Dashboard	Dashboard Summary Viewing	PASSED
		Dashboard Analytics and Monitoring	PASSED
4	Home Page	Home Page Navigation	PASSED
5	Accomplishment Reporting	Report Submission	PASSED
		Report Review	PASSED
6	Budget and Financial Documentation	Budget Submission	PASSED
		Financial Document Upload	PASSED
		Submitted Financial Record Viewing	PASSED
7	Document Management and Archiving	Archive Record Availability	PASSED
		Archived Record Retrieval	PASSED
		Archive Record Preservation	PASSED
8	Announcements	Announcement Creation	PASSED
		Announcement Management	PASSED
		Announcement Viewing	PASSED
9	Module Management	Submission Slot Status Tracking	PASSED
		Submission Slot Creation	PASSED
		Submission Slot Management	PASSED
		LYDP Public Document Management	PASSED
10	Report Consolidation	Report Consolidation Monitoring and Quality Documentation	PASSED
		Barangay Submission Detail Viewing	PASSED
11	Notifications	Notification Generation	PASSED
		Notification Viewing	PASSED
12	Activity Calendar and Scheduling	Schedule Creation	PASSED
		Schedule Management	PASSED
		Schedule Viewing	PASSED
13	Chat	Message/Communication Function	PASSED
14	Meetings	Virtual Meeting Participation	PASSED
		Meeting Management	PASSED
		Meeting Attendance and Conference Finalization	PASSED
15	Leadership Information	Leadership Information Management	PASSED
		Leadership Information Viewing	PASSED
16	Point System Ranking	Point Assignment	PASSED
		Ranking Generation	PASSED
		Ranking Viewing	PASSED

17	Public View	Public Information Viewing	PASSED
		Comment and Feedback	PASSED
		Feedback Management	PASSED
18	Profile Settings	Profile and Account Settings Management	PASSED
[SYSTEM/APPLICATION] Mobile			
1	Mobile Login	Mobile User Login	PASSED
2	Mobile Home Page	Mobile Home Page Display	PASSED
3	Mobile Accomplishment Reporting	Mobile Report Submission	PASSED
4	Mobile Budget Documents	Mobile Financial Document Upload	PASSED
		Mobile Financial Record Viewing	PASSED
5	Mobile Consolidation	Mobile Consolidation and Quality Documentation Monitoring	PASSED
6	Mobile Module Management	Mobile Submission Slot Creation	PASSED
		Mobile Submission Slot Management	PASSED
7	Mobile View Reports	Mobile Submitted Report Viewing	PASSED
8	Mobile Announcement	Mobile Announcement Creation	PASSED
		Mobile Announcement Management	PASSED
		Mobile Announcement Viewing	PASSED
9	Mobile Leadership Profiles	Mobile Leadership Information Viewing	PASSED
10	Mobile Chat	Mobile Chat/Messaging	PASSED
11	Mobile Meetings	Mobile Meeting Participation	PASSED
		Mobile Meeting Management	PASSED
12	Mobile Activity Calendar	Mobile Event Creation	PASSED
		Mobile Event Management	PASSED
		Mobile Schedule Viewing	PASSED
13	Mobile Rankings	Mobile Ranking Viewing	PASSED
14	Mobile Profile	Mobile Profile Management	PASSED
15	Mobile Notification	Mobile Notification Viewing	PASSED

As shown in Table 45, five (5) IT professionals conducted unit testing on the individual functions and system components of the SK 360° web and mobile applications. The testing verified authentication, user management, report submission, budget documentation, notifications, scheduling, communication, report consolidation, public information access, and profile management. Functions with errors or unexpected behavior were revised and retested. The final results showed that the evaluated functions operated according to their expected results, with no failed test cases recorded. Detailed procedures, test cases, expected and actual results, statuses, issues, and corrective actions are presented in Appendix G.

Functionality Testing

Functionality testing was conducted to verify that the web and Android mobile applications of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City operated according to the defined functional requirements and intended user roles. The testing used a scenario-based manual testing method in which each system function was evaluated using prepared test cases, test data, expected results, validation rules, and role-based access requirements. The web application was tested through its supported web environment, while the Android mobile application was tested through compatible Android devices. Each test case examined whether the system accepted valid inputs, properly handled invalid or incomplete inputs, performed the required processes, restricted unauthorized actions, and produced the expected output. The actual results of each test were compared with the expected results and recorded together with the corresponding remark or status to determine whether each function operated as intended.

Table 46 Summary Result of Web Application Functionality Testing

Module No.	Module Name	Test Code	Function Name	Remark/Status
1	Login	001	User Login	PASSED
2	User Management	002	Account Management	PASSED
		003	Leadership Term Management	PASSED
3	Dashboard	004	Dashboard Summary Viewing	PASSED
		005	Dashboard Analytics and Monitoring	PASSED
4	Home Page	006	Home Page Navigation	PASSED
5	Accomplishment Reporting	007	Report Submission	PASSED
		008	Report Review	PASSED
6	Budget and Financial Documentation	009	Budget Submission	PASSED
		010	Financial Document Upload	PASSED
		011	Submitted Financial Record Viewing	PASSED
7	Document Management and Archiving	012	Archive Record Availability	PASSED
		013	Archived Record Retrieval	PASSED
		014	Archive Record Preservation	PASSED
8	Announcements	015	Announcement Creation	PASSED
		016	Announcement Management	PASSED
		017	Announcement Viewing	PASSED
9	Module Management	018	Submission Slot Status Tracking	PASSED
		019	Submission Slot Creation	PASSED
		020	Submission Slot Management	PASSED
		021	LYDP Public Document Management	PASSED

10	Report Consolidation	022	Report Consolidation Monitoring and Quality Documentation	PASSED
		023	Barangay Submission Detail Viewing	PASSED
11	Notification	024	Notification Generation	PASSED
		025	Notification Viewing	PASSED
12	Activity Calendar and Scheduling	026	Schedule Creation	PASSED
		027	Schedule Management	PASSED
		028	Schedule Viewing	PASSED
13	Chat	029	Message/Communication Function	PASSED
14	Meetings	030	Virtual Meeting Participation	PASSED
		031	Meeting Management	PASSED
		032	Meeting Attendance and Conference Finalization	PASSED
15	Leadership Information	033	Leadership Information Management	PASSED
		034	Leadership Information Viewing	PASSED
16	Point System Ranking	035	Point Assignment	PASSED
		036	Ranking Generation	PASSED
		037	Ranking Viewing	PASSED
17	Public View	038	Public Information Viewing	PASSED
		039	Comment and Feedback	PASSED
		040	Feedback Management	PASSED
18	Profile Settings	041	Profile and Account Settings Management	PASSED

As shown in Table 46, the web application testing was conducted by five (5) IT professionals who evaluated the system's functional and non-functional requirements. The testing results showed that the evaluated modules operated according to their intended functions and documented requirements. The testing covered authentication, user management, dashboard access, report submission, budget and financial documentation, report approval, report consolidation, announcements, notifications, scheduling, communication, leadership information, Point System Ranking, public information access, and profile management. These modules support the administrative, reporting, and coordination activities of the SK Federation of Lipa City and participating barangay SK councils. Minor issues identified during the initial testing were corrected and subjected to retesting. The retesting process helped verify whether the corrections addressed the previously identified issues. No critical defects remained unresolved, and the final results indicated that the web application successfully performed its required functions. These findings provide supporting evidence that the application met the evaluated requirements under the conditions used during testing. See the full details of the report in Appendix H – Detailed Functional Test Report section of the manuscript.

Table 47 Summary Result of Mobile Application Functionality Testing

Module No.	Module Name	Test Code	Function Name	Remark/Status
1	Mobile Login	042	Mobile User Login	PASSED
2	Mobile Home Page	043	Mobile Home Page Display	PASSED
3	Mobile Accomplishment Reporting	044	Mobile Report Submission	PASSED
4	Mobile Budget Documents	045	Mobile Financial Document Upload	PASSED
		046	Mobile Financial Record Viewing	PASSED
5	Mobile Consolidation	047	Mobile Consolidation and Quality Documentation Monitoring	PASSED
6	Mobile Module Management	048	Mobile Submission Slot Creation	PASSED
		049	Mobile Submission Slot Management	PASSED
7	Mobile View Reports	050	Mobile Submitted Report Viewing	PASSED
8	Mobile Announcement	051	Mobile Announcement Creation	PASSED
		052	Mobile Announcement Management	PASSED
		053	Mobile Announcement Viewing	PASSED
9	Mobile Leadership Profiles	054	Mobile Leadership Information Viewing	PASSED
10	Mobile Chat	055	Mobile Chat/Messaging	PASSED
11	Mobile Meetings	056	Mobile Meeting Participation	PASSED
		057	Mobile Meeting Management	PASSED
12	Mobile Activity Calendar	058	Mobile Event Creation	PASSED
		059	Mobile Event Management	PASSED
		060	Mobile Schedule Viewing	PASSED
13	Mobile Rankings	061	Mobile Ranking Viewing	PASSED
14	Mobile Profile	062	Mobile Profile Management	PASSED
15	Mobile Notification	063	Mobile Notification Viewing	PASSED

As shown in Table 47, the mobile application testing results showed that the Flutter-based Android application operated according to its intended functions and documented requirements. The testing covered mobile login, home page, accomplishment reporting, budget document submission, report monitoring, module management, report viewing, announcements, leadership profiles, chat, meetings, activity calendar, rankings, profile settings, and notifications. Minor issues identified during the initial testing were corrected and retested. No critical defects remained unresolved, and the final results indicated that the mobile application successfully performed its required functions. See the full details of the report in Appendix H – Detailed Functional Test Report section of the manuscript.

Integration Testing

Integration testing was conducted to verify that the interconnected modules of the SK 360° web and Android mobile applications operate correctly as a unified system. The testing focused on validating the flow of data, compatibility of interfaces, and proper interaction among related modules and system functions. Scenario-based integration testing was performed by executing connected workflows and observing whether information created or processed in one module was correctly transmitted, stored, retrieved, and reflected in other applicable modules. The purpose of the testing was to identify errors that may occur when independently developed functions interact with one another and to ensure that the integrated components operate consistently according to the defined system requirements.

Table 48 Summary Result of Web Application Integration Testing

Module No.	Module Name	Overall Remarks
WEB-01	Login	PASSED
WEB-02	User Management	PASSED
WEB-03	Dashboard	PASSED
WEB-04	Home Page	PASSED
WEB-05	Accomplishment Reporting	PASSED
WEB-06	Budget and Financial Documentation	PASSED
WEB-07	Document Management and Archiving	PASSED
WEB-08	Announcements	PASSED
WEB-09	Module Management	PASSED
WEB-10	Report Consolidation	PASSED
WEB-11	Notifications	PASSED
WEB-12	Activity Calendar and Scheduling	PASSED
WEB-13	Chat	PASSED
WEB-14	Meetings	PASSED
WEB-15	Leadership Information	PASSED
WEB-16	Point System Ranking	PASSED
WEB-17	Public View	PASSED
WEB-18	Profile Settings	PASSED

As shown in Table 48, the web application successfully passed the integration testing across all eighteen (18) evaluated modules. The testing verified the interaction and data flow among authentication, user management, dashboard, reporting, budget

documentation, document archiving, announcements, module management, report consolidation, notifications, scheduling, communication, meetings, leadership information, Point System Ranking, public view, and profile settings. No critical integration defects were identified, and all modules were marked as PASSED after verifying their connected workflows and system interactions. These results indicate that the web application operates consistently as an integrated system. See the full details of the report in Appendix I – Detailed Integration Test Report section of the manuscript.

Table 49 Summary Result of Mobile Application Integration Testing

Module No.	Module Name	Overall Remarks
MOB-01	Mobile Login	PASSED
MOB-02	Mobile Home Page	PASSED
MOB-03	Mobile Accomplishment Reporting	PASSED
MOB-04	Mobile Budget Documents	PASSED
MOB-05	Mobile Consolidation	PASSED
MOB-06	Mobile Module Management	PASSED
MOB-07	Mobile View Reports	PASSED
MOB-08	Mobile Announcement	PASSED
MOB-09	Mobile Leadership Profiles	PASSED
MOB-10	Mobile Chat	PASSED
MOB-11	Mobile Meetings	PASSED
MOB-12	Mobile Activity Calendar	PASSED
MOB-13	Mobile Rankings	PASSED
MOB-14	Mobile Profile	PASSED
MOB-15	Mobile Notification	PASSED

As shown in Table 49, the Flutter-based Android mobile application successfully passed the integration testing across all fifteen (15) evaluated modules. The testing verified the interaction and data flow among mobile login, home page, accomplishment reporting, budget documents, consolidation monitoring, module management, report viewing, announcements, leadership profiles, chat, meetings, activity calendar, rankings, profile settings, and notifications. The results showed that the mobile modules properly communicated with the centralized backend and operated according to their expected workflows. No critical integration defects were identified, and all modules were marked as PASSED. See the full details of the report in Appendix I – Detailed Mobile Integration Test Report section of the manuscript.

Systems Performance Assessment

Web Application Performance Report

The Web Application Performance Assessment was conducted to evaluate the speed, responsiveness, and overall performance of the SK 360° web application under normal operating conditions. The assessment consisted of five execution cycles. During each cycle, all major web modules were evaluated using Google Lighthouse under consistent testing conditions. Google Lighthouse measured the Performance Score, Speed Index, First Contentful Paint (FCP), Largest Contentful Paint (LCP), Total Blocking Time (TBT), and Cumulative Layout Shift (CLS). The results from the five execution cycles were recorded and averaged for each module to determine the overall performance of the web application and identify areas that may require further optimization.

Table 50 Summary Result of Web Application Performance

Modules	OVERALL PERFORMANCE					
	Performance Score	**Speed Index** (sec)	**FCP** (sec)	**LCP** (sec)	**TBT** (ms)	**CLS**
Login	100	0.6s	0.6s	0.6	0ms	0
User Management	98.8	0.82s	0.82s	0.88s	0ms	0
Dashboard	98.8	0.84s	0.82s	0.82s	8ms	0
Home Page	99	0.72s	0.72s	0.74s	0ms	0.001
Accomplishment Reporting	98.9	0.72s	0.72s	0.76s	0ms	0
Budget and Financial Documentation	98.4	0.8s	0.78s	0.72s	0ms	0
Document Management and Archiving	99.2	0.7s	0.7s	0.76s	2ms	0
Announcement	99	0.72s	0.72s	0.8s	0ms	0
Module Management	98.8	0.76s	0.76s	0.84s	2ms	0.001
Report Consolidation	99.6	0.72s	0.7s	0.72s	0ms	0.001
Notification	88.6	1.5s	1.44s	1.6s	0ms	0
Activity Calendar and Scheduling	86.2	0.92	0.9s	0.98s	26ms	0.2316
Chat	95.6	0.9s	0.82s	1.32	18ms	0.059
Meetings	98	0.9s	0.9s	0.9s	0ms	0.001
Leadership Information	98.8	0.78s	0.78s	0.82s	2ms	0

Point System Ranking	97.8	0.84s	0.84s	0.96s	0ms	0.003
Public View	98.6	0.68s	0.68s	0.94	0ms	0.003
Profile Settings	98.8	0.74s	0.74s	0.82s	0ms	0
Overall Mean	**97.38**	**0.8144**	**0.8022**	**0.8878**	**3.22**	**0.0167**

Table 50 presents the summary results of the web application performance testing of SK 360° across eighteen (18) modules. The application obtained an average performance score of 97.38 out of 100, with mean values of 0.8144 seconds for Speed Index, 0.8022 seconds for First Contentful Paint (FCP), and 0.8878 seconds for Largest Contentful Paint (LCP). These results indicate that page content generally appeared quickly under the tested conditions. The average Total Blocking Time (TBT) of 3.22 milliseconds also indicates minimal blocking of the main thread, while the average Cumulative Layout Shift (CLS) of 0.0167 reflects generally stable page layouts. However, the Activity Calendar and Scheduling module recorded the lowest performance score of 86.2 and the highest CLS of 0.2316, identifying visual stability as an area for improvement. The Notification module obtained the second-lowest performance score of 88.6 and recorded the longest Speed Index, FCP, and LCP values among the evaluated modules. Although these findings identify areas requiring optimization, the reported measurements alone do not establish the presence or absence of critical defects. Overall, the testing effort demonstrated favorable web application performance under the tested conditions and identified specific modules for further refinement. See full details of the report in Appendix J – System Performance Assessment Report section of the manuscript.

Mobile Application Performance Report

The Mobile Application Performance Assessment was conducted to evaluate the speed, responsiveness, and overall performance of the SK 360° Android mobile application during normal use. The testing involved performing representative mobile functions through multiple execution runs and measuring key performance indicators such as application load time, response time, and screen rendering time. Since the SK 360° Android mobile application is developed using Flutter, Flutter DevTools was used to monitor and assess the application's

performance. The recorded results were averaged to determine the overall performance of the mobile application and to identify functions that may require further optimization.

Table 51 Summary Result of Mobile Application Performance

Test Cycle(s)	Mobile Performance Criteria			REMARKS
	Load Time (seconds)	Response Time (seconds)	Screen Rendering Time (seconds)	
1	1.18	1.67	1.22	PASSED
2	0.83	1.536	1.26	PASSED
3	0.75	1.477	1.28	PASSED
4	0.75	1.489	1.19	PASSED
5	0.73	1.501	1.26	PASSED
			Overall Mean	**1.21s**

Table 51 presents the performance testing results of the SK 360° Android mobile application across five (5) test cycles, covering application load time, response time, and screen rendering time. The application recorded an average load time of 0.848 seconds, an average response time of 1.5346 seconds, and an average screen rendering time of 1.242 seconds, resulting in an overall mean of 1.21 seconds across the three measured criteria. Load time decreased from 1.18 seconds in the first cycle to 0.73 seconds in the fifth cycle, while response time ranged from 1.477 to 1.67 seconds. Screen rendering time remained relatively consistent, ranging from 1.19 to 1.28 seconds. All five test cycles were marked as Passed, indicating a successful outcome based on the recorded assessment. No critical performance defects were reported in the summary, although the timing measurements alone do not establish the absence of defects beyond the tested scenarios. Among the evaluated criteria, response time had the highest average, making it a potential focus for further optimization. Additional testing across different Android devices, network conditions, and data volumes would help determine whether these results remain consistent in broader use. Overall, the results demonstrate responsive application behavior under the tested conditions and provide a baseline for subsequent performance assessments. See full details of the report in Appendix J – System Performance Assessment Report section of the manuscript.

Systems Accessibility Assessment

Web Application Accessibility Report

The web application's accessibility was assessed using the Web Accessibility Test Document and Chrome DevTools Lighthouse. The assessment recorded accessibility scores, detected errors, passed audits, non-applicable audits, and remarks for eighteen (18) modules across five recorded execution runs per module. The scores were averaged to summarize the application's performance in automated accessibility checks and identify areas requiring improvement.

Table 52 Summary Result of Web Application Accessibility Compliance

User Interfaces (Main Modules)	Web Accessibility Compliance Score
Login	98
User Management	93
Dashboard	93
Home Page	93
Accomplishment Reporting	91
Budget and Financial Documentation	85
Document Management and Archiving	94
Announcement	89
Module Management	93
Report Consolidation	90
Notification	95
Activity Calendar and Scheduling	96
Chat	82
Meetings	90
Leadership Information	88

Point System Ranking	85
Public View	100
Profile Settings	93
Overall Mean	**91.56**

Table 52 presents the web accessibility assessment results of SK 360° across eighteen (18) modules, with an overall mean score of 91.56 out of 100. Public View obtained the highest score of 100, followed by Login at 98 and Activity Calendar and Scheduling at 96. These modules demonstrated strong results in the automated accessibility checks, with Public View recording no failed audits in the documented assessment. Chat recorded the lowest score of 82, followed by Budget and Financial Documentation and Point System Ranking at 85 each, identifying these interfaces as priorities for further improvement. User Management obtained a score of 93, matching Dashboard, Home Page, Module Management, and Profile Settings. Each module recorded identical scores across its five listed execution runs, indicating consistent recorded outcomes under the tested conditions. The documented findings included missing or insufficient control labels, improper ARIA attributes, focusable elements within hidden content, insufficient color contrast, and issues with heading structure and page landmarks. These concerns may affect text readability, keyboard navigation, and access through assistive technologies. The report did not assign severity classifications to the findings; therefore, it does not establish whether any identified issue constitutes a critical defect. Overall, the testing effort provided a useful assessment of automated accessibility performance and identified specific barriers requiring correction and retesting. The overall score should not be interpreted as proof of complete accessibility compliance, as manual accessibility evaluation remains necessary. Further evaluation using keyboard navigation and screen readers would help determine how the identified concerns affect actual user tasks. See full details of the report in Appendix K – System Accessibility Assessment Report section of the manuscript.

Systems Supportability and Compatibility Assessment

Web Application Supportability and Compatibility Report

The Web Application Supportability and Compatibility Assessment was conducted to determine whether SK 360° operated properly across the selected web browsers within the intended deployment environment. Manual testing was performed by loading the web application, navigating its major modules, and checking page rendering, user interactions, and system functions for browser-related inconsistencies. The browser versions were identified through the version information displayed in each browser's About or settings page, while the Safari testing environment included its corresponding iOS version. These version details were recorded to identify the specific browser environments covered by the assessment. Browser developer tools were used when necessary to inspect rendering and compatibility issues. The tested browsers, their recorded versions, and compatibility results are summarized in Table 53.

Table 53 Summary Result of Web Application Browser Compatibility

Web Browser	Version	Compatibility
Edge	155.0.4283.45	Compatible
Firefox	157.0.1	Compatible
Safari	26.2 (iOS 26.2)	Compatible
Opera	136.0.6008.22	Compatible
Chrome	154.0.8037.98 (64-bit)	Compatible
Brave	1.97.56 (Chromium 155.0.8059.40)	Compatible

Table 53 presents the results of the Web Application Supportability and Compatibility Assessment of SK 360°. All six evaluated browsers—Microsoft Edge, Mozilla Firefox, Safari, Opera, Google Chrome, and Brave—were recorded as Compatible, resulting in a 100% compatibility rate among the tested browser environments. Manual testing indicated that the evaluated pages and functions operated properly, with no significant browser-related issues reported in interface rendering, navigation, and user interactions. Safari was evaluated on iOS 26.2, providing evidence of compatibility within the tested mobile browser environment, while the other recorded results covered the selected desktop browsers. No critical compatibility defects were reported in the assessment summary. Overall, the testing effort supported the application's compatibility with the browser versions listed in Table 53. These findings apply to the specific versions, devices, and functions tested; compatibility with other environments or future browser updates requires further verification.

Mobile Application Supportability and Compatibility Report

The Mobile Application Supportability and Compatibility Assessment was conducted to determine whether the SK 360° Android application operated properly across the Android versions included in the assessment. The assessment aimed to examine whether differences in operating system environments affected the application's installation, accessibility, interface presentation, and execution of its intended functions. Manual testing was performed by installing and launching the application, navigating its screens, and checking the behavior of its major functions within each tested environment. The assessment covered screen display, user interactions, authentication, document handling, notifications, communication, and permission handling. These checks focused on whether users could access the available features, move between screens, and complete the selected tasks without compatibility-related interruptions. Particular attention was given to interface elements and permission requests that could affect access to application features across different Android versions. The observed behavior was compared with the expected results to identify installation failures, display inconsistencies, crashes, or functions affected by differences in operating system

versions. This process provided a basis for assessing the application's suitability for the evaluated environments and identifying areas that might require adjustments to maintain consistent operation. The scope of the assessment was limited to the devices, operating system versions, and usage scenarios included in the manual testing. The Android versions, corresponding SDK/API levels, codenames, and compatibility results are summarized in Table 54.

Table 54 Summary Result of Mobile Application OS Compatibility

Version	SDK/API Level	Codename	Compatibility
Android 16	Level 36	Baklava	**Compatible**
Android 15	Level 35	Vanilla Ice Cream	**Compatible**
Android 14	Level 34	Upside Down Cake	**Compatible**
Android 13	Level 33	Tiramisu	**Compatible**
Android 12	Level 32	Snow Cone	**Compatible**
	Level 31		**Compatible**
Android 11	Level 30	Red Velvet Cake	**Compatible**

Table 54 presents the results of the manual compatibility assessment of the SK 360° Android application across Android 11, Android 12, Android 13, Android 14, Android 15, and Android 16. All listed operating system versions were marked as Compatible, indicating that the application met the compatibility checks performed within the tested environments. These results support the application's suitability for the evaluated Android versions and suggest that operating system differences did not prevent completion of the tested tasks. No critical compatibility defects were reported in the summary, although the findings are limited to the devices, configurations, and scenarios included in the assessment. Manual testing provided a practical means of observing application behavior and identifying issues that could affect users during installation, navigation, and interaction with system functions. Overall, the testing effort yielded favorable compatibility results across the listed operating systems. These findings provide a basis for defining the application's supported Android environments, while additional testing remains useful when new operating system versions, device configurations, or application updates are introduced.

Systems Security Compliances

Web Application Security Compliances

Pre-vulnerability Assessment Report

The Web Application Pre-Vulnerability Assessment was conducted to evaluate the existing security configurations of SK 360° and identify potential vulnerabilities and security weaknesses that could affect the protection of system information. The proponents utilized SecurityHeaders.com, an online security assessment tool that examines HTTP response headers to determine the implementation of recommended web security protections. The assessment involved scanning the deployed web application at https://sk360lipacity.org and reviewing the generated security report, including the overall security rating, configured security headers, and missing security headers. The findings were examined and classified according to their potential severity and impact on the web application. The identified configuration weaknesses provided a baseline for evaluating the effectiveness of subsequent corrective security measures, with the results summarized in Table 55.

Table 55 Summary Result of Web Application Pre-vulnerability Assessment

Web Application Risks	Severity	Count	Description
Missing Strict-Transport-Security	Medium	1	The website lacked an HSTS header to instruct browsers to enforce HTTPS for subsequent connections.
Technology Information Disclosure	Low	1	The X-Powered-By header disclosed the PHP version used by the server.
Total		**2**	**Two security configuration findings were observed.**



Figure 55 Security Headers Compliance Pre-Assessment Report

The pre-vulnerability assessment of the SK 360° Web Application (https://sk360lipacity.org) achieved an A Grade Rating from SecurityHeaders.com, indicating that five of the six evaluated HTTP security response headers were successfully implemented. These included X-Content-Type-Options, X-Frame-Options, Referrer-Policy, Permissions-Policy, and Content-Security-Policy, which provide browser-level protections against security threats such as clickjacking, MIME-sniffing, and certain insecure content handling practices. However, the assessment identified two security configuration findings: the absence of the Strict-Transport-Security header, classified as medium severity, and the disclosure of the server's PHP version through the X-Powered-By header, classified as low severity. No critical or high-severity findings were identified within the scope of the assessment. The results indicated that the web application had implemented several recommended security measures but required further configuration improvements to strengthen its security header compliance. The detailed assessment results and raw HTTP response headers are provided in the Systems Security Compliance Report appendix.

Post Vulnerability Assessment Report

The Web Application Post-Vulnerability Assessment was conducted to evaluate the effectiveness of the security improvements implemented in SK 360° following the pre-vulnerability assessment. The proponents utilized SecurityHeaders.com to reassess the deployed web application at https://sk360lipacity.org by examining its HTTP security response headers and reviewing the generated security report. The assessment focused on verifying the implementation of the previously missing Strict-Transport-Security header, identifying any remaining security configuration weaknesses, and comparing the results with the pre-assessment findings. The reassessment also aimed to determine whether the implemented security configurations strengthened browser-level protection against potential web-based security threats. The results were documented to determine the effectiveness of the corrective measures and the overall improvement in security header compliance, as summarized in Table 56.

Table 56 Summary Result of Web Application Post Vulnerability Assessment

Web Application Risks	Severity	Count	Description
Missing Strict-Transport-Security	Resolved	0	The Strict-Transport-Security header was successfully implemented, enabling browsers to enforce HTTPS connections for subsequent requests.
Technology Information Disclosure	Low	1	The X-Powered-By header continued to disclose the server's PHP version.
Total Remaining Findings		**1**	**One low-severity security configuration finding remained after the assessment.**

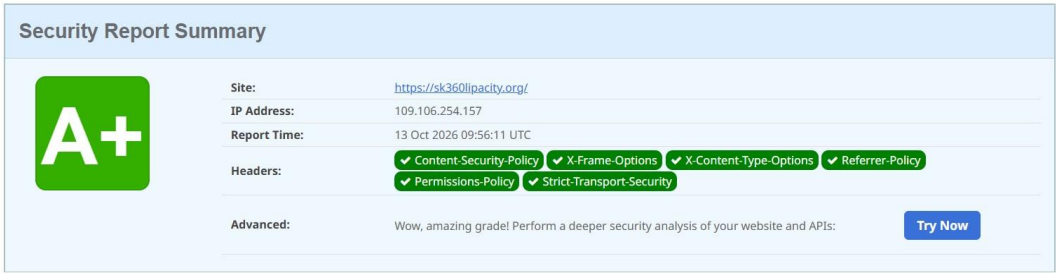


Figure 56 Security Headers Compliance Post Assessment Report

The post-vulnerability assessment of the SK 360° Web Application (https://sk360lipacity.org) achieved the highest A+ Grade Rating from SecurityHeaders.com, demonstrating an improvement from the A Grade Rating obtained during the selected pre-assessment. The results confirmed that all six evaluated HTTP security response headers, namely X-Content-Type-Options, X-Frame-Options, Referrer-Policy, Permissions-Policy, Content-Security-Policy, and Strict-Transport-Security, were successfully implemented. These headers provide browser-level security protections that help mitigate risks associated with clickjacking, MIME-sniffing, insecure connections, and certain content-related vulnerabilities. The previously identified missing Strict-Transport-Security header was resolved, while one low-severity configuration finding remained due to the disclosure of the server's PHP version through the X-Powered-By header. Overall, the assessment demonstrated the effectiveness of the implemented corrective measures in improving the web application's security header compliance. The detailed assessment results and raw HTTP response headers are provided in the Systems Security Compliance Report appendix.

Mobile Application Security and Data Privacy Compliance

The Mobile Application Security and Data Privacy Compliance Assessment was conducted to evaluate the security controls and data privacy practices implemented in the SK 360° Android mobile application. The proponents employed Static Application Security Testing (SAST), configuration review, and functional security testing using the OWASP Mobile Application Security Verification Standard (MASVS) as the assessment framework. Mobile Security Framework (MobSF) was used to analyze the Android application package (APK) for potential security weaknesses, while Android Debug Bridge (ADB), Visual Studio Code, and PowerShell were utilized to examine application permissions, security configurations, and relevant security implementations. The assessment covered secure data storage, cryptography, authentication, network communication, platform interaction, code quality, resilience, and privacy practices. The findings were documented and classified according to the application's observed compliance with the identified secure coding practices.

Table 57 Mobile Secure Coding Practices Compliance

OWASP MASVS	Secure Coding Practices	Compliance Status
STORAGE	The app securely stores sensitive data.	Compliant
	The app prevents leakage of sensitive data.	Compliant
CRYPTO	The app employs current strong cryptography and uses it according to industry best practices.	Compliant
	The app performs key management according to industry best practices.	Compliant
AUTH	The app uses secure authentication and authorization protocols and follows the relevant best practices.	Compliant
	The app performs local authentication securely according to the platform best practices.	Not Compliant
	The app secures sensitive operations with additional authentication.	Compliant
NETWORK	The app secures all network traffic according to the current best practices.	Compliant
	The app performs identity pinning for all remote endpoints under the developer's control.	Non-Compliant
PLATFORM	The app uses IPC mechanisms securely.	Compliant
	The app uses WebView securely.	Compliant
	The app uses the user interface securely.	Compliant

CODE	The app requires an up-to-date platform version.	Non Compliant
	The app has a mechanism for enforcing app updates.	Non-Compliant
	The app only uses software components without known vulnerabilities.	Compliant
	The app validates and sanitizes all untrusted inputs.	Compliant
RESILIENCE	The app validates the integrity of the platform.	Non-Compliant
	The app implements anti-tampering mechanisms.	Partially Compliant
	The app implements anti-static analysis mechanisms.	Partially Compliant
	The app implements anti-dynamic analysis techniques.	Partially Compliant
PRIVACY	The app minimizes access to sensitive data and resources.	Compliant
	The app prevents identification of the user.	Partially Compliant
	The app is transparent about data collection and usage.	Partially Compliant
	The app offers user control over their data.	Compliant

The results of the Mobile Application Security and Data Privacy Compliance Assessment presented in Table 57 indicate that the SK 360° Android mobile application complied with most of the evaluated secure coding practices based on the OWASP Mobile Application Security Verification Standard (MASVS). The assessment identified satisfactory security controls in sensitive data storage, cryptography, authentication, platform interaction, and input validation. However, several security weaknesses were identified, particularly the absence of certificate pinning, mechanisms for enforcing application updates, and platform integrity validation. The application also did not fully meet the requirement for an up-to-date minimum supported Android version. Partial compliance was observed in anti-tampering, anti-static analysis protection, anti-dynamic analysis, user identification protection, and transparency regarding data collection and usage. These findings indicate that the application implements several essential security controls while retaining areas that require further improvement in security, resilience, and data privacy protection.

User Acceptance and Systems Usability Evaluation

The overall evaluation of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City used a quantitative user acceptance and systems usability evaluation approach. The evaluation determined the respondents' assessment of the system's software quality, including functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The evaluation included thirty-six (36) respondents composed of one (1) SK Federation President, ten (10) SK Chairmen, ten (10) SK Secretaries, ten (10) public users, and five (5) IT professionals. The SK officials evaluated the authenticated system functions assigned to their roles, while public users assessed the publicly accessible features of the web application. The IT professionals also completed the evaluation questionnaire and conducted separate functional and non-functional technical testing.

Evaluation Instrument

The evaluation questionnaire was adopted and contextualized from the ISO/IEC 25010-based evaluation instrument presented in Management Information System of Barangay Daang Bukid, Bacoor City, Cavite by Santiago et al. (2022). The questionnaire was modified to suit the functions, user roles, access permissions, and web and Android mobile platforms of SK 360°. The evaluation retained the eight software product quality characteristics established under ISO/IEC 25010:2011: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The questionnaire evaluates eight software quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Each characteristic contains applicable subcriteria used to assess the respondents' evaluation of the system. The questionnaire uses a five-point rating scale ranging from 5 – Excellent to 1 – Poor and will be administered through Google Forms. The adapted questionnaire and completed survey documents are included in the appendix under the User Evaluation Survey Report.

Sample Size and Sampling Technique

The study will use a role-based usability evaluation approach to identify the respondents. The respondents will be selected according to their relationship to the intended users, beneficiaries, and technical evaluation requirements of SK 360°. The selection will be based on the respondents' roles and their ability to evaluate the functions and software quality characteristics relevant to SK 360°.

A total of thirty-six (36) respondents will participate in the system evaluation. The respondent distribution consists of one (1) SK Federation President, ten (10) SK Chairmen, ten (10) SK Secretaries, ten (10) public users, and five (5) IT professionals. The computation of the respondents is expressed as follows:

Total respondents = 1 SK Federation President + 10 SK Chairmen + 10 SK Secretaries + 10 Public Users + 5 IT Professionals

Total respondents = 36

No Slovin's formula will be applied because the respondents are identified according to predetermined user roles and technical evaluation requirements. The SK Federation President will assess administrative, monitoring, approval, consolidation, and management functions. The SK Chairmen and SK Secretaries will assess reporting, budget and financial documentation, communication, notifications, scheduling, leadership information, and other functions permitted by their roles. Public users evaluated the web application's announcements, calendar, leadership information, annual budget, published LYDP, and feedback feature with email OTP verification.

The five (5) IT professionals will also complete the evaluation questionnaire. In addition, they will conduct separate functional and non-functional testing using prepared test cases to identify errors, inconsistencies, performance issues, compatibility concerns, and other technical problems. Their questionnaire responses will be included in the survey results, while their technical testing results will be analyzed separately.

Data Analysis Strategy

The data analysis strategy was used to determine the respondents' overall assessment of SK 360°. Data were collected through an online questionnaire administered using Google Forms. A five-point rating scale was used to measure the respondents' assessment of the eight ISO/IEC 25010 software quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The collected responses were summarized using the weighted mean to determine the overall assessment for each statement and criterion. The standard deviation was also computed to determine the consistency or variation of the respondents' ratings. Higher mean scores indicate a more favorable assessment of the system, while lower scores indicate a less favorable assessment.

The technical testing results of the five (5) IT professionals were analyzed separately from their questionnaire responses. Functional test results were assessed using pass-or-fail outcomes, while non-functional results were examined using recorded measurements, scores, and applicable acceptance criteria. These findings supported the assessment of whether the web and Android mobile applications operated according to their documented requirements and expected behavior.

Numerical Rating	Range	Interpretation
5	4.21–5.00	Excellent
4	3.41–4.20	Very Good
3	2.61–3.40	Good
2	1.81–2.60	Fair
1	1.00–1.80	Poor

Table 58 Likert Scale

Table 58 presents the five-point evaluation scale used to interpret the questionnaire results. A weighted mean ranging from 4.21 to 5.00 indicates an excellent assessment. A mean ranging from 3.41 to 4.20 indicates a very good assessment, while a mean between 2.61 and 3.40 indicates a good assessment. Scores from 1.81 to 2.60 indicate a fair assessment, while scores from 1.00 to 1.80 indicate a poor assessment. The scale was applied to the eight ISO/IEC 25010

software quality characteristics evaluated in the questionnaire. The resulting ratings were used to describe the respondents' perceptions of the quality and acceptability of SK 360°.

Evaluation Metrics

The weighted mean and standard deviation were used to analyze the respondents' assessment of SK 360°. These statistical measures provided a quantitative interpretation of the respondents' experiences and evaluations based on the eight ISO/IEC 25010 software quality characteristics. The evaluation criteria included functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

A. Weighted Mean

The weighted mean was calculated using the frequency of each numerical rating.

Formula:

$$\bar{x}_w = \frac{\sum(wx)}{\sum w}$$

Where:

- $\bar{x}_w$ = Weighted Mean
- w = Frequency or weight assigned to each response
- x = Numerical value of the response
- $\sum w$ = Total number of responses

The overall assessment for each evaluation criterion was determined using the weighted mean. A higher weighted mean indicates a more favorable evaluation of SK 360°, while a lower weighted mean indicates a less favorable evaluation.

The evaluation criteria were based on the questionnaire drawn from Santiago et al. (2022) and used to assess SK 360°. The computed results were compared with the interpretation ranges presented in Table 58. These results were used to identify the system's perceived strengths and areas requiring improvement.

B. Standard Deviation

The sample standard deviation was calculated to describe the variation in the respondents' ratings.

Formula:

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n - 1}}$$

Where:

- s = Standard Deviation
- x_i = Individual response value
- $\bar{x}$ = Mean of the responses
- n = Total number of responses

A low standard deviation indicates that the respondents provided similar ratings, while a high standard deviation indicates greater variation in their assessments. This variation may reflect differences in the respondents' experiences with SK 360°. The reported standard deviations summarize the combined questionnaire responses of the thirty-six (36) respondents across the evaluated software quality characteristics rather than separate respondent-group results.

System Evaluation Results Discussions

This section presents and discusses the results of the user acceptance and systems usability evaluation of SK 360°. It includes the demographic profile of the thirty-six (36) respondents, the combined evaluation results for each of the eight ISO/IEC 25010-based software quality characteristics, and the overall system evaluation. The questionnaire responses, including those of the five (5) IT professionals, were analyzed using the weighted mean and standard deviation. Technical testing findings were considered separately from the survey ratings. The questionnaire and anonymized completed evaluation documents are included in the Appendix under the User Evaluation Survey Report.

Table 59 Demographic Characteristics of Respondents

Demographics	Frequency	Percentage
Respondents		
IT Professionals	5	13.89%
SK Federation President	1	2.78%
SK Chairmen	10	27.78%
SK Secretaries	10	27.78%
Public Users	10	27.78%
TOTAL	**36**	**100.00%**

Table 59 presents the distribution of the thirty-six (36) respondents who completed the system evaluation questionnaire. SK Chairmen, SK Secretaries, and public users each accounted for ten (10) respondents or 27.78% of the total. Five (5) IT professionals represented 13.89%, while the SK Federation President accounted for 2.78%. This distribution included perspectives from administrative users, barangay-level officials, public users, and technical evaluators. The IT professionals also conducted separate technical testing. These categories provided different perspectives on SK 360°, with assessments interpreted according to the respondents' roles and experience with the system.

Table 60 Demographic Characteristics of Respondents by Profession

Demographics	Frequency	Percentage
IT Profession		
Faculty Member	2	40.00%
QA Consultant	1	20.00%
Web Developer	1	20.00%
BI Analyst / Software Engineer / Web Developer	1	20.00%
TOTAL	5	100.00%

Table 60 presents the professional profile of the five (5) IT professionals who participated in the system evaluation. Two (2) respondents were faculty members, representing 40% of the group. The remaining respondents included one (1) QA consultant, one (1) web developer, and one (1) respondent identified as a BI analyst, software engineer, and web developer, each representing 20%. Their professional backgrounds provided academic and industry perspectives for evaluating SK 360°. These respondents completed the evaluation questionnaire and also participated in separate technical testing of the system.

Table 61 Respondents Evaluation Summary Results for Functionality Suitability

Statements	Mean	Standard Deviation	Interpretation
Functional Completeness: Degree to which the set of functions covers all the specified tasks and intended users' objectives.	4.78	0.485	Excellent
Functional Correctness: Degree to which a product or system provides accurate results when used by intended users.	4.72	0.513	Excellent
Functional Appropriateness: Degree to which the functions facilitate the accomplishment of specified tasks and objectives.	4.75	0.554	Excellent
Grand Mean	**4.75**		**Excellent**

Table 61 presents the functional suitability evaluation of SK 360° based on the responses of thirty-six (36) respondents. The system obtained a grand mean of 4.75, interpreted as Excellent. Functional completeness received the highest mean of 4.78, followed by functional appropriateness at 4.75 and functional correctness at 4.72. These findings indicate that respondents favorably assessed

the system's coverage of required functions, accuracy of results, and support for accomplishing intended tasks. Although functional correctness received the lowest mean, all evaluated subcriteria remained within the Excellent range.

Table 62 Respondents Evaluation Summary Results for Performance Efficiency

Statements	Mean	Standard Deviation	Interpretation
Time Behaviour: Degree to which the response time and throughput rates of a product or system, when performing its functions, meet requirements.	4.69	0.525	Excellent
Resource Utilization: Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.	4.67	0.586	Excellent
Capacity: Degree to which the maximum limits of a product or system parameter meet requirements.	4.58	0.604	Excellent
Grand Mean	**4.65**		**Excellent**

Table 62 presents the performance efficiency evaluation of SK 360° based on the responses of thirty-six (36) respondents. The system obtained a grand mean of 4.65, interpreted as Excellent. Time behaviour received the highest mean of 4.69, followed by resource utilization at 4.67 and capacity at 4.58. These findings indicate favorable assessments of the system's responsiveness, resource use, and ability to accommodate operational demands. Capacity received the lowest mean but remained within the Excellent range. These survey findings reflect respondents' perceptions and complement the actual performance testing results.

Table 63 Respondents Evaluation Summary Results for Compatibility

Statements	Mean	Standard Deviation	Interpretation
Co-existence – Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.	4.67	0.535	Excellent
Interoperability – Degree to which two or more systems, products or components can exchange information and use the information that has been exchanged.	4.69	0.577	Excellent
Grand Mean	**4.68**		**Excellent**

Table 63 presents the compatibility evaluation of SK 360° based on the responses of thirty-six (36) respondents. The system obtained a grand mean of 4.68, interpreted as Excellent. Interoperability received the higher mean of 4.69, while co-existence obtained 4.67. These findings indicate that respondents favorably assessed the system's ability to exchange information among connected components and operate alongside other applications. Both subcriteria received Excellent interpretations, suggesting a positive assessment of compatibility. These results reflect respondents' perceptions and complement the findings from technical compatibility and integration testing.

Table 64 Respondents Evaluation Summary Results for Usability

Statements	Mean	Standard Deviation	Interpretation
Appropriateness Recognizability – Degree to which users can recognize whether a product or system is appropriate for their needs.	4.67	0.586	Excellent
Learnability – Degree to which a product or system can be used by specified users to achieve specified goals of learning to use the product or system with effectiveness, efficiency, freedom from risk, and satisfaction in a specified context of use.	4.78	0.540	Excellent
Operability – Degree to which a product or system has attributes that make it easy to operate and control.	4.78	0.540	Excellent
User Error Protection – Degree to which a system protects users against making errors.	4.67	0.586	Excellent
User Interface Aesthetics – Degree to which a user interface enables pleasing and satisfying interaction for the user.	4.75	0.500	Excellent
Accessibility – Degree to which a product or system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.	4.78	0.540	Excellent
Grand Mean	**4.74**		**Excellent**

Table 64 presents the usability evaluation of SK 360° based on the responses of thirty-six (36) respondents. The system obtained a grand mean of 4.74, interpreted as Excellent. Learnability, operability, and accessibility received the highest mean of 4.78 each, followed by user interface aesthetics at 4.75. Appropriateness recognizability and user error protection both obtained 4.67, remaining within the Excellent range. These findings suggest that respondents

generally perceived the system as easy to learn and operate, with a pleasing interface and support for their needs. The accessibility rating reflects respondents' perceptions and does not independently establish accessibility compliance.

Table 65 Respondents Evaluation Summary Results for Reliability

Statements	Mean	Standard Deviation	Interpretation
Maturity – Degree to which a system, product or component meets needs for reliability under normal operation.	4.72	0.615	Excellent
Availability – Degree to which a system, product or component is operational and accessible when required for use.	4.67	0.632	Excellent
Fault Tolerance – Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.	4.72	0.513	Excellent
Recoverability – Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.	4.69	0.577	Excellent
Grand Mean	**4.70**		**Excellent**

Table 65 presents the reliability evaluation of SK 360° based on the responses of thirty-six (36) respondents. The system obtained a grand mean of 4.70, interpreted as Excellent. Maturity and fault tolerance received the highest mean of 4.72 each, followed by recoverability at 4.69 and availability at 4.67. These findings indicate that respondents favorably assessed the system's dependability, accessibility when needed, and perceived ability to handle interruptions and recover from failures. Although availability received the lowest mean, all subcriteria remained within the Excellent range. These assessments reflect respondents' perceptions and should be considered alongside technical testing findings on system reliability.

Table 66 Respondents Evaluation Summary Results for Security

Statements	Mean	Standard Deviation	Interpretation
Confidentiality – Degree to which a product or system ensures that data are accessible only to those authorized to have access.	4.72	0.513	Excellent
Integrity – Degree to which a system, product or component prevents	4.75	0.500	Excellent

unauthorized access to, or modification of, computer programs or data.			
Non-repudiation – Degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.	4.78	0.485	Excellent
Accountability – Degree to which the actions of an entity can be traced uniquely to the entity.	4.78	0.485	Excellent
Authenticity – Degree to which the identity of a subject or resource can be proved to be the one claimed.	4.78	0.485	Excellent
Grand Mean	**4.76**		**Excellent**

Table 66 presents the security evaluation of SK 360° based on the responses of thirty-six (36) respondents. The system obtained a grand mean of 4.76, interpreted as Excellent. Non-repudiation, accountability, and authenticity received the highest mean of 4.78 each, followed by integrity at 4.75 and confidentiality at 4.72. These findings indicate favorable perceptions of the system's protection of restricted information, prevention of unauthorized changes, traceability of actions, and verification of user identity. Although confidentiality received the lowest mean, all subcriteria remained within the Excellent range. These survey results reflect respondents' assessments and complement technical security testing rather than independently establish the effectiveness of security controls.

Table 67 Respondents Evaluation Summary Results for Maintainability

Statements	Mean	Standard Deviation	Interpretation
Modularity – Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.	4.69	0.624	Excellent
Reusability – Degree to which an asset can be used in more than one system, or in building other assets.	4.72	0.513	Excellent
Analyzability – Degree of effectiveness and efficiency with which it is possible to assess the impact of an intended change, diagnose deficiencies or causes of failures, or identify parts to be modified.	4.69	0.624	Excellent
Modifiability – Degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.	4.72	0.513	Excellent
Testability – Degree of effectiveness and efficiency with which test criteria can be established and tests can be	4.61	0.599	Excellent

performed to determine whether those criteria have been met.			
Grand Mean	**4.69**		**Excellent**

Table 67 presents the maintainability evaluation of SK 360° based on the responses of thirty-six (36) respondents. The system obtained a grand mean of 4.69, interpreted as Excellent. Reusability and modifiability received the highest mean of 4.72 each, followed by modularity and analyzability at 4.69 each. Testability obtained the lowest mean of 4.61 but remained within the Excellent range. These findings indicate favorable respondent assessments across the five maintainability subcriteria. However, because maintainability involves technical characteristics that may not be directly observable to all users, these ratings should be interpreted alongside the IT professionals' technical findings and evidence from system development and maintenance activities.

Table 68 Respondents Evaluation Summary Results for Portability

Statements	Mean	Standard Deviation	Interpretation
Adaptability – Degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software, or other operational or usage environments.	4.78	0.485	Excellent
Installability – Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.	4.72	0.513	Excellent
Replaceability – Degree to which a product can replace another specified software product for the same purpose in the same environment.	4.72	0.513	Excellent
Grand Mean	**4.74**		**Excellent**

Table 68 presents the portability evaluation of SK 360° based on the responses of thirty-six (36) respondents. The system obtained a grand mean of 4.74, interpreted as Excellent. Adaptability received the highest mean of 4.78, while installability and replaceability both obtained 4.72. These findings indicate favorable respondent assessments of the system's ability to adapt to different operating environments, support installation procedures, and serve as an alternative to software with similar purposes. All three subcriteria remained within the Excellent range. These ratings reflect respondents' perceptions and should be

considered alongside actual installation and compatibility testing within the system's supported environments.

Table 69 Overall Summary of Respondents' System Evaluation Results

Evaluation Criteria	Mean	Interpretation
Functional Suitability	4.75	Excellent
Performance Efficiency	4.65	Excellent
Compatibility	4.68	Excellent
Usability	4.74	Excellent
Reliability	4.70	Excellent
Security	4.76	Excellent
Maintainability	4.69	Excellent
Portability	4.74	Excellent
Grand Mean	**4.71**	**Excellent**

Table 69 presents the overall evaluation results of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City based on the assessments of thirty-six (36) respondents across eight software quality characteristics. The system obtained a grand mean of 4.71, interpreted as Excellent, indicating an overall favorable assessment among the participating SK officials, public users, and IT professionals.

Security received the highest mean of 4.76, followed by functional suitability at 4.75. These results indicate favorable respondent perceptions of the system's protection of information, management of authorized access, and provision of functions that support intended tasks. Usability and portability both obtained mean scores of 4.74, reflecting positive assessments of the system's ease of use and suitability for supported operating environments.

Reliability obtained a mean of 4.70, followed by maintainability at 4.69 and compatibility at 4.68. These scores reflect favorable assessments of the system's dependability, capacity for modification and maintenance, and interaction with other components or applications. Performance efficiency received the lowest mean of 4.65 but remained within the Excellent range. Although the differences among the criterion means were small, the comparatively lower performance efficiency score identifies an area for further examination, particularly in relation to responsiveness, resource utilization, and capacity.

The evaluation findings also support the assessment of the project's stated objectives. The developed web and Android mobile applications incorporate centralized accomplishment and budget reporting, document management and archiving, federation-level report consolidation, budget allocation and liquidation documentation, announcements, scheduling, real-time communication, leadership management, and public information access. The implementation of Point System Ranking, descriptive analytics, visualization dashboards, and Key Performance Indicators (KPIs) further addresses the objectives related to participation monitoring, reporting compliance, budget utilization, and data-driven decision-making. These implemented functions demonstrate the alignment of the developed system with its functional objectives, while the evaluation results address the objective of assessing the system using the eight ISO/IEC 25010-based software quality characteristics. The overall Excellent rating indicates that respondents positively assessed the developed system in relation to the evaluated quality criteria.

Furthermore, the participation of the SK Federation President, SK Chairmen, SK Secretaries, public users, and IT professionals provided different perspectives on the system’s overall quality and acceptability. The consistently Excellent ratings across the eight software quality characteristics indicate favorable respondent assessments, while emphasizing the importance of continued monitoring, user feedback, and periodic evaluation to address evolving operational and technological requirements.

Overall, the findings support a favorable assessment of SK 360° among the selected respondents. However, the survey results represent their perceptions and experiences during the evaluation and do not independently establish the effectiveness of every technical characteristic. The results should therefore be interpreted alongside the functional and non-functional testing findings. Together, these sources of evidence support the assessment of the project's objectives and provide a basis for further system refinement, helping the proponents prioritize improvements that address the operational needs of the SK Federation of Lipa City..

CHAPTER 5
SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY

The study focused on the design, development, and evaluation of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, a web and Android mobile-based system intended to support the administrative, operational, and coordination requirements of the SK Federation of Lipa City and its seventy-two (72) barangay SK councils. The project was developed to address challenges associated with manual report submissions, fragmented record management, inconsistent financial documentation, delayed communication, and limited centralized monitoring of barangay SK activities. The operational requirements were identified through an interview with the SK Federation President, which provided information regarding existing processes, organizational concerns, and required system functionalities. These findings served as the basis for developing an integrated platform that supports centralized documentation, communication, transparency, and monitoring within the defined scope of the project.

The developed system incorporated centralized accomplishment and budget reporting, document management and archiving, federation-level report review and consolidation, and budget allocation and liquidation documentation. Accomplishment reports were submitted through completed document uploads, while budget reporting supported direct data entry or document upload through the web application and document upload through the Android mobile application. Budget-related records were organized according to monthly, quarterly, semi-annual, and annual reporting categories to support record organization and monitoring. The web application also provided consolidated PDF generation for selected accomplishment and budget reports, while the mobile application supported the monitoring of submission statuses and available quality documentation. These functionalities addressed the objectives related to centralized reporting, documentation management, financial record organization, and archival preservation.

The system further integrated announcements, an activity calendar, meeting scheduling and participation, real-time messaging and virtual meetings, leadership information management, and account and leadership transition support. A Point System

Ranking was implemented to record positive and negative points based on documentation, compliance, late submissions, and official meeting participation. Descriptive analytics, visualization dashboards, and Key Performance Indicators (KPIs) were incorporated to present information concerning submission performance, participation, completed and pending reports, budget utilization, and yearly comparisons. The web application also provided public access to selected announcements, calendar information, leadership information, annual budget information, and the published Local Youth Development Plan (LYDP), together with email OTP-verified feedback and commenting. These implemented functions demonstrated the alignment of the developed platform with the objectives concerning communication, coordination, participation monitoring, public information access, and data-driven monitoring.

The system evaluation involved thirty-six (36) respondents consisting of one (1) SK Federation President, ten (10) SK Chairmen, ten (10) SK Secretaries, ten (10) public users, and five (5) IT professionals. The questionnaire evaluated eight ISO/IEC 25010-based software quality characteristics, namely functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The system obtained an overall grand mean of 4.71, interpreted as Excellent. Security received the highest mean of 4.76, followed by functional suitability at 4.75, while usability and portability both obtained 4.74. Reliability received 4.70, maintainability obtained 4.69, compatibility received 4.68, and performance efficiency obtained 4.65. All eight characteristics were interpreted as Excellent, indicating favorable assessments of the system among the participating respondents.

In addition to the questionnaire evaluation, functional and non-functional technical testing was conducted to examine the behavior and technical characteristics of the developed applications. The web application security assessment demonstrated an improvement in HTTP security header compliance from an A rating to an A+ rating following the implementation of corrective measures. However, a low-severity information disclosure finding remained. The Android mobile application security assessment also identified implemented security controls alongside areas requiring further improvement, including certificate pinning, application update enforcement, and platform integrity validation. These findings provided supplementary evidence concerning the system's technical condition and identified areas for continued enhancement.

Overall, the project addressed its six specific objectives through the development of the identified system functionalities and the completion of the software quality evaluation. The findings indicate that SK 360° provides an integrated platform capable of supporting the documented reporting, financial documentation, communication, coordination, monitoring, and public information requirements of the SK Federation of Lipa City. However, the favorable evaluation results represent the perceptions and experiences of the selected respondents, while technical testing findings are limited to the conditions and environments examined. The results therefore provide a basis for assessing the developed system within its defined scope and identifying improvements for its continued operational use.

5.2 CONCLUSION

Based on the development and evaluation findings, the proponents concluded that SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City addressed the identified need for a centralized platform that supports the administrative and operational processes of the SK Federation and barangay SK councils. The integration of accomplishment and budget reporting, document archiving, report consolidation, communication, scheduling, leadership management, and monitoring functions demonstrated that multiple youth governance activities could be organized within a single web and Android mobile-based system. The developed platform provides a structured alternative to fragmented documentation and communication practices by establishing common digital workflows and centralized access to authorized records and information.

The implementation of budget documentation, Point System Ranking, and descriptive analytics further demonstrated the applicability of digital monitoring tools to local youth governance. By presenting submission statuses, participation records, budget information, and operational summaries, the system provides SK officials with information that can support administrative monitoring and decision-making. The availability of selected public information and OTP-verified feedback through the web application also establishes a digital channel for public access and participation. Nevertheless, the system functions as a documentation, coordination, and monitoring platform rather than an automated financial auditing or regulatory compliance verification mechanism.

Responsibility for validating financial information and complying with applicable government reporting requirements remains with the authorized SK officials.

The overall evaluation grand mean of 4.71, interpreted as Excellent, indicates that the developed system received favorable assessments across all eight evaluated software quality characteristics. These results suggest that the participating respondents positively perceived the system's functional suitability, performance, compatibility, usability, reliability, security, maintainability, and portability. However, the evaluation does not independently establish long-term operational effectiveness, measurable reductions in administrative workload, or complete technical compliance. The identified security and performance considerations also indicate that additional improvements and continuous assessment remain necessary to support reliable and secure system operation.

The study contributes to the application of information technology in local youth governance by presenting a centralized digital platform designed around the operational requirements of the SK Federation of Lipa City. Its integration of reporting, monitoring, communication, leadership continuity, and public information functions provides a possible reference for similar digital governance initiatives. The findings demonstrate the potential of an integrated system to support more organized administrative processes and accessible information, while emphasizing the importance of role-based access, user-centered functionality, technical testing, and continuous improvement. Future studies may further determine the platform's long-term operational impact and its applicability to other local government environments.

Furthermore, the development of SK 360° highlights the importance of institutional continuity and organized information management in youth governance. Through centralized document archiving, leadership profile management, and account transition support, the system provides a structured means of preserving records and maintaining access to relevant information across successive SK administrations. These functionalities support the continuity of administrative processes by reducing dependence on individual officials for storing and managing organizational records. The study therefore emphasizes the potential contribution of digital governance platforms in strengthening documentation practices, promoting organizational accountability, and supporting the long-term sustainability of youth governance operations within the SK Federation of Lipa City.

5.3 RECOMMENDATION

Based on the findings, evaluation results, and identified limitations of SK 360°: A Centralized Digital Platform for Youth Governance in Lipa City, the proponents recommend continued implementation and systematic monitoring of the developed platform within the SK Federation of Lipa City. Authorized SK officials should be encouraged to use the system consistently for accomplishment and budget report submission, record management, announcements, scheduling, communication, and monitoring. The Federation may establish standardized procedures for preparing, submitting, reviewing, and maintaining digital records to promote consistent use across barangay SK councils. Regular orientation and hands-on training sessions are also recommended to familiarize users with the system's role-based functionalities, reporting procedures, and document management processes.

The proponents further recommend establishing clear operational responsibilities for system administration, account management, data verification, and record maintenance. Authorized administrators should regularly review user accounts, access permissions, and leadership information, particularly during changes in SK administration. Periodic database backups, recovery testing, system maintenance, and monitoring of operational records should be conducted to support data availability, continuity, and protection against accidental information loss. The Federation should also maintain appropriate procedures for verifying submitted financial documents and ensuring compliance with existing government reporting requirements, since the platform does not replace official financial auditing procedures or mandatory physical documentation.

For technical enhancement, future developers are encouraged to strengthen the security mechanisms of both the web and Android mobile applications. Improvements may include addressing remaining server information disclosure, implementing appropriate certificate pinning, introducing mechanisms for enforcing application updates, and strengthening platform integrity and application resilience controls. Continued security testing, vulnerability assessment, dependency review, and validation of access permissions should be conducted to identify weaknesses introduced by future system updates. These activities would help maintain the protection of sensitive records and improve the system's security posture as its operational use expands.

Further optimization of system performance is also recommended, particularly because performance efficiency obtained the lowest mean among the eight evaluated software quality characteristics, despite remaining within the Excellent range. Future development may focus on improving page and screen responsiveness, optimizing database queries, managing document storage efficiently, and examining resource utilization during simultaneous user activities. Additional performance and load testing under representative operational conditions should be conducted to determine the system's capacity to support increasing numbers of active users, submitted reports, uploaded documents, and communication activities across the participating barangays.

Future researchers and developers may explore additional functionalities beyond the current project scope, subject to operational requirements, feasibility, data privacy considerations, and approval from the appropriate authorities. Possible enhancements include integration with authorized government information systems, expanded reporting and data visualization capabilities, improved mobile document management, and carefully evaluated offline functionality for selected tasks. Advanced analytical methods may also be investigated to provide additional insights into reporting patterns, participation, and budget-related information. However, such enhancements should undergo separate requirements analysis, security review, and technical evaluation before implementation.

Finally, future studies are encouraged to evaluate SK 360° over a longer period of actual operational use to determine its effects on reporting timeliness, document retrieval, administrative workload, participation monitoring, and user satisfaction. Additional research involving a larger number of SK officials and barangay councils may provide broader evidence concerning the system's effectiveness and usability. Comparative studies with existing manual or fragmented digital workflows may also help establish measurable operational improvements. These future evaluations would provide a stronger basis for assessing the long-term value of SK 360° and determining whether its approach could be adapted for other youth governance organizations or local government settings.

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